

115.109

Introduction to Business Data Analysis

Topic 2: Summarising & Dashboards

Dr Andrew Murphy

(Please contact me via Communication Tools in Stream)

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assess ments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Summarising Data

- Sources of information
- Constructing tables
- Line graph, pie charts, bar charts, pictograms
- Frequency distributions
- Misusing statistics
- Net Promoter Score
- Dashboards

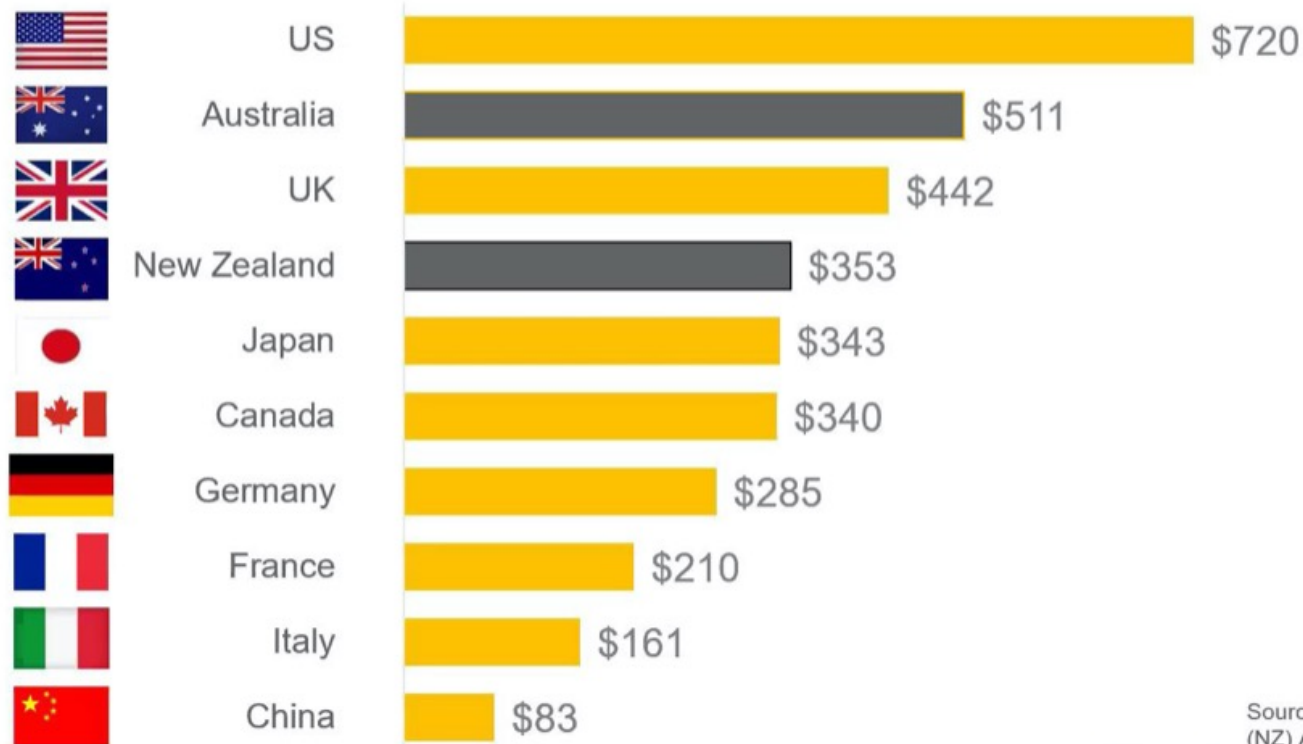
Data Sources

- A variety of sources of available information include:
 - Company data (often confidential)
 - Official sites (eg Statistics NZ)
 - Semi-official sites (eg Nielsen)
 - Survey reports: vox-pop and ‘proper’ market research results published by research houses (eg Datamonitor) and media
 - Results from experiments, surveys
 - Academic journals

Brands across all industry sectors invest heavily to engage Australian and New Zealand shoppers and consumers



Media Ad Spend per Person (\$USD) | 2019



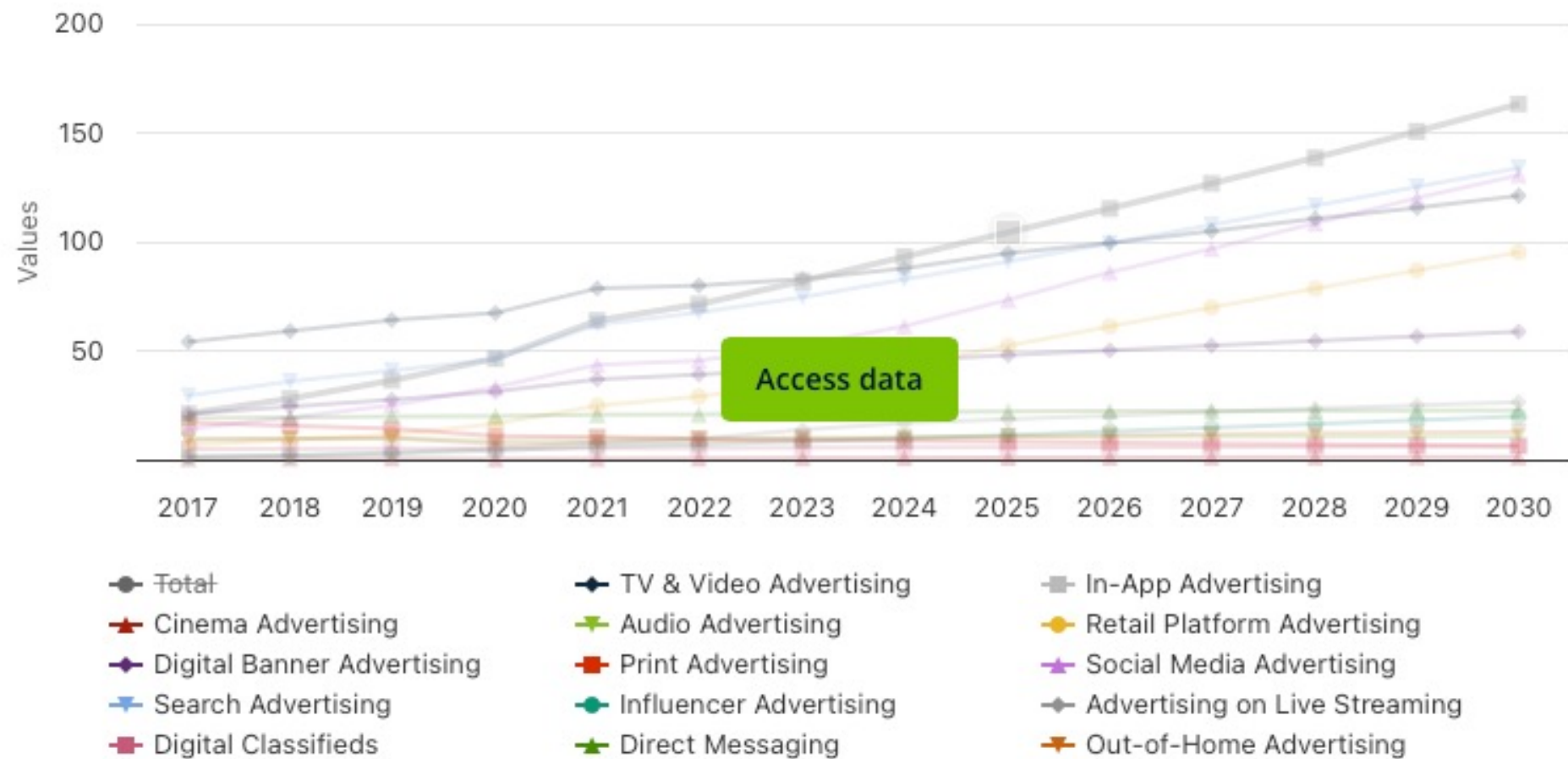
Source (Australia) eMarketer 2019
(NZ) Advertising Standards Authority



Worldwide Advertising per capita

<https://www.statista.com/outlook/amo/advertising/worldwide>

Average Total Ad Spending per Capita



Most recent update: Apr 2026

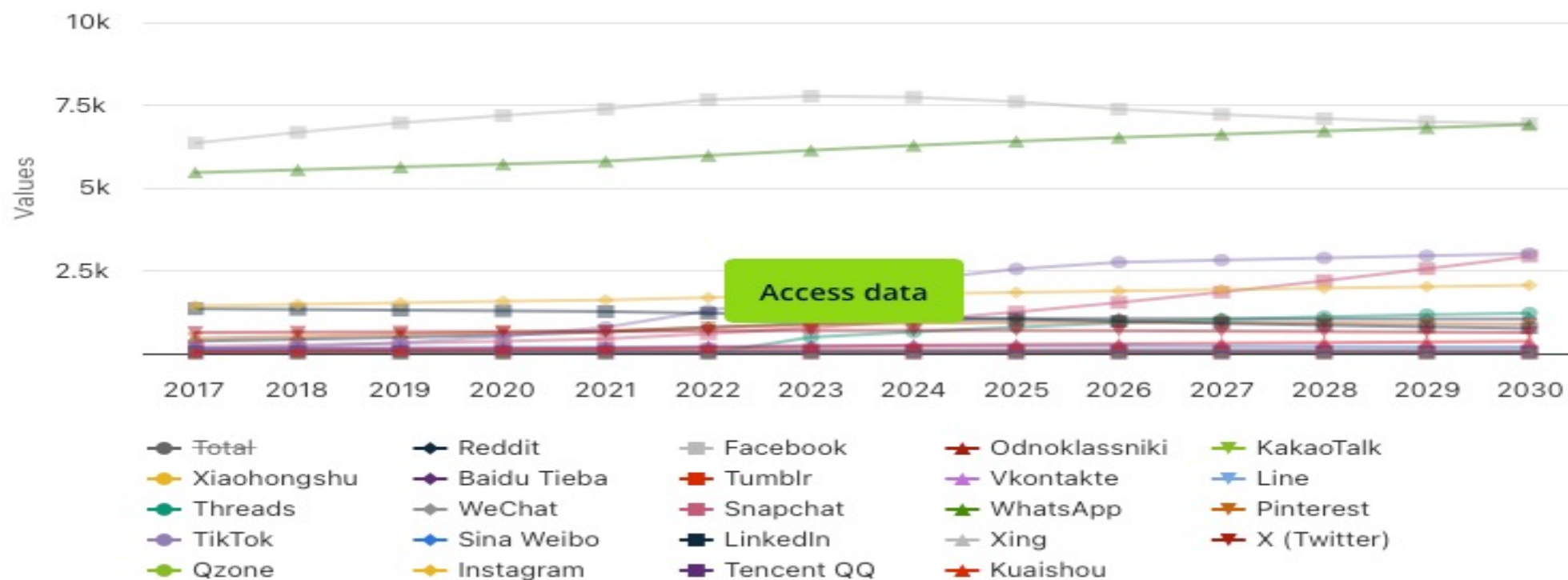
Source: Statista Market Insights



New Zealand Digital Advertising

<https://www.statista.com/outlook/amo/advertising/influencer-advertising/new-zealand#reach>

Reach by social network

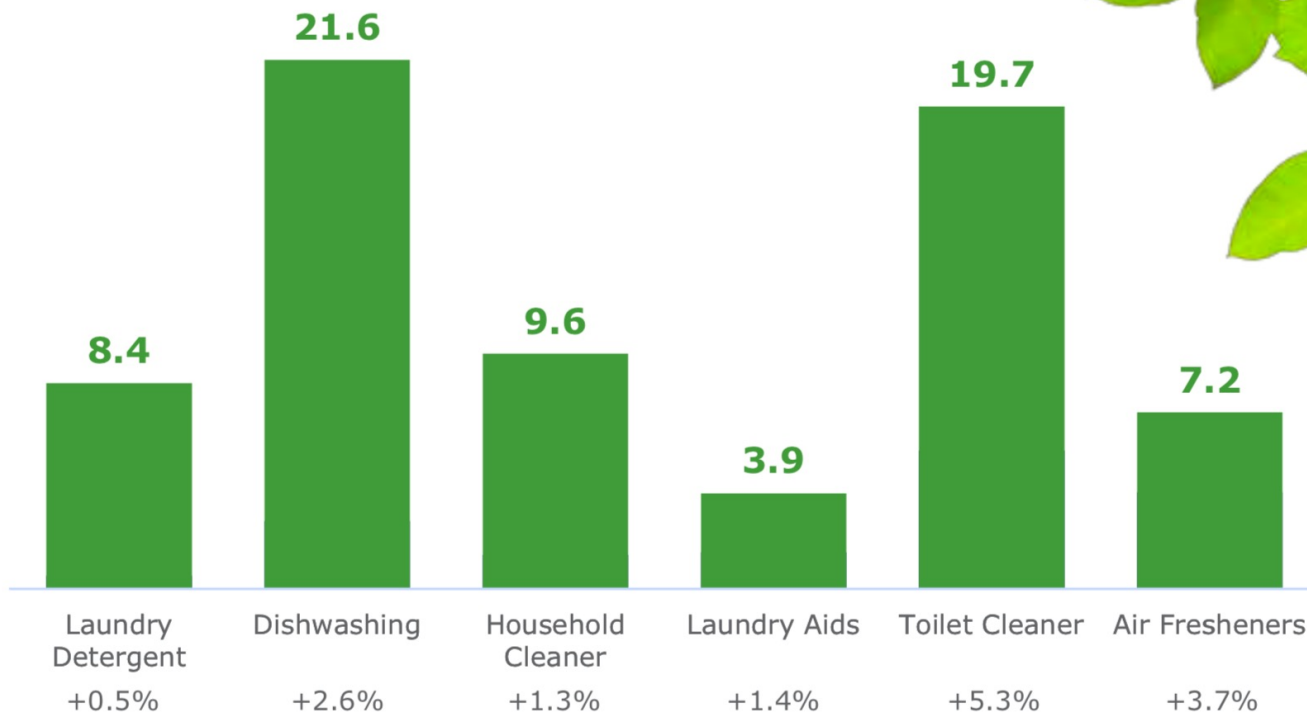


Most recent update: Apr 2026

Source: Statista Market Insights

SHOPPERS MIGRATE TO ECO-FRIENDLY PRODUCTS

Value Growth % vs YA | 2017



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Better for Me
Oi Products

Our 100% pure certified organic cotton pads, panty liners and tampons are naturally absorbent, safe, comfortable and hypoallergenic to provide comfort and protection you can trust.



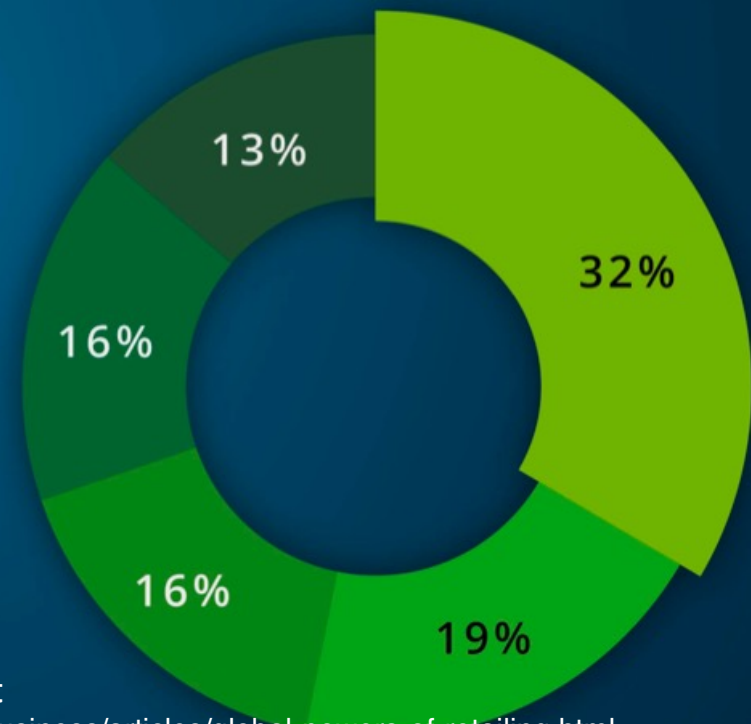
#3 value growth in feminine hygiene

Consumer demand sparks sustainability movement in retail

55% of consumers purchased a sustainable product or service within the past 4 weeks*

Of those surveyed....

- 32% Paid significantly more than an alternative
- 19% Waited longer for deliver or to obtain it
- 16% Bought a used or upcycled product
- 16% Accepted a lower quality product/service
- 13% Agreed to participate in a special process to dispose of it.



* October 2021

Deloitte's Global Powers of Retailing 2022 report

<https://www2.deloitte.com/global/en/pages/consumer-business/articles/global-powers-of-retailing.html>



SO WHO HAS DELIVERED ON THIS?

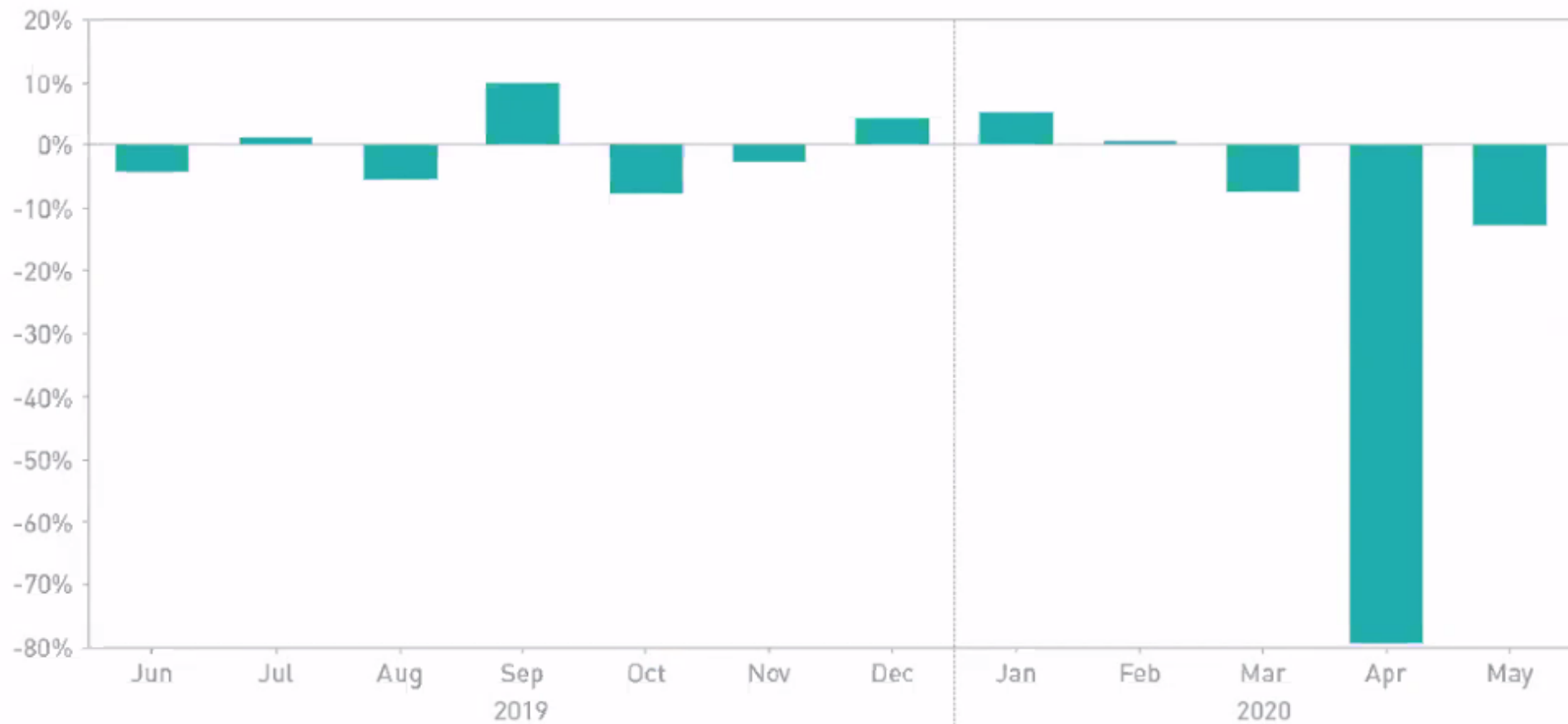
TOP 25

Rank	This Year	+/- \$ vs YA	Rank Change	Growth % MAT
1	Total Fonterra Brands NZ	\$621 M		
		+\$14.3M	—	2.4%
2	Total Goodman Fielder	\$574 M		
		+\$14.8M	—	2.6%
3	Total Heinz Watties	\$437 M		
		+\$10.1M	—	2.4%
4	Total CCA	\$405 M		
		+\$7.2M	—	1.8%
5	Total Lion Nathan Liquor	\$340 M		
		+\$4.7M	—	1.4%
6	Total Nestle NZ	\$267 M		
		+\$9.3M	—	3.6%
7	Total Unilever	\$250 M		
		-\$3.5M	—	-1.4%
8	Total Mondelez International	\$247 M		
		+\$7.3M	↑	3.0%
9	Total Frucor Suntory	\$237 M		
		-\$5.6M	↓	-2.3%
10	Total Griffins Foods	\$229 M		
		+\$2.8M	—	1.3%
11	Total Mars NZ	\$218 M		
		+\$14.3M	↑	7.0%
12	Total Pernod Ricard	\$216 M		
		+\$10.1M	—	4.9%
13	Total Db Breweries	\$205 M		
		-\$2.1M	↓	-1.0%
14	Total RB	\$200 M		
		-\$3.6M	—	-1.8%
15	Total Bluebird Foods	\$160 M		
		+\$14.2M	—	9.8%
16	Total George Weston Foods	\$150 M		
		+\$12.0M	—	8.7%
17	Total Hellers NZ	\$139 M		
		+\$3.6M	↑	2.7%
18	Total Colgate Palmolive	\$138 M		
		+\$2.5M	↓	1.8%
19	Total Kimberly Clark	\$136 M		
		+\$10.2M	↑	8.1%
20	Total Sanitarium P/L	\$132 M		
		-\$3.4M	↓	-2.5%
21	Total Cerebos Greggs	\$126 M		
		+\$3.4M	—	2.8%
22	Total Arnotts NZ Ltd	\$124 M		
		+\$4.4M	—	3.7%
23	Total Asaleo Care	\$115 M		
		-\$0.5M	—	-0.4%
24	Total Johnson & Johnson	\$112 M		
		-\$1.6M	—	-1.4%
25	Total Nestle Purina	\$103 M		
		+\$4.9M	—	5.0%

All manufacturers excl. P/L & Tobacco, top 25 by value



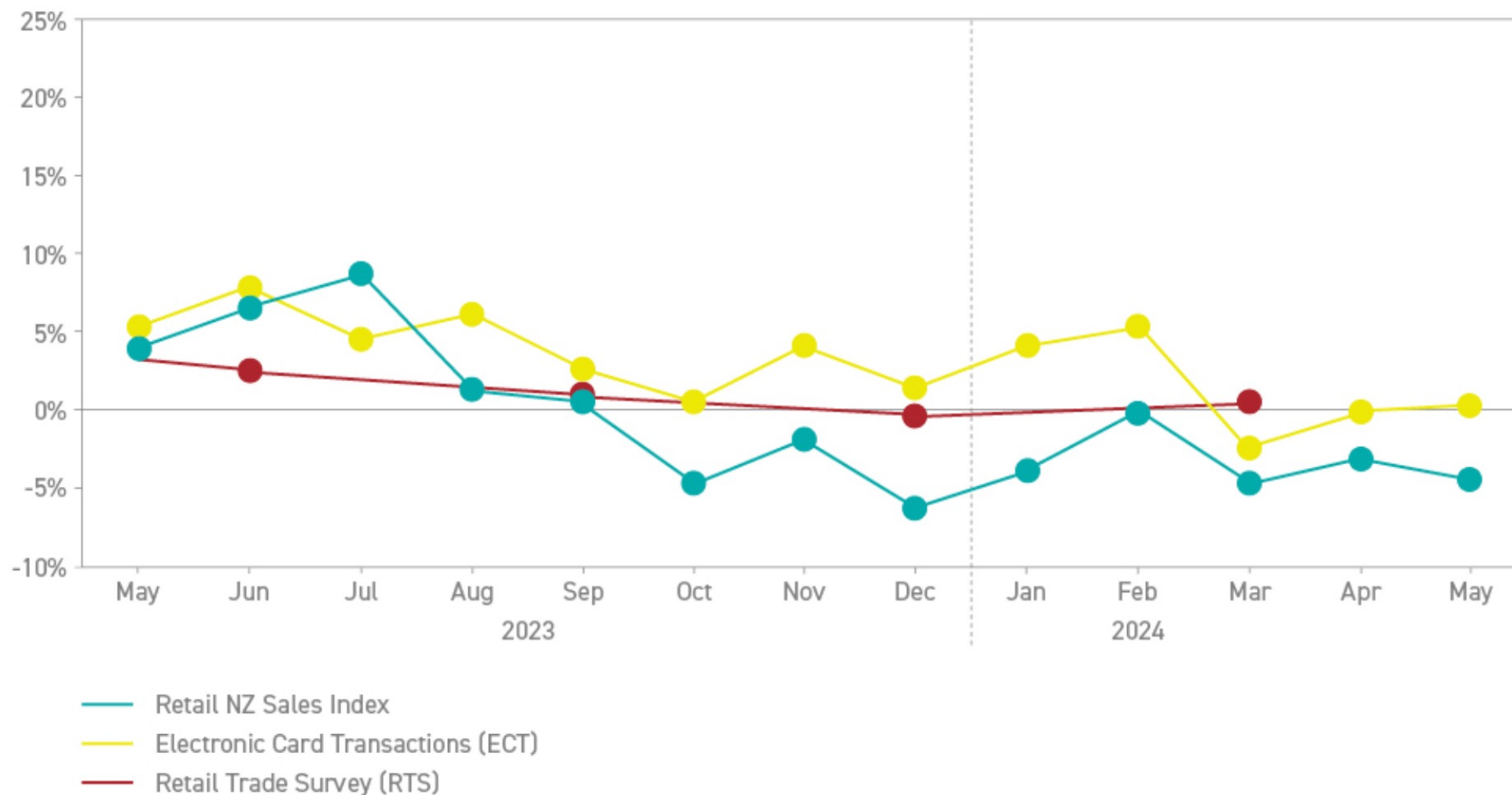
The big picture – Retail NZ Sales Index



Source: Retail NZ Sales Index

RetailNZ Sales Index

New Zealand spending compared to the previous year



Research and data to make progress against the world's largest problems

3579 charts across 297 topics
All free: open access and open source

TRUSTED IN RESEARCH AND MEDIA

Science nature PNAS  **BBC** The New York Times CNN
FT theguardian THE WALL STREET JOURNAL.  The Washington Post Vox

USED IN TEACHING

 HARVARD UNIVERSITY Stanford Berkeley  UNIVERSITY OF CALIFORNIA  UNIVERSITY OF CAMBRIDGE  UNIVERSITY OF OXFORD  MIT

Featured work

Coronavirus Data Explorer by Our World in Data



COVID-19 Data Explorer

Explore all of our data on COVID-19 vaccinations, cases, excess mortality, and much more.

By Our World in Data team

SDG Tracker

We launched a new version of our Sustainable Development Goals Tracker

With our SDG Tracker you can track global and country-by-country progress toward the United Nations Sustainable Development Goals (SDGs).

By Our World in Data team

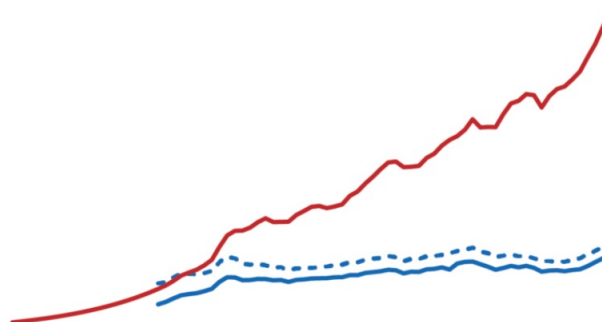
Population Growth by Our World in Data

We did a major overhaul of our work on Population Growth

Explore global and country data on population growth, demography, and how this is changing.

By Hannah Ritchie and Lucas Rod s-Guirao

Economic Inequality by Our World in Data



Mental Health by Our World in Data

Measuring progress towards the Sustainable Development Goals

The United Nations [Sustainable Development Goals](https://sdg-tracker.org) (SDGs) are targets for global development adopted in September 2015, set to be achieved by 2030. All countries of the world have agreed to work towards achieving these goals.

<https://sdg-tracker.org>



Global Powers of Retailing Top 250, FY2023 (2025 report)

<https://www.deloitte.com/content/dam/assets-shared/docs/industries/consumer/2025/deloitte-global-powers-of-retailing-2025.pdf>



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FY2023 retail revenue rank	Rank change from FY2022 ¹	Name of company	Country of origin	FY2023 retail revenue (US\$m)	FY2023 parent company/ group revenue ² (US\$m)	Dominant operational format	# countries of operation	FY2018- 2023 retail revenue CAGR ³	FY2022- 2023 retail revenue growth	FY2023 Net profit margin ⁴
1	● 0	Walmart Inc.	United States	648 125	648 125	Hypermarket/ supercenter	19	4.7%	6.0%	2.5%
2	● 0	Amazon.com, Inc.	United States	251 902	574 785	Non-store	23	12.4%	5.4%	5.3%
3	● 0	Costco Wholesale Corporation	United States	242 290	242 290	Cash & carry/ warehouse club	14	11.3%	6.8%	2.6%
4	● 0	Schwarz Group	Germany	177 009	181 015	Discount store	32	9.4%	8.9%	n/a
12	▲ 1	Ahold Delhaize	Netherlands	97 837 **	97 837 **	Supermarket	10	7.1%	1.9%	2.1%
13	▲ 1	Carrefour S.A.	France	90 803	93 708	Hypermarket/ supercenter	37	1.8%	2.2%	1.9%
14	▼ -2	Lowe's Companies, Inc.	United States	86 377	86 377	Home improvement	1	3.9%	-11.0%	8.9%
15	▲ 1	Tesco PLC	United Kingdom	85 218	86 243	Hypermarket/ supercenter	5	1.4%	4.2%	1.7%
16	▲ 1	Albertsons Companies, Inc.	United States	79 238	79 238	Supermarket	1	5.5%	2.0%	1.6%
17	▲ 1	Edeka-Verbund	Germany	75 930 e**	78 027 **	Supermarket	1	5.5%	6.5%	n/a
29	● 0	The IKEA Group (INGKA Holding B.V.)	Netherlands	42 960	48 180	Other specialty	31	2.6%	0.0%	3.4%
30	▼ -2	Woolworths Limited	Australia	42 005 **	45 303 **	Supermarket	2	1.6%	4.8%	0.2%

Data Sources

- There are basic rules that should be followed once a set of data has been collected:
 - the data must be factual and relevant
 - the data source is reliable (c.f. Wikipedia)
 - the data has been accurately coded / transcribed
 - that tables and figures are relevant to the problem:
no ‘chart junk’

Summarising: Tables

Australia has a total of 971 400 people between 15 and 34 years of age who participate in swimming/diving, outdoor soccer or cricket, comprising 473 300 swimming/diving, 309 300 soccer and 188 800 cricket. Of these, there are 204 100 aged 15–17 (71 300 swimming, 81 500 soccer and 51 300 cricket), 312 700 aged 18–24 (141 900 swimming, 120 200 soccer and 50 600 cricket) and 454 600 aged 25– 34 (260 100 swimming, 107 600 soccer and 86 900 cricket).

Table 11.1 The number of adults aged between 15 and 34 years of age in Australia who participate in swimming, soccer or cricket

Age (years)	Swimming/ diving	Soccer	Cricket	Total
15 – 17	71 300	81 500	51 300	204 100
18 – 24	141 900	120 300	50 600	312 700
25 – 34	260 100	107 600	86 900	454 600
Total	473 600	309 300	188 800	971 400

Source: Australian Bureau of Statistics, *Participation in Sport and Physical Education*, 2009-10, Cat. No. 4177.0

Summarising: Tables

When constructing a table, it is important to determine which elements and relationships should be emphasised: across columns (comparing sports for a given age) or rows (comparing ages for a given sport)? Absolute or relative (eg %, % change)?

Table 11.1 The share of adults aged 15-34 years of age in Australia, by participation in swimming, soccer or cricket [% across rows]

Age (years)	Swimming/ diving	Soccer	Cricket	Total
15 – 17	34.93%	39.93%	25.13%	204 100
18 – 24	45.36%	38.47%	16.18%	312 700
25 – 34	57.22%	23.67%	19.12%	454 600
Total	473 600	309 300	188 800	971 400

Source: Australian Bureau of Statistics, *Participation in Sport and Physical Education*, 2009-10, Cat. No. 4177.0

Standardised by age (row totals) – compare across sport (columns)

Summarising: Tables

When constructing a table, it is important to determine which elements and relationships should be emphasised: across columns (comparing sports for a given age) or rows (comparing ages for a given sport)? Absolute or relative (eg %, % change)?

Table 11.1 The share of adults who participate in swimming, soccer or cricket in Australia, by age group [% within columns]

Age (years)	Swimming/ diving	Soccer	Cricket	Total
15 – 17	15.05%	26.35%	27.17%	204 100
18 – 24	29.96%	38.89%	26.80%	312 700
25 – 34	54.92%	34.79%	46.03%	454 600
Total	473 600	309 300	188 800	971 400

Source: Australian Bureau of Statistics, *Participation in Sport and Physical Education*, 2009-10, Cat. No. 4177.0

Standardised by sport (column totals) – compare across age (rows)

Tables: TWL Sales

Sales Measures (Store) - Detail

Saturday 25 Jul 2020

TWL Daily Sales Report

Week 52

Store View - Daily

Region	Branch	Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Sales \$		Conversion Var to LY (pts)
		TY	TY	LY	Budget	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY	LY		
REGION 01	WHANGAPARAQA	30,659	27,126	29,905	3,533	754	13.0%	2.6%	0.6%	10.2%	4.4%	10,846	(221)	-2.0%	36.4%	-1.63	0.6%	1,047	-3.4%	\$29.28	8.1%	58.0%	-0.02	
	WHANGAREI	124,509	112,461	103,938	12,048	20,571	10.7%	19.8%	2.3%	7.5%	3.4%	47,868	8,817	22.6%	36.4%	0.87	2.6%	2,808	4.4%	\$44.20	15.0%	55.6%	-0.02	
	KAITIAI	40,708	30,363	30,624	10,346	10,084	34.1%	32.9%	0.5%	7.6%	2.8%	13,514	2,281	20.3%	33.2%	-3.48	0.7%	1,019	10.8%	\$39.65	19.1%	65.1%	-1.55	
	DARGAVILLE	32,702	26,293	26,392	6,409	6,310	24.4%	23.9%	0.6%	12.4%	4.7%	8,022	(1,070)	-11.8%	24.5%	-9.92	0.4%	905	9.0%	\$36.12	14.9%	63.5%	4.93	
	SNELLS BEACH	30,799	30,430	31,394	369	(595)	1.2%	-1.9%	0.6%	9.4%	3.5%	9,468	(2,186)	-18.6%	30.7%	-6.32	0.5%	928	-8.9%	\$33.21	8.0%	64.8%	2.51	
	WAIKAPA	66,334	55,058	57,903	11,276	8,431	20.5%	14.6%	1.2%	12.4%	1.9%	22,021	553	2.6%	33.2%	-3.88	1.2%	1,385	-7.6%	\$47.48	23.0%	58.5%	0.77	
	SILVERDALE (SWAS)	75,000	66,267	66,237	8,733	5,763	13.2%	8.3%	1.4%	11.1%	5.4%	25,321	(500)	-1.9%	33.8%	-3.53	1.4%	2,206	10.0%	\$33.96	-1.6%	51.6%	-2.61	
	KAIKOHE	20,674	17,106	17,484	3,568	3,190	20.9%	18.2%	0.4%	7.2%	2.6%	7,371	989	15.5%	36.7%	-0.86	0.4%	639	13.1%	\$32.35	4.5%	45.9%	-2.25	
	Total	421,386	383,164	386,878	56,282	54,596	15.4%	14.9%	7.8%	9.6%	3.6%	144,429	8,681	6.4%	34.3%	-2.73	7.7%	10,937	3.1%	\$38.42	11.4%	56.9%	-0.50	
	REGION 02	WESTCITY	52,656	50,821	51,677	1,835	979	3.6%	1.9%	1.0%	11.1%	4.4%	18,032	(952)	-5.0%	34.2%	-2.49	1.0%	1,588	4.3%	\$33.16	-0.6%	46.3%	0.73
NEW LYNN		56,169	51,140	51,592	5,029	4,577	9.8%	8.9%	1.0%	9.5%	4.3%	17,878	(1,611)	-8.3%	31.8%	-5.95	1.0%	1,586	0.6%	\$35.05	7.2%	48.8%	-10.70	
BIRKENHEAD		0	28,494	30,068	(28,494)	(30,068)	-100.0%	-100.0%	0.0%			0	(11,102)	-100.0%		-36.92	0.0%	0	-100.0%		-100.0%		-52.90	
ALBANY		101,014	77,617	87,241	23,397	13,773	30.1%	15.8%	1.9%	11.5%	3.3%	34,272	623	1.9%	33.9%	-4.84	1.8%	2,862	10.2%	\$37.79	4.8%	49.0%	1.01	
MILFORD		30,177	26,733	27,721	3,443	2,456	12.9%	8.9%	0.6%	11.1%	4.2%	10,888	1,358	14.2%	36.1%	1.70	0.6%	1,107	1.9%	\$27.27	6.8%	48.5%	-6.93	
LINCOLN CENTRE		63,720	47,767	50,305	15,933	13,415	33.3%	28.7%	1.2%	6.7%	5.5%	21,431	2,315	12.1%	33.6%	-4.37	1.1%	1,718	4.1%	\$37.14	21.7%	55.1%	0.54	
WESTGATE		99,123	82,094	81,338	17,029	7,785	8.6%	9.6%	1.7%	9.3%	3.1%	32,046	845	2.7%	36.0%	-2.40	1.7%	2,167	-6.8%	\$41.14	10.4%	53.0%	-0.10	
GLENFIELD MALL (SWAS)		54,481	50,561	49,030	3,920	4,851	8.9%	9.8%	1.0%	8.8%	3.3%	20,195	1,007	5.2%	37.1%	-1.59	1.1%	2,056	9.9%	\$28.43	-0.9%	35.6%	-0.19	
Total		447,339	415,667	429,572	31,672	17,767	7.6%	4.1%	8.3%	10.0%	3.9%	154,751	(7,517)	-4.6%	34.6%	-3.18	8.3%	12,882	-4.4%	\$34.64	8.9%	47.0%	-1.85	
REGION 03		NEWMARKET	30,854	25,854	26,810	5,000	(8,956)	19.3%	-22.5%	0.6%	8.4%	6.8%	9,075	(5,991)	-39.6%	29.4%	-8.43	0.5%	1,133	-36.9%	\$26.90	4.8%	54.2%	-2.19
	PAKURANGA FC	47,749	45,832	45,124	1,917	(1,375)	4.2%	-2.8%	0.9%	8.6%	7.1%	14,609	(2,703)	-15.6%	30.6%	-4.04	0.8%	1,517	-4.3%	\$31.49	1.6%	47.0%	-0.06	
	SYLVIA PARK	95,581	83,134	104,302	32,427	(8,741)	51.4%	-8.4%	1.8%	10.8%	4.8%	32,409	(8,112)	-20.0%	33.9%	-4.94	1.7%	2,590	-12.6%	\$36.90	8.7%	42.6%	-3.59	
	ST LUKES	33,152	27,384	28,953	5,768	4,199	21.1%	14.5%	0.6%	13.1%	3.2%	11,500	432	3.9%	34.7%	-3.54	0.6%	1,186	5.7%	\$27.96	8.3%	46.8%	0.23	
	ROYAL OAK	75,621	58,788	65,275	16,733	10,246	28.9%	16.7%	1.4%	11.0%	2.8%	27,271	2,746	11.2%	36.1%	-1.46	1.5%	1,913	-1.2%	\$39.48	18.0%	55.8%	0.77	
	LUNN AVE (SWAS)	45,282	52,477	0	(7,195)	45,282	-13.7%		0.8%	8.2%	2.6%	17,354	17,354	36.3%	38.32	0.9%	1,278		\$36.03		55.2%	55.19		
	MT ROSKILL	43,383	36,942	36,513	6,441	4,970	17.4%	12.6%	0.8%	11.6%	3.4%	15,805	1,935	14.0%	36.4%	0.42	0.8%	1,455	3.0%	\$29.71	11.0%	53.1%	-2.44	
	ATRIUM	19,621	24,419	26,361	(4,899)	(8,840)	-22.1%	-25.9%	0.4%	12.7%	4.1%	6,241	(3,989)	-39.0%	32.0%	-6.84	0.3%	974	-22.8%	\$20.04	-4.3%	42.0%	-42.03	
	Total	351,022	334,829	332,337	56,193	38,685	16.8%	11.6%	7.3%	10.5%	4.3%	134,263	1,671	1.3%	34.3%	-3.30	7.2%	12,044	1.8%	\$32.48	9.7%	49.8%	-0.64	
	REGION 04	PUKEKOHE	94,954	100,575	86,842	(5,622)	8,112	-5.6%	9.3%	1.8%	9.6%	2.8%	35,923	3,483	10.7%	37.8%	0.45	1.9%	2,329	4.5%	\$40.35	4.1%	57.3%	0.62
TAKANINI		83,715	80,726	82,101	2,989	1,614	3.7%	2.0%	1.6%	7.4%	4.7%	30,193	(39)	-0.1%	36.1%	-0.76	1.6%	2,207	-1.2%	\$37.91	3.5%	53.3%	0.14	
BOTANY DOWNS		74,293	71,969	70,040	2,325	4,253	3.2%	6.1%	1.4%	10.6%	4.2%	24,942	(2,957)	-10.6%	33.6%	-0.26	1.3%	2,209	2.0%	\$33.49	3.5%	46.9%	-0.15	
CLENDON		40,320	34,478	35,464	5,842	4,856	16.9%	13.7%	0.8%	13.5%	3.1%	14,311	308	2.2%	36.5%	-3.99	0.8%	1,255	1.5%	\$31.91	11.5%	49.3%	-1.06	
AIRPORT (SWAS)		49,389	46,076	50,662	3,313	(1,274)	7.2%	-2.8%	0.9%	7.0%	4.5%	17,304	141	0.8%	36.0%	1.16	0.9%	1,219	-14.9%	\$40.61	14.6%	48.1%	0.73	
MANUKAU		124,182	125,505	119,540	(1,343)	4,822	-1.1%	3.9%	2.3%	8.0%	3.7%	41,756	(852)	-1.3%	33.6%	-1.78	2.2%	2,819	-6.7%	\$43.58	10.3%	45.6%	-3.65	



thewarehouse.co.nz

WHS warehouse university

noel leeming

Torpedo

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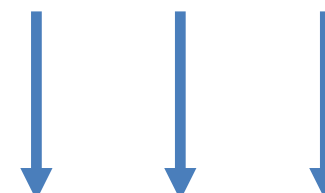


Tables: Guidelines

- “Goldilocks” principle: not too simple, not too complex (multiple tables OK)
- Should have a clear identifier (eg Table 11.1)
- A title that describes the types of information given
- Row and column labels should be precise and unambiguous
- Categories should not overlap (“mutually exclusive”)
- The units of measurement must be clearly stated
- Any unimportant figures should be combined or omitted
- Any subheadings should be labelled clearly
- Any relevant totals, subtotals, percentages should be shown
- Use same number of decimal places throughout and align vertically on decimal point (may require dummy columns)

Retail Trade Survey

(dummy columns)



	A	B	C	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
6	Actual retail sales values⁽¹⁾																				
7	By quarter by industry			1. Figures exclude GST.										\$(million)							
8																					
9				2023	2024					2025											
10	Industry			Dec	Mar	Jun	Sep	Dec	Mar	June	Sep	Dec									
12	Supermarket and grocery stores			7,076	6,790	6,507	6,566	7,196	6,943	6,736	6,796	7,375									
13	Specialised food			544	475	469	456	549	499	475	484	590									
14	Liquor			593	545	501	503	612	522	480	472	600									
15	Non-store and commission-based retailing			597	479	461	489	544	462	480	484	560									
16	Department stores			2,190	1,562	1,686	1,531	2,173	1,573	1,706	1,615	2,190									
17	Furniture, floor coverings, houseware, textiles			863	721	748	753	871	738	760	772	910									
18	Hardware, building, and garden supplies			2,926	2,525	2,527	2,565	2,859	2,465	2,492	2,600	2,922									
19	Recreational goods			944	774	700	647	968	788	737	696	1,064									
20	Clothing, footwear, and accessories			1,499	1,191	1,231	1,113	1,500	1,234	1,234	1,121	1,552									
21	Electrical and electronic goods			1,000	845	840	847	1,007	847	907	969	1,163									
22	Pharmaceutical and other store-based retailing			2,215	1,991	2,006	1,954	2,304	2,141	2,139	2,155	2,590									
23	Accommodation			1,418	1,656	1,139	1,094	1,525	1,710	1,208	1,170	1,607									
24	Food and beverage services			4,124	3,965	3,821	3,825	4,254	4,004	3,904	4,049	4,431									
25	Core industries total			25,988	23,520	22,635	22,344	26,364	23,925	23,257	23,383	27,552									
26	Motor vehicles and parts			3,918	3,905	3,750	3,875	3,882	4,016	4,058	4,341	4,277									
27	Fuel			2,662	2,531	2,398	2,252	2,384	2,431	2,189	2,169	2,402									
28	All industries total			32,569	29,956	28,783	28,471	32,629	30,372	29,504	29,893	34,232									

<https://www.stats.govt.nz/information-releases/retail-trade-survey-december-2025-quarter/>

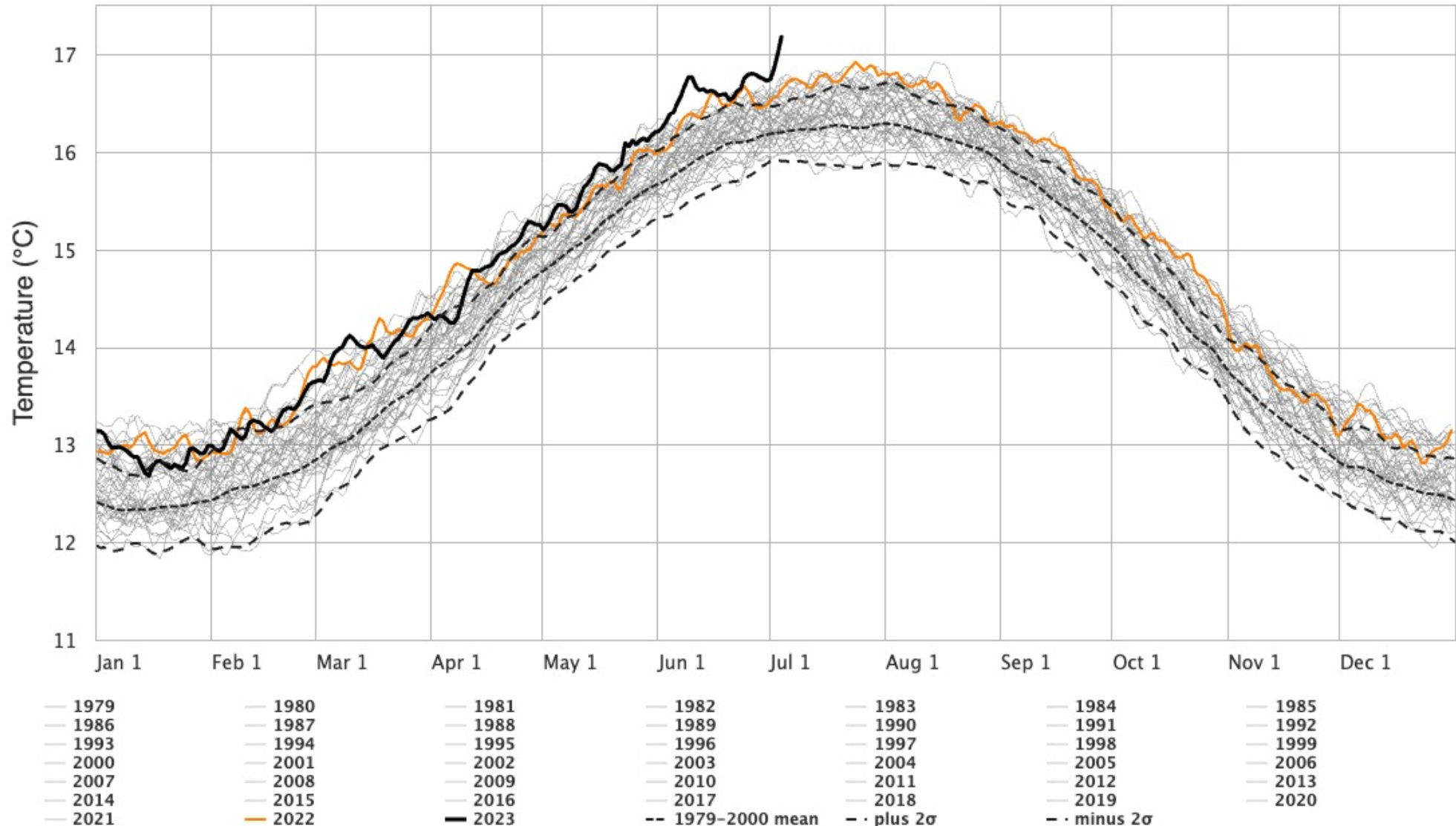
Graphs / Figures / Charts

1. Give the graph a clear title
2. Label the axes clearly
3. Do not put too many curves on the same graph
4. Identify sources of data
5. If possible, accompany the graph with the table of the data that are represented by the graph (eg in appendix)
6. Use a zero on the scale of the vertical axis if possible
7. Avoid “false 3D” and unnecessary symbols (“chart junk”)

Graph Example: Temps

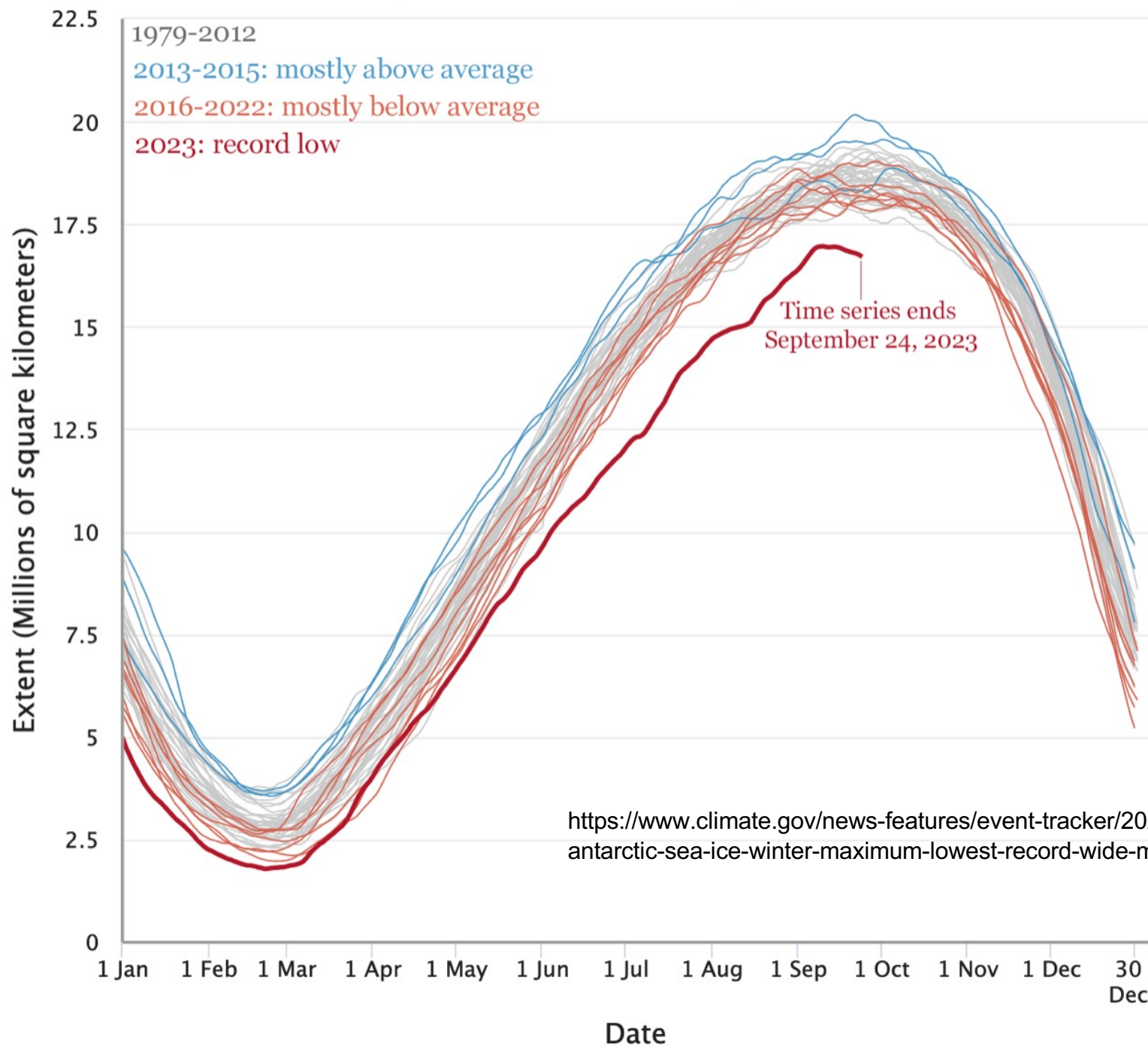
2m Temperature World (90°S–90°N, 0–360°E)

NCEP CFSV2/CFSR | ClimateReanalyzer.org, Climate Change Institute, University of Maine



Antarctic Sea Ice Extent

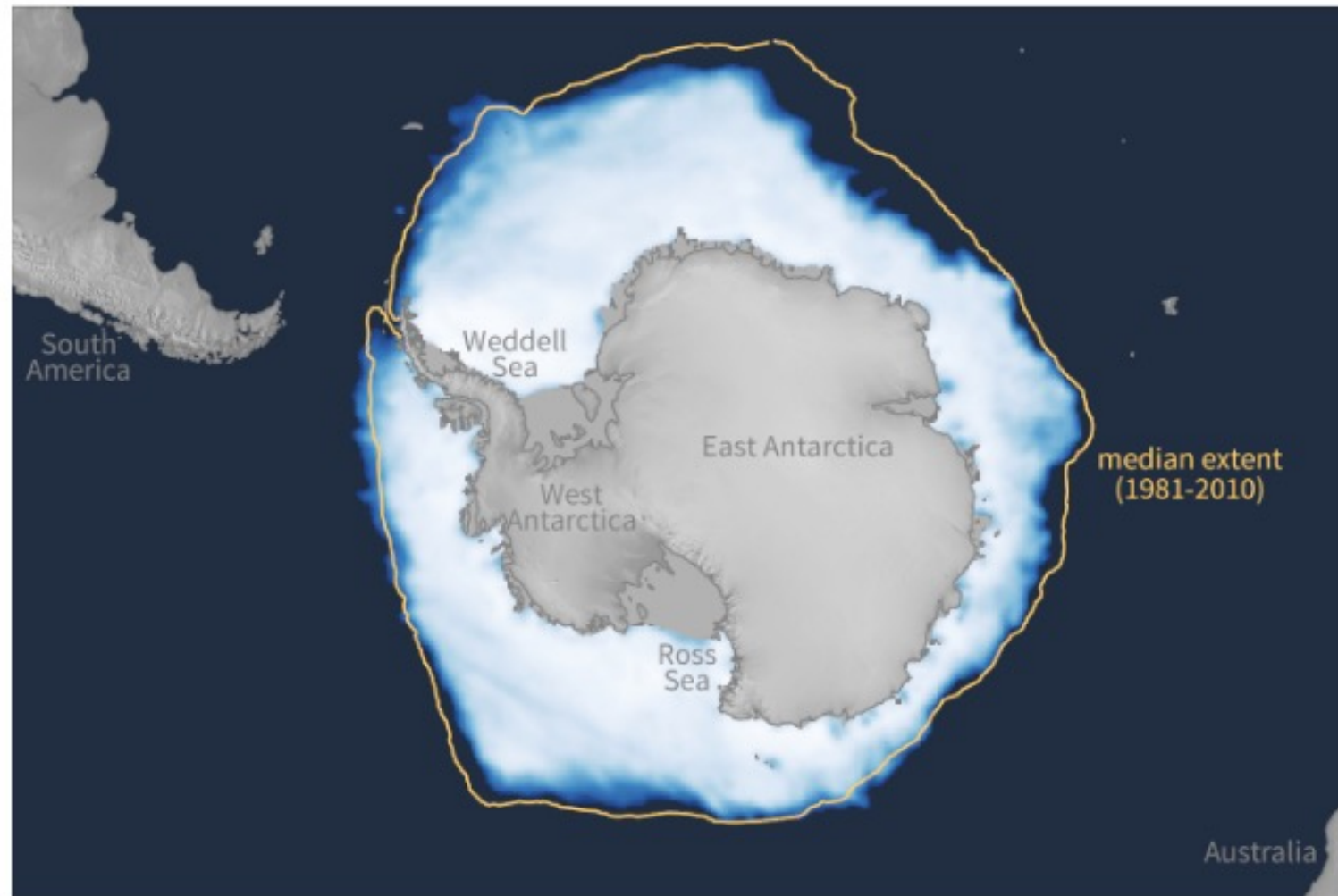
(Area of ocean with at least 15% sea ice)



Graph Example: Time

2024 WINTER MAXIMUM

<https://www.climate.gov/maps-data/data-snapshots/data-source/antarctic-sea-ice-extent-maximum>



September 2024

Sea ice concentration (percent)

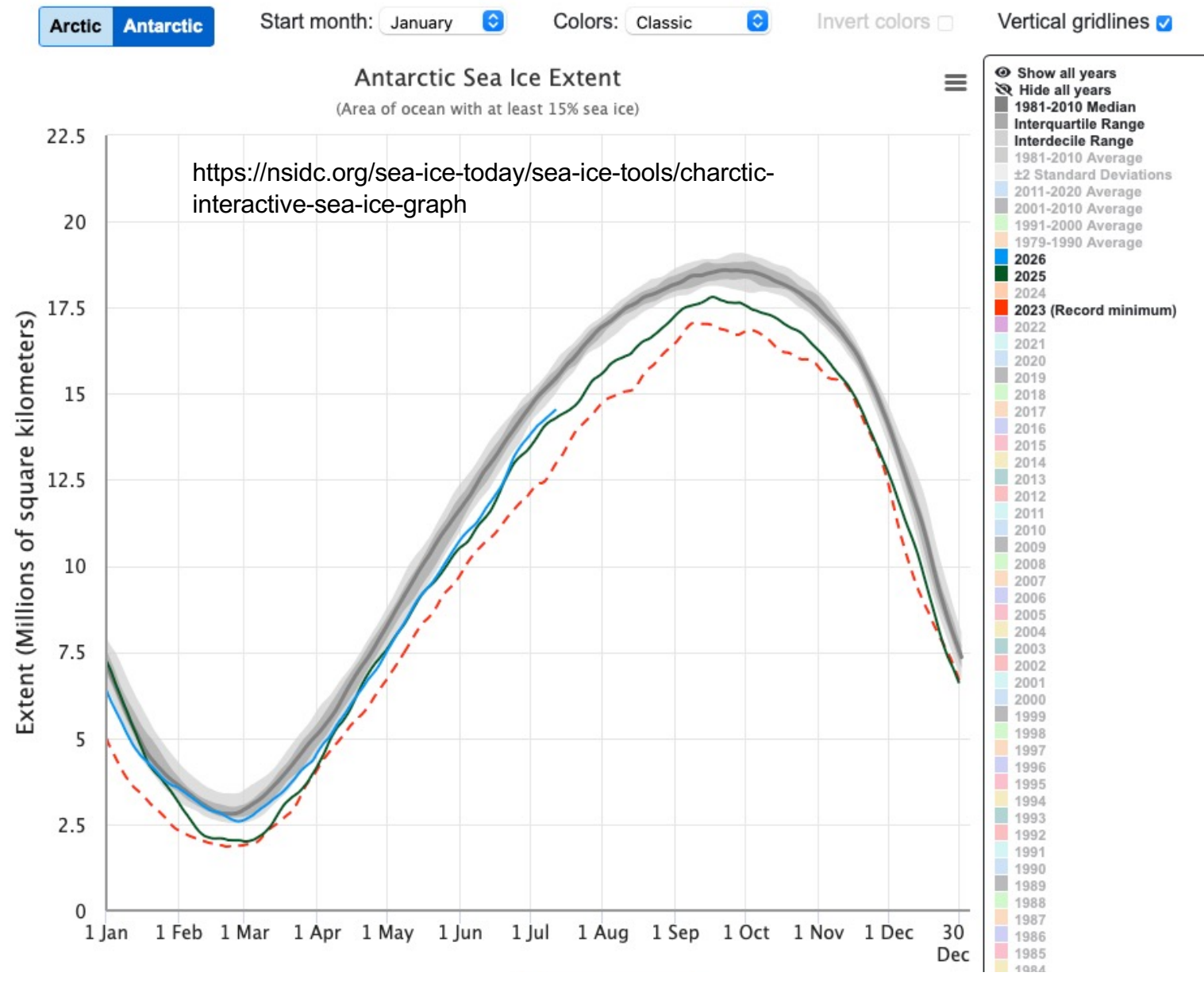


NOAA Climate.gov
Data: NSIDC

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Graph Example: Time

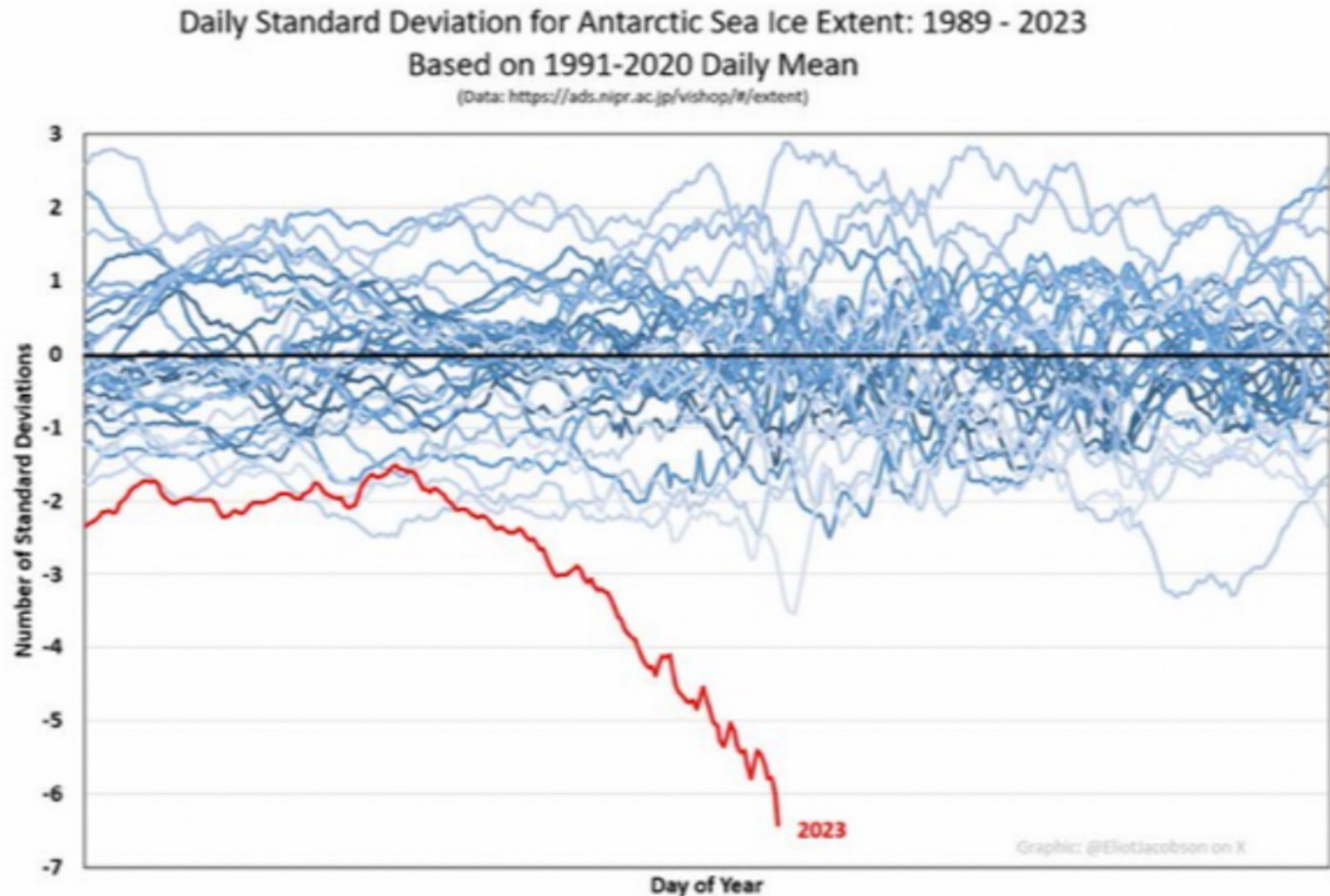


Graph Example: Time



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Graph Example

<https://taxfoundation.org/taxing-high-income-2019/>

Top Effective Marginal Tax Rates in Europe

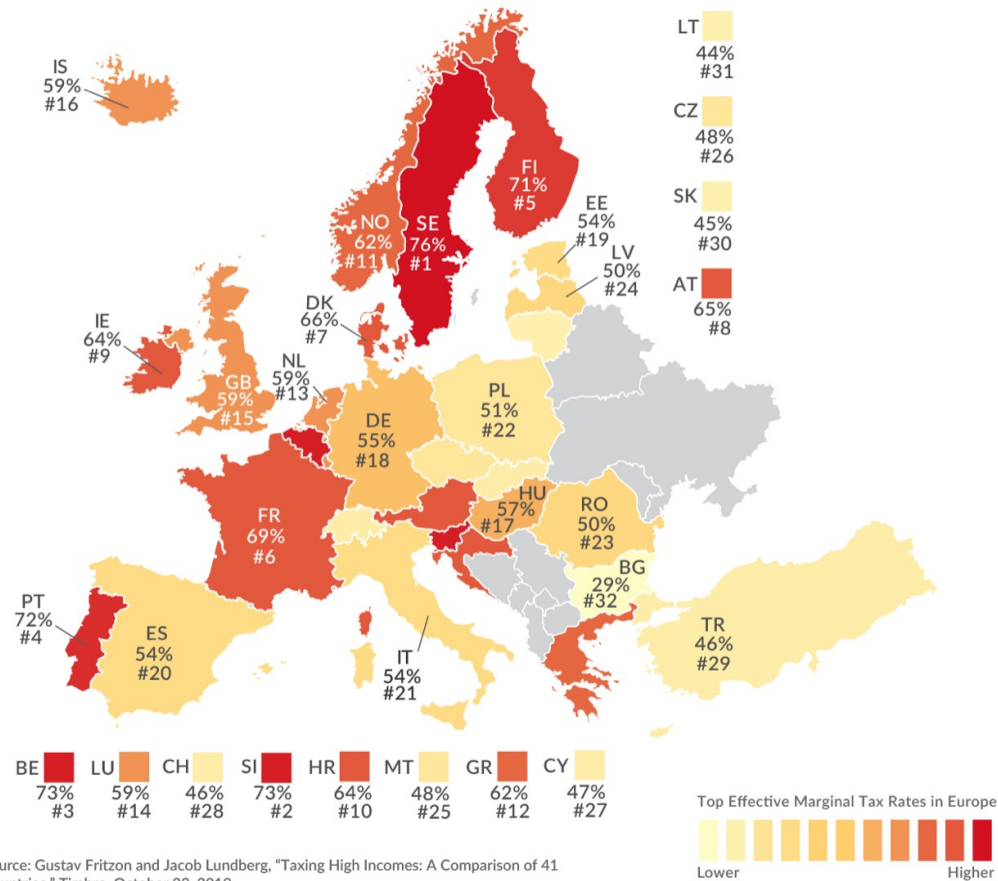
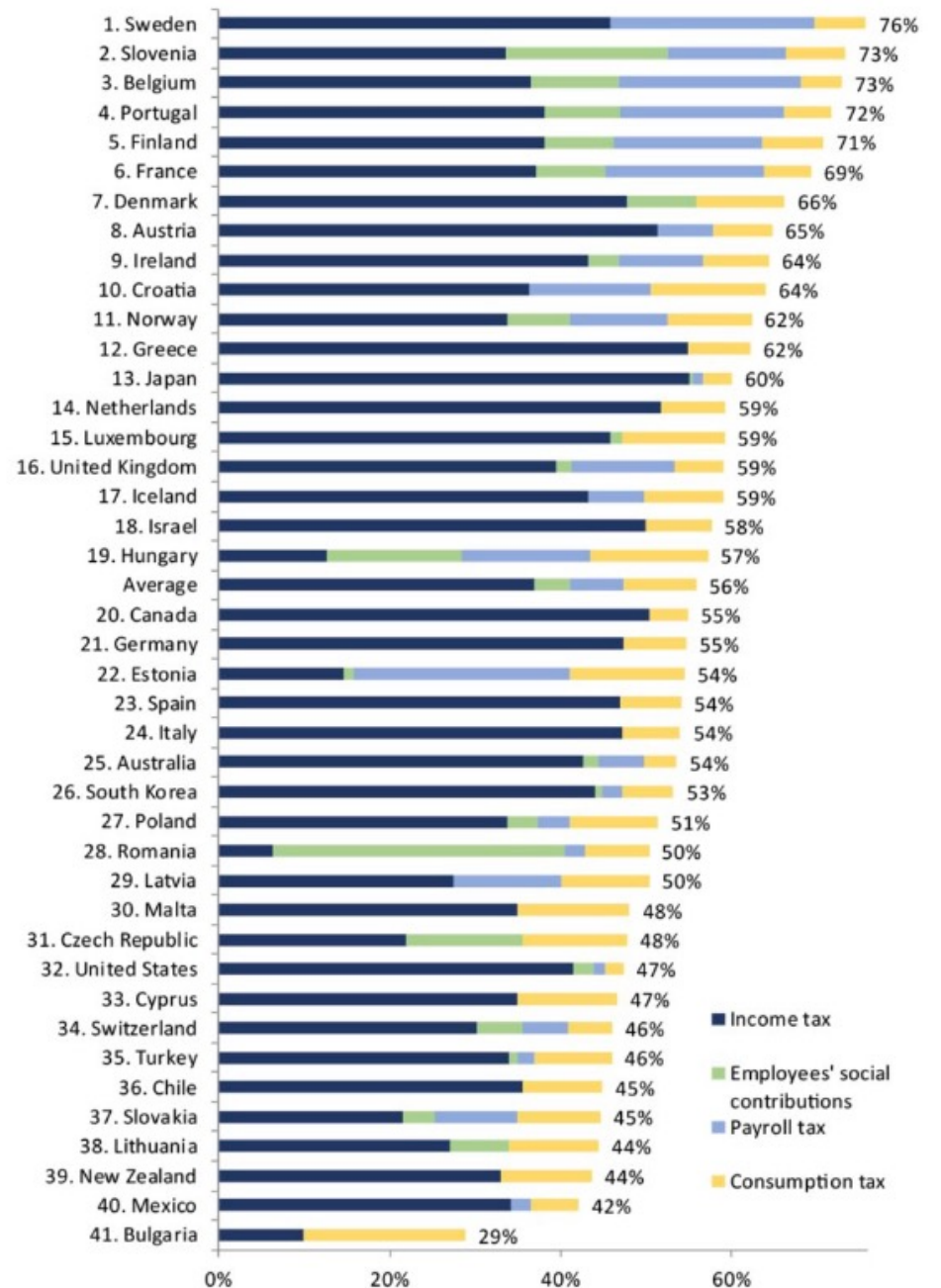


Figure 1 Top Effective Marginal Tax Rates in 2019 and Their Composition

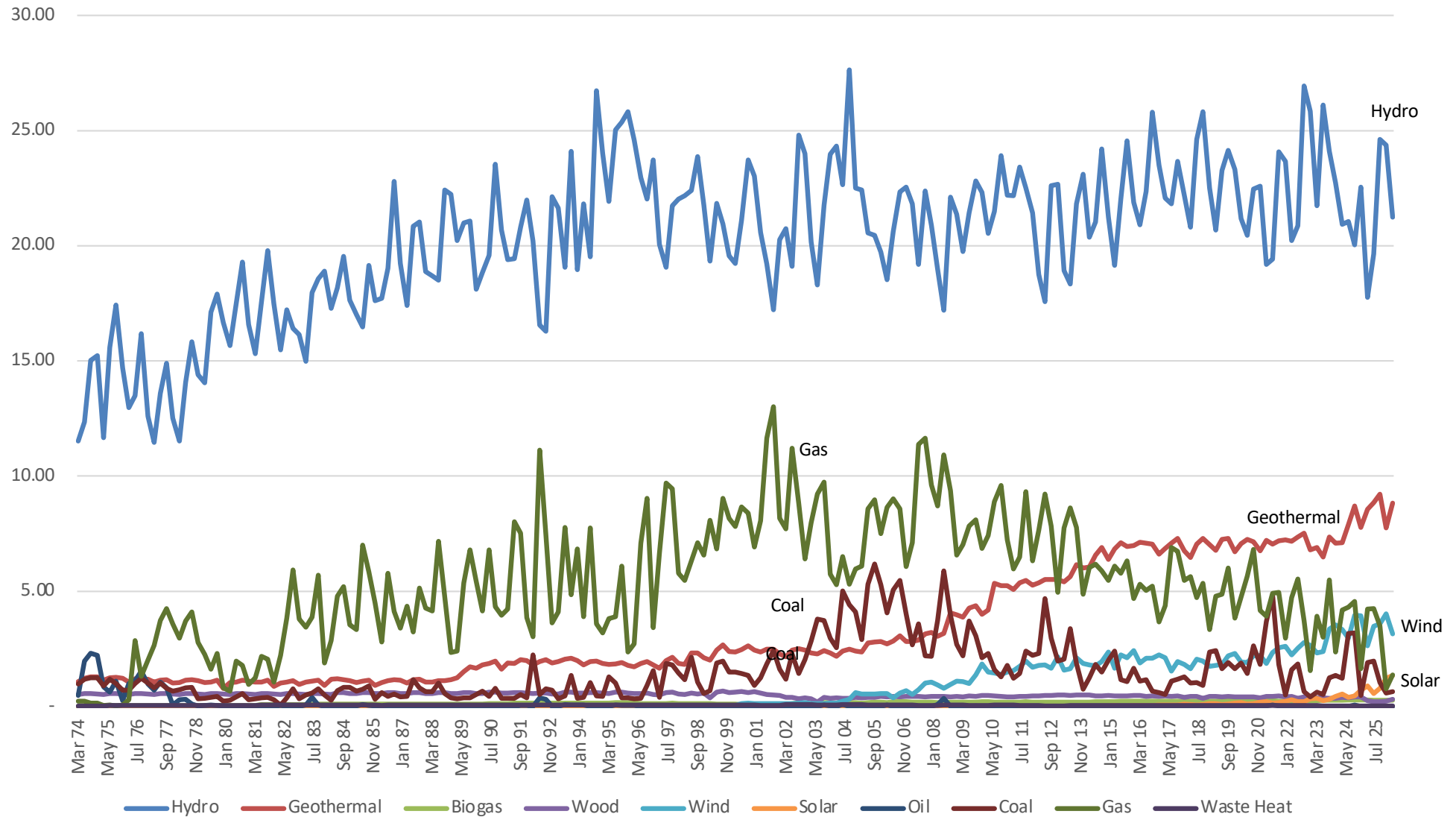


Line Chart

<https://www.mbie.govt.nz/building-and-energy/energy-and-natural-resources/energy-statistics-and-modelling/energy-statistics/electricity-statistics>

PetaJoules

Electricity Generation (Quarterly) in NZ



Bar Chart

<https://www.transpower.co.nz/system-operator/live-system-and-market-data/consolidated-live-data>

% Renewables (as at): 14 Jul 2026

14:30

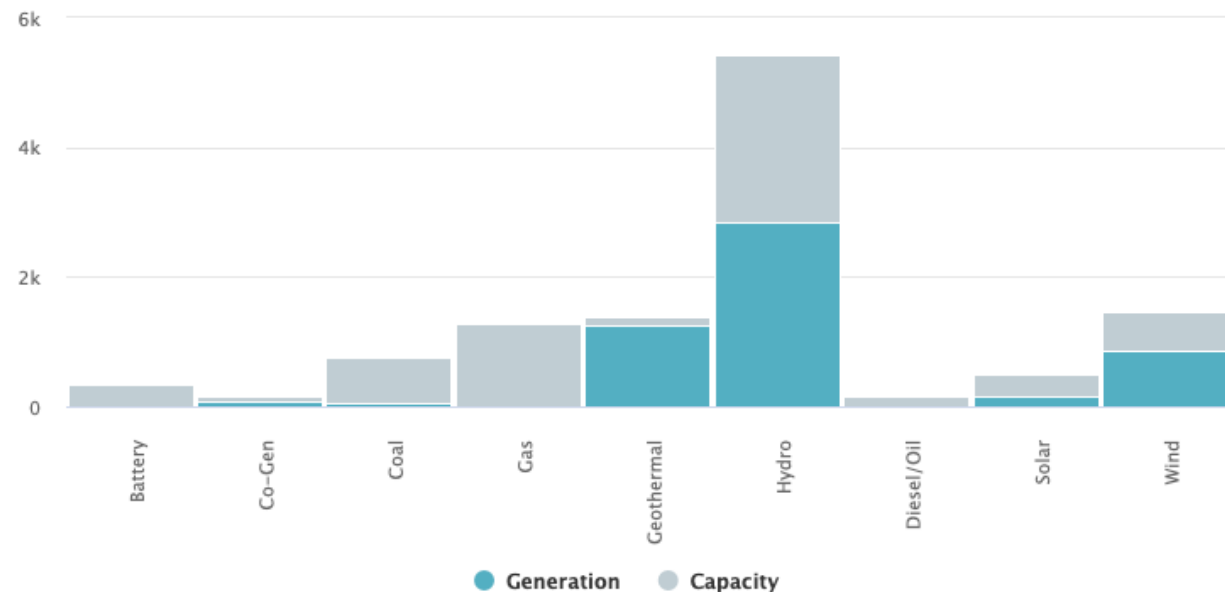
Current Generation (MW)

Power Generation (as at) 14 Jul 2026 14:30

Battery	0 MW
Co-Gen	72 MW
Coal	52 MW
Gas	0 MW
Geothermal	1242 MW
Hydro	2838 MW
Diesel/Oil	0 MW
Solar	147 MW
Wind	864 MW

Current Generation (MW)

Click and drag in the plot area to zoom in.



Total generation (as at) 14 Jul 2026 14:30

5216 MW

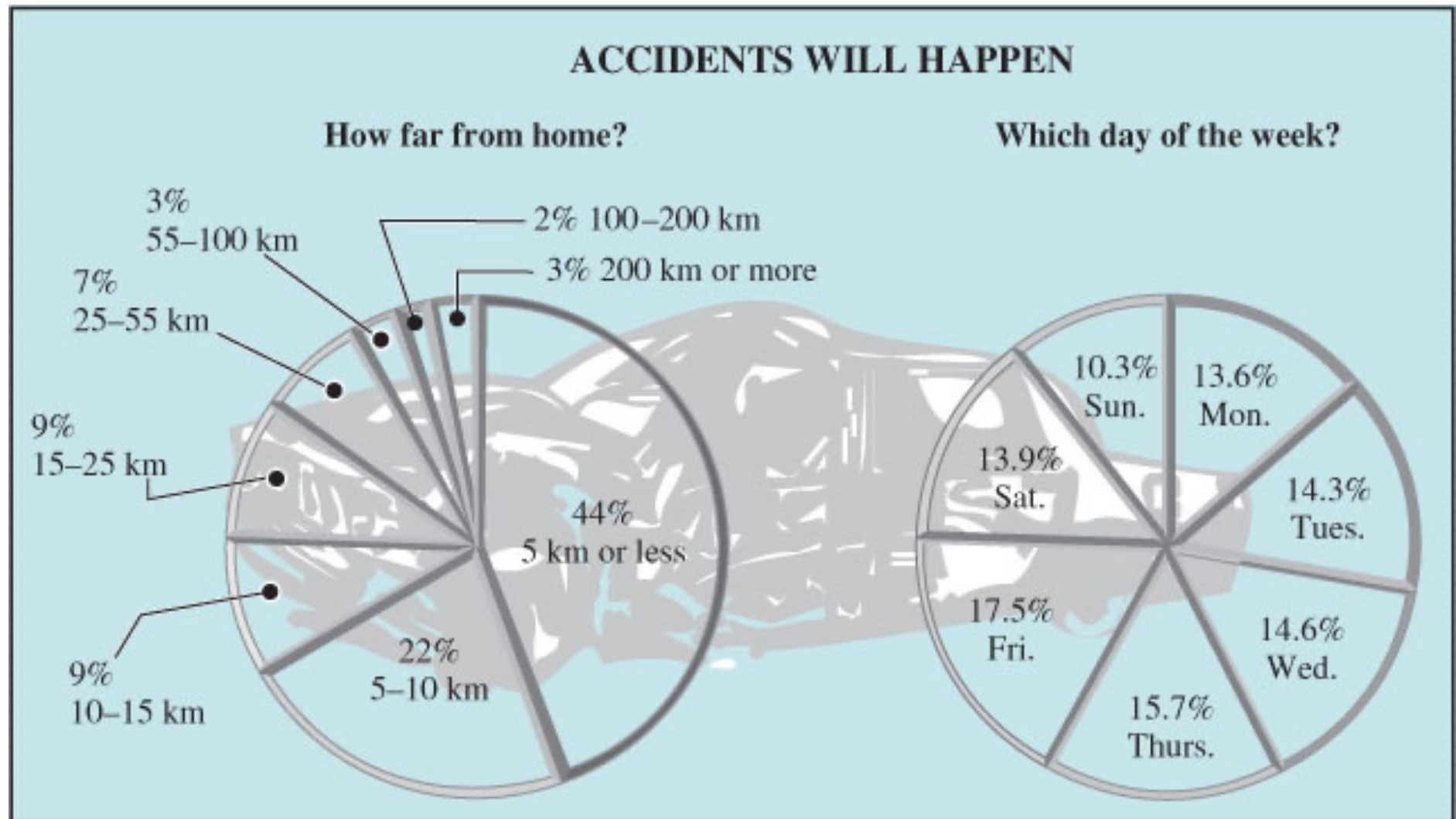


Pie Chart

- A *pie chart* is often used to give a visual presentation of data to indicate the proportions (percentages) that make up a given total
- The pie chart is a circle that is divided by radial lines into sectors in such a way that the area of each sector is proportional to the size of the quantity represented by that sector
- In a case where the sectors are large enough to do so, the percentage is often written inside the sector. Otherwise it is written outside the chart

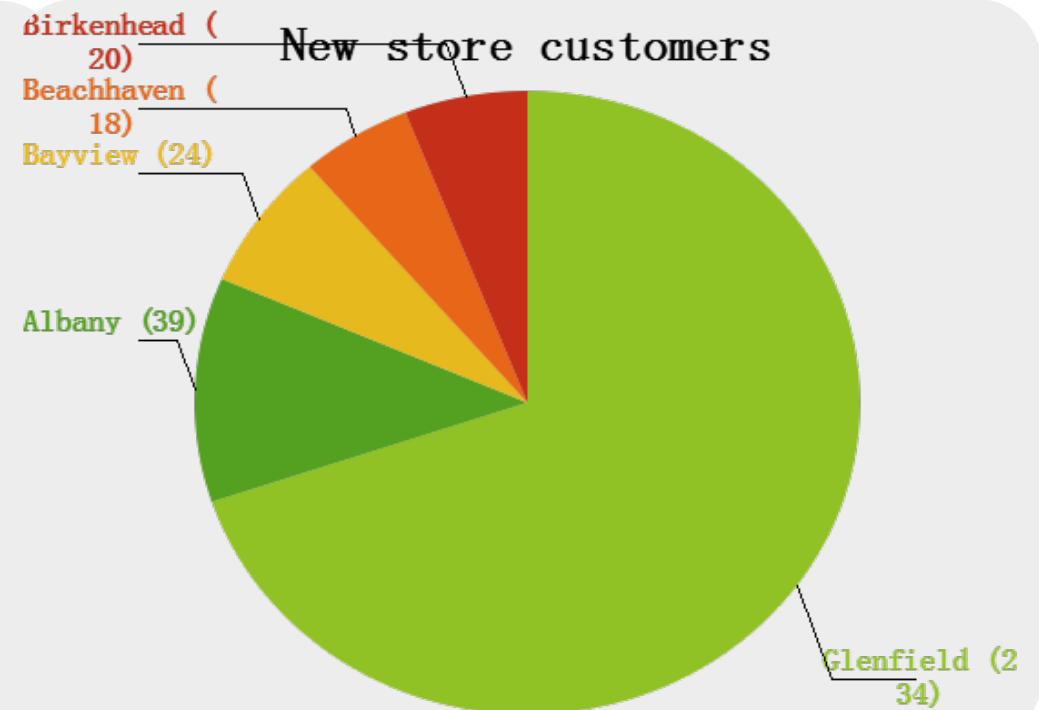
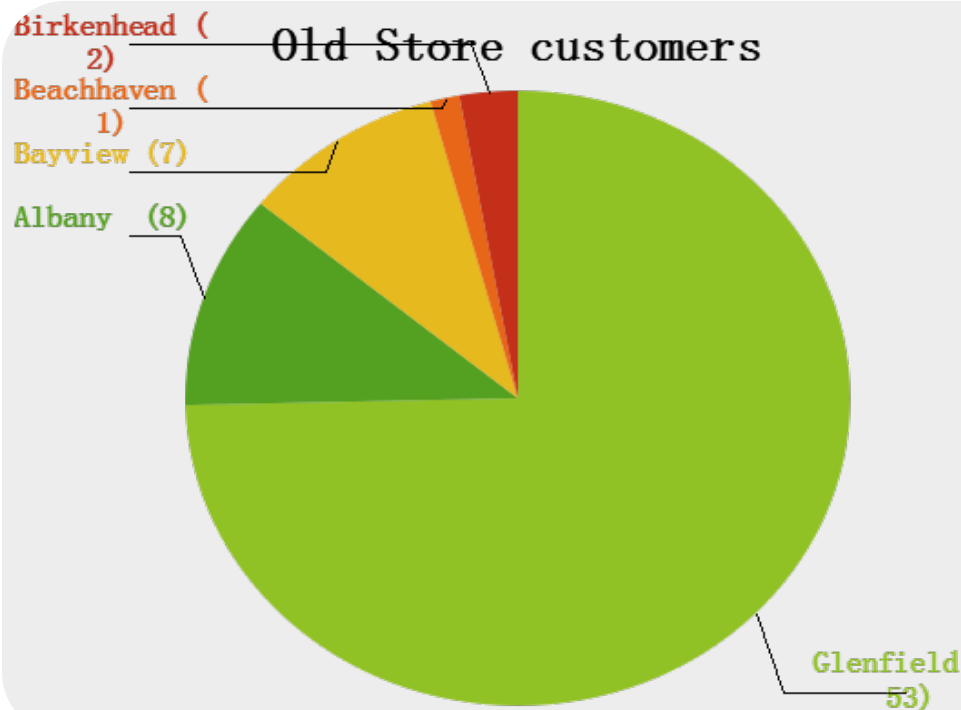
Pie Charts

Figure 11.8: How far from home motorists are when involved in traffic accidents, and percentage that occur on each day of the week



Pie Charts

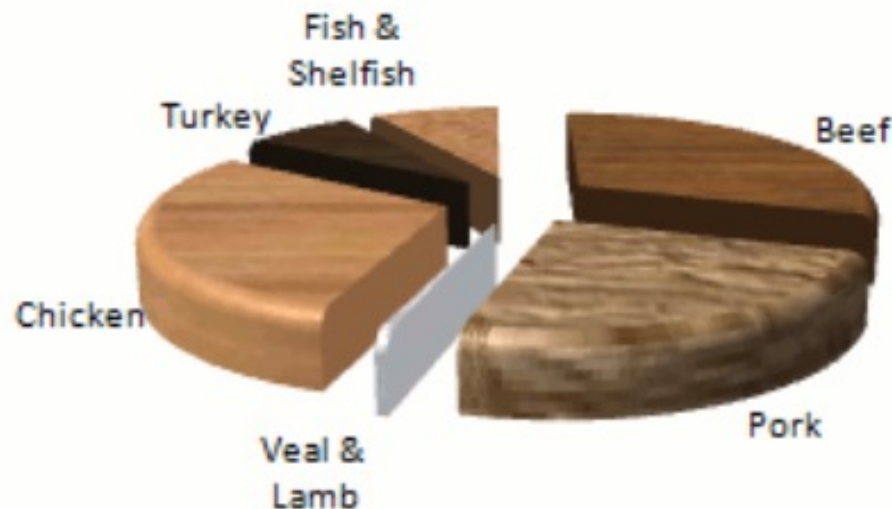
Origins of Customers of Countdown Glenfield ('old' vs 'new' store designs)



Pie Chart Guidelines

<http://chandoo.org/wp/2009/02/20/better-pie-charts/>

- **Too many data points.** A pie chart is probably effective with up to 4 values, anything more and you better have a strong story behind it for people to read and interpret.
- **Difficult to interpret.** Means avoiding all the flashy formatting, 3D pie charts, 3D donuts and layered donuts etc.
- Due to a parallax effect, the slice “Pork” seems to be the largest, but that’s not the case. And because the chart is “exploded”, it’s even harder to compare angles because they no longer share the same center.
(See “Abusing Pies” [Carnegie: Excel charts: pie charts])

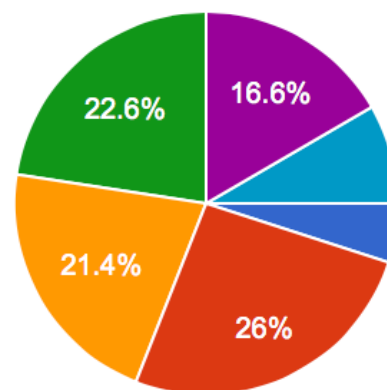


Pie Chart Guidelines

<http://chandoo.org/wp/2009/02/20/better-pie-charts/>

- **Poor use of color.** The purpose of color in most visualizations is to bring contrast. ... But most programs that are used to generate pie charts have poor (and often scary) color choices.
- Whenever possible, try using data labels, using contrasting colours and if nothing works, try using other types of charts for comparison.
- **Difficult to compare** across series, for the simple reason that you can't use more than one variable in a pie. If you want to compare two or more series you have to make two or more pie charts. But if you can hardly compare slices within a pie, using multiple pies is beyond forgiveness.

Age Group



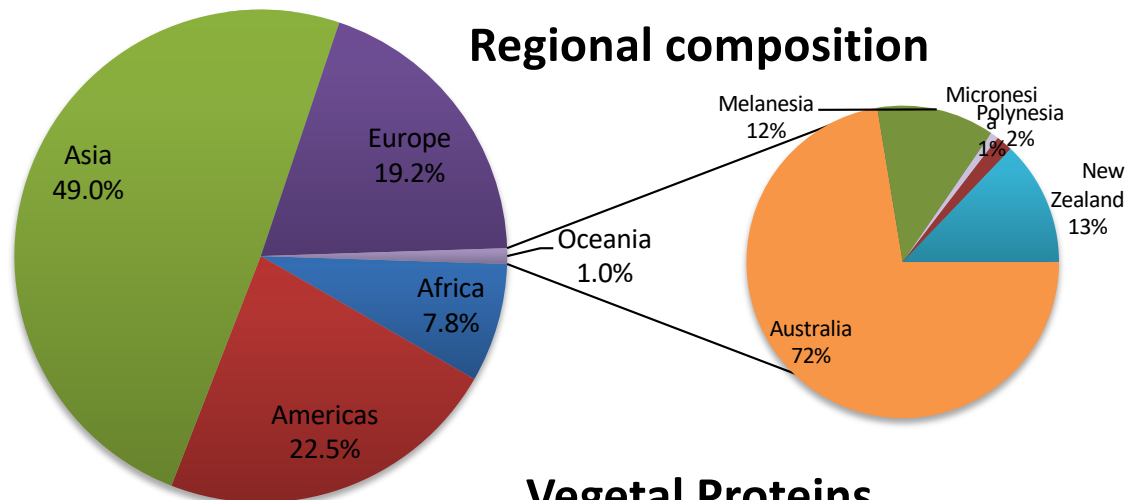
0 – 18	35	4.8%
19 – 34	183	25.3%
35 – 44	151	20.9%
45 – 54	159	22%
55 – 64	117	16.2%
65 +	59	8.1%



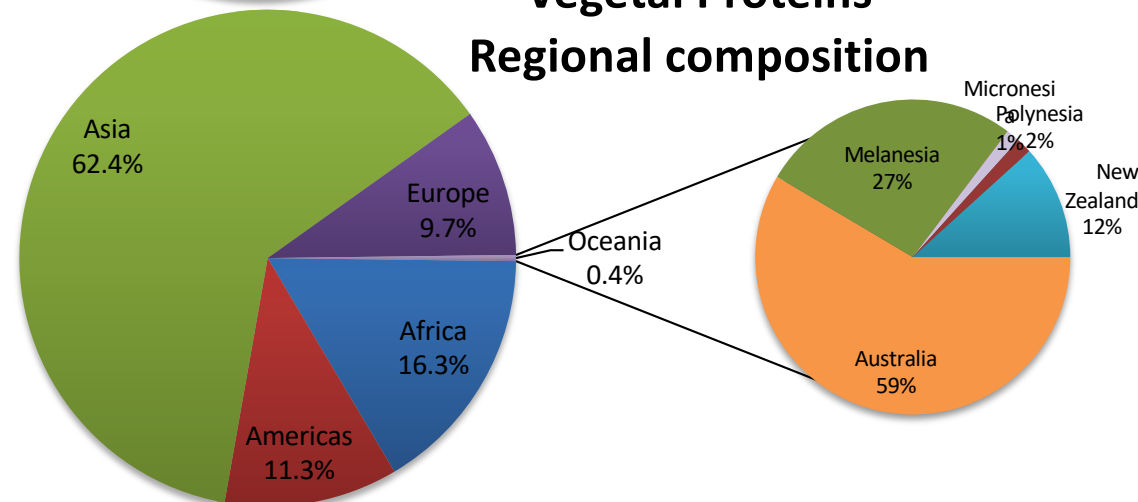
Pie Charts

Animal & Vegetal Protein Consumption by World Region (Source: FAOSTAT)

Animal Proteins
Regional composition

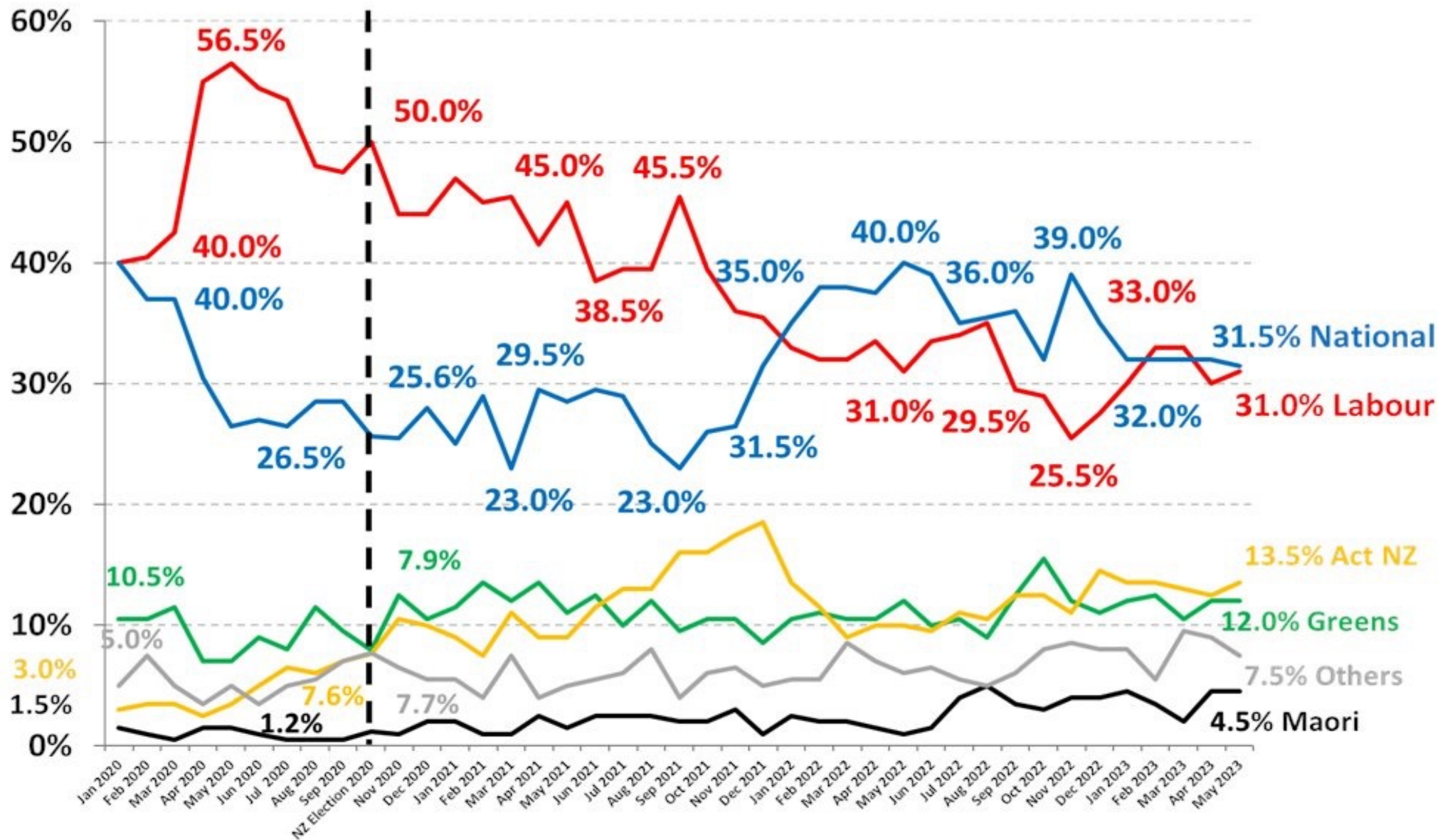


Vegetal Proteins
Regional composition

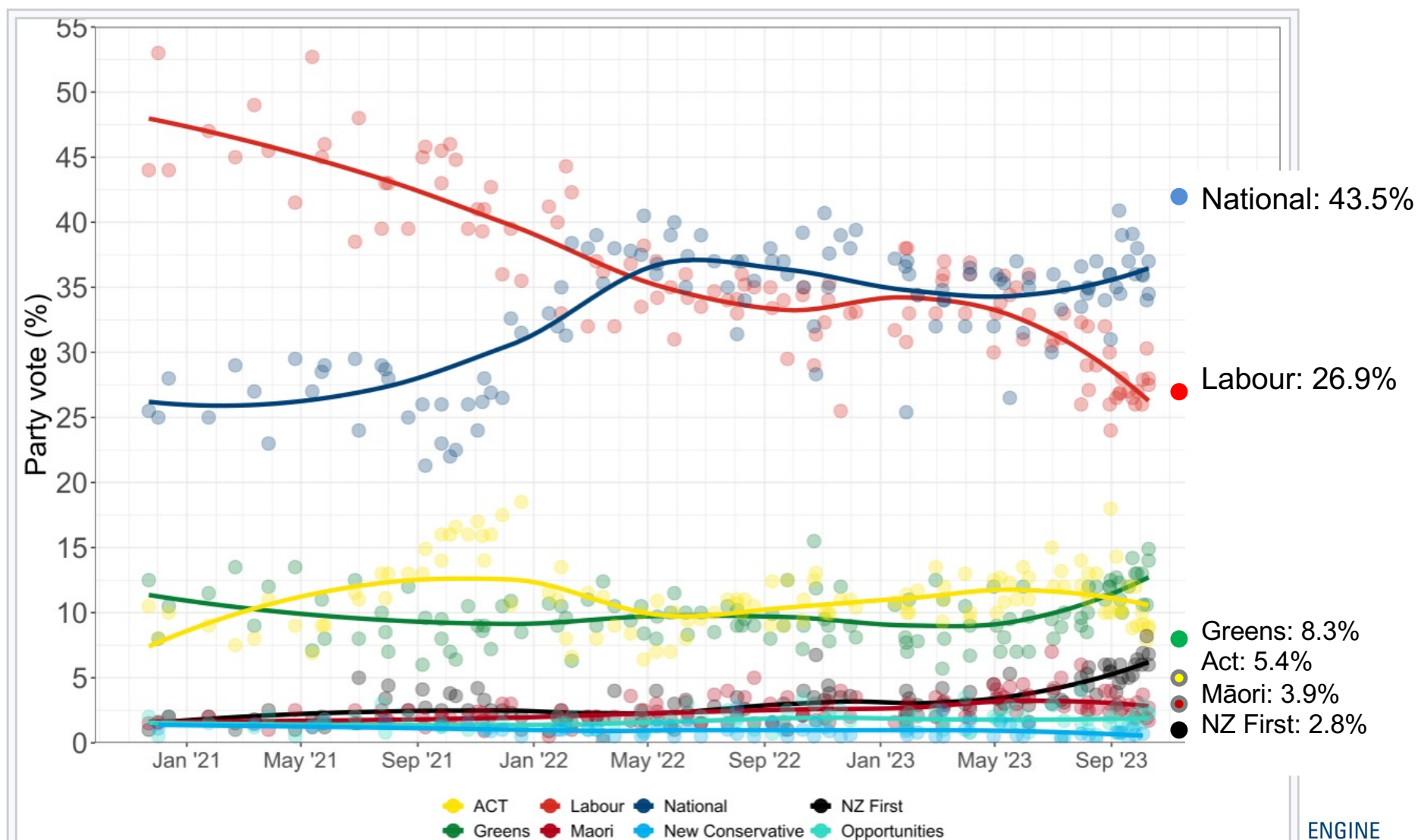


Line Charts

- allow simple depictions of trends and comparisons

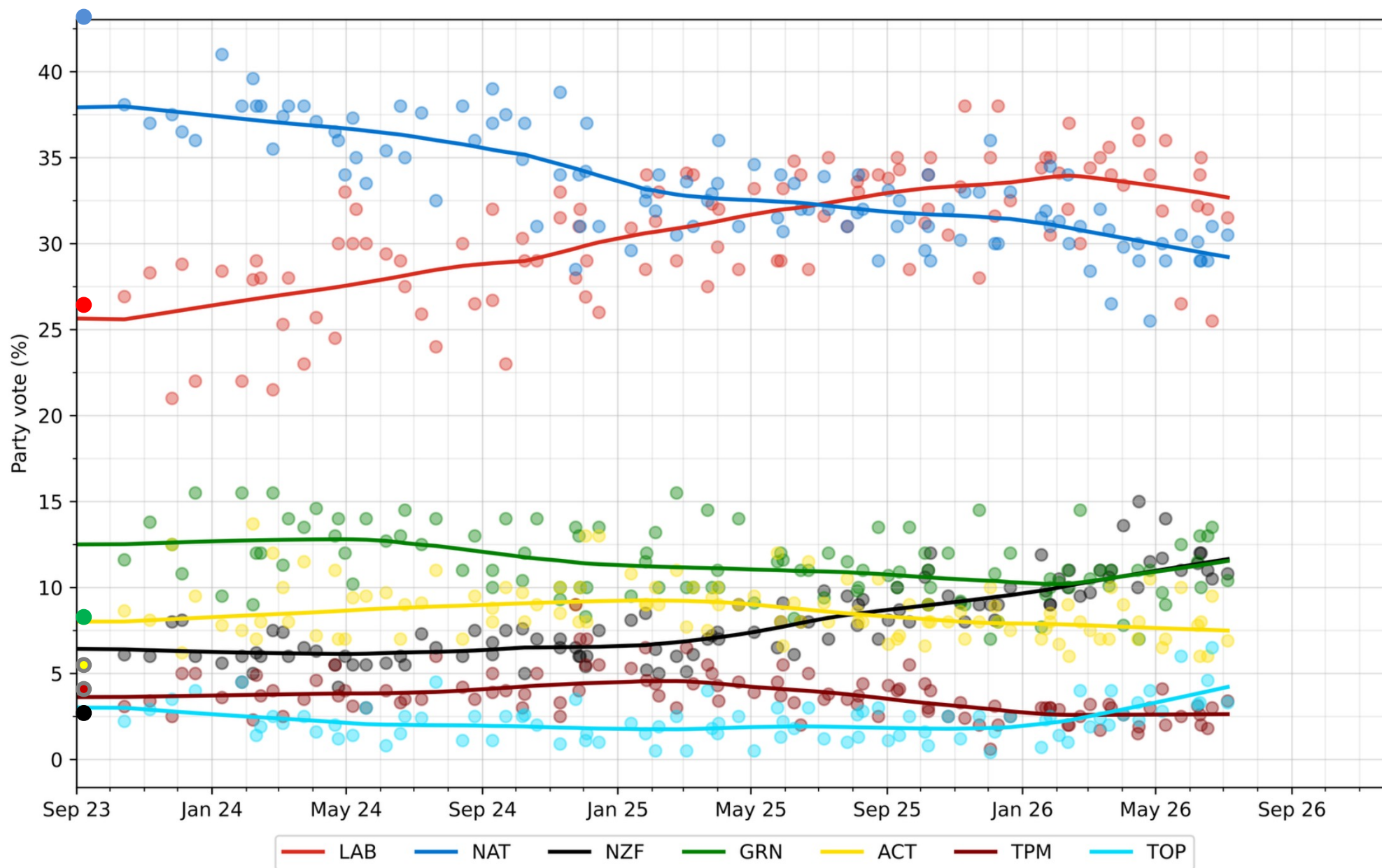


Party vote [\[edit \]](#)



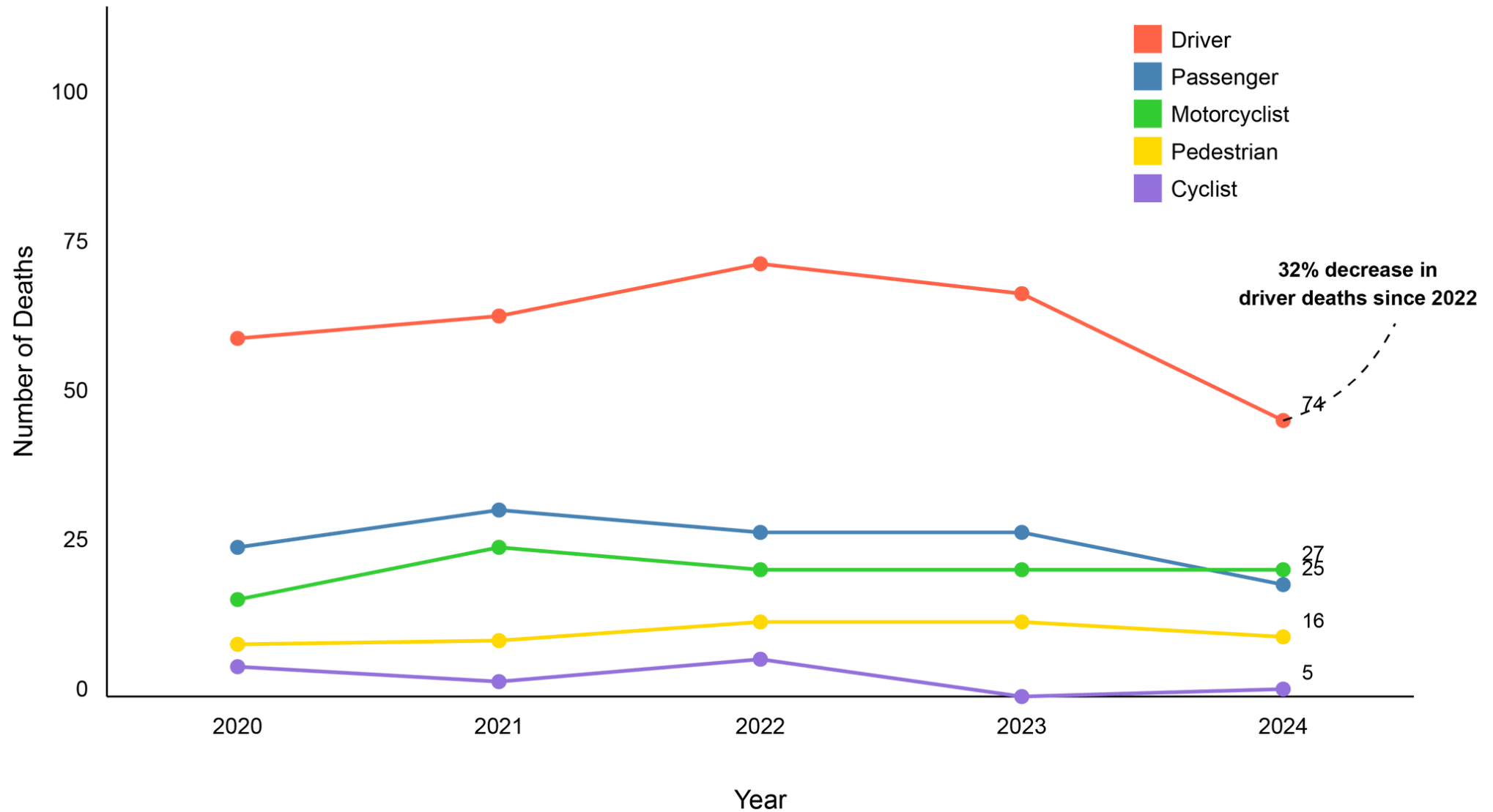
Graph of opinion polls conducted (as at 10 October 2023). Smoothing is set to span 65%.





Line Chart

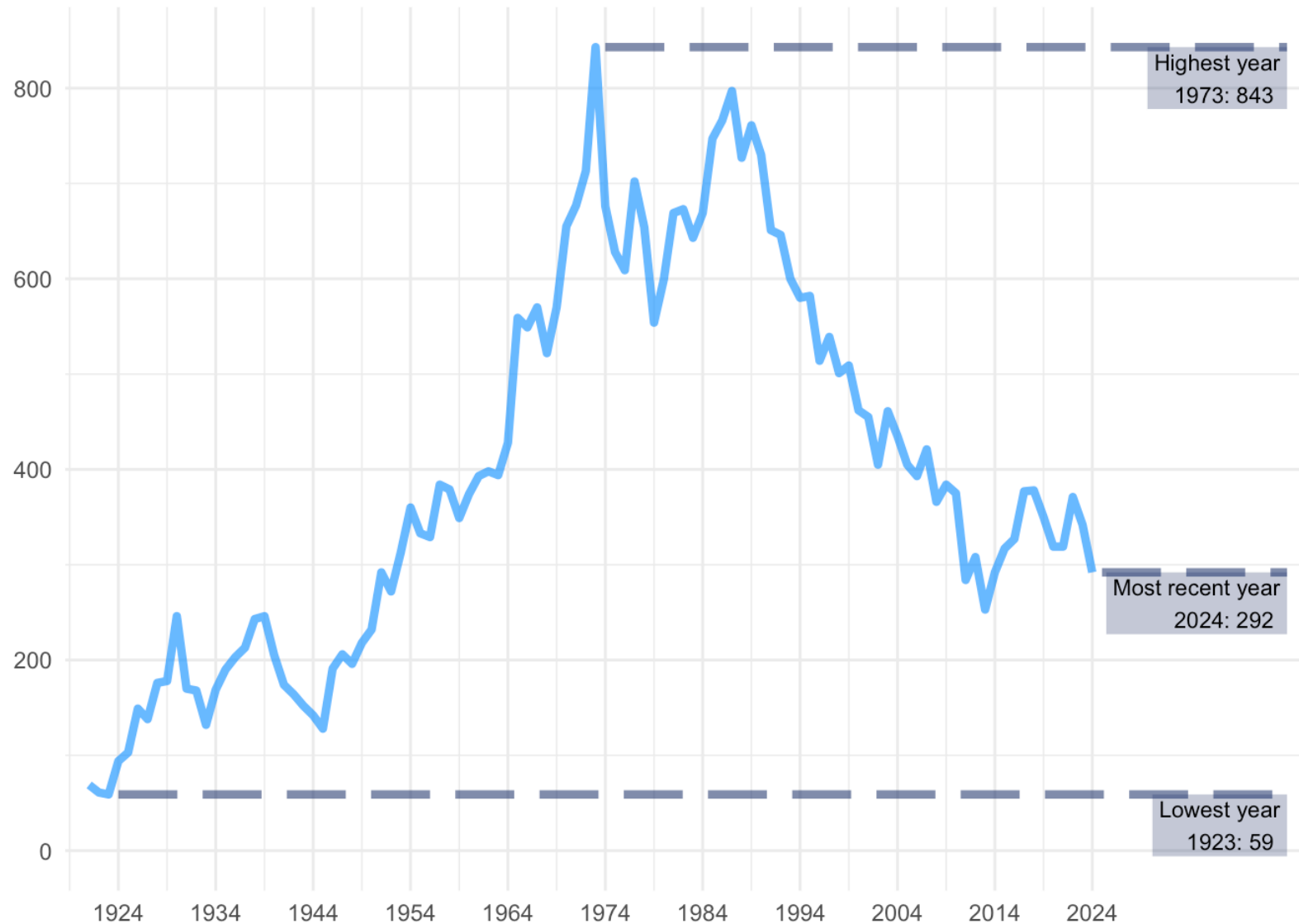
Provisional Road Deaths in New Zealand (Up to 23 July Each Year)



Line Chart

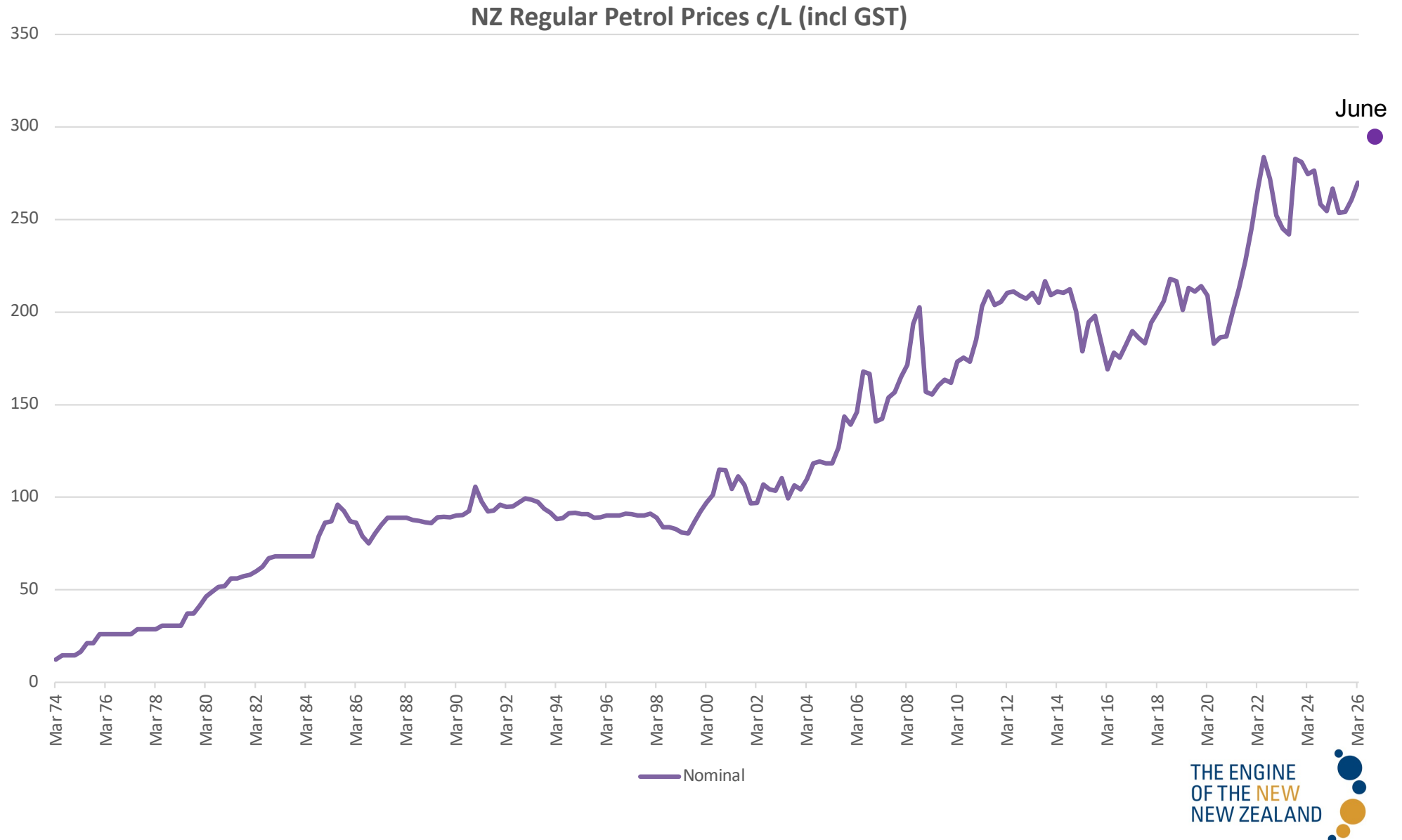
Death on NZ roads since 1921

<https://www.transport.govt.nz/statistics-and-insights/safety-road-deaths/death-on-nz-roads-since-1921/>



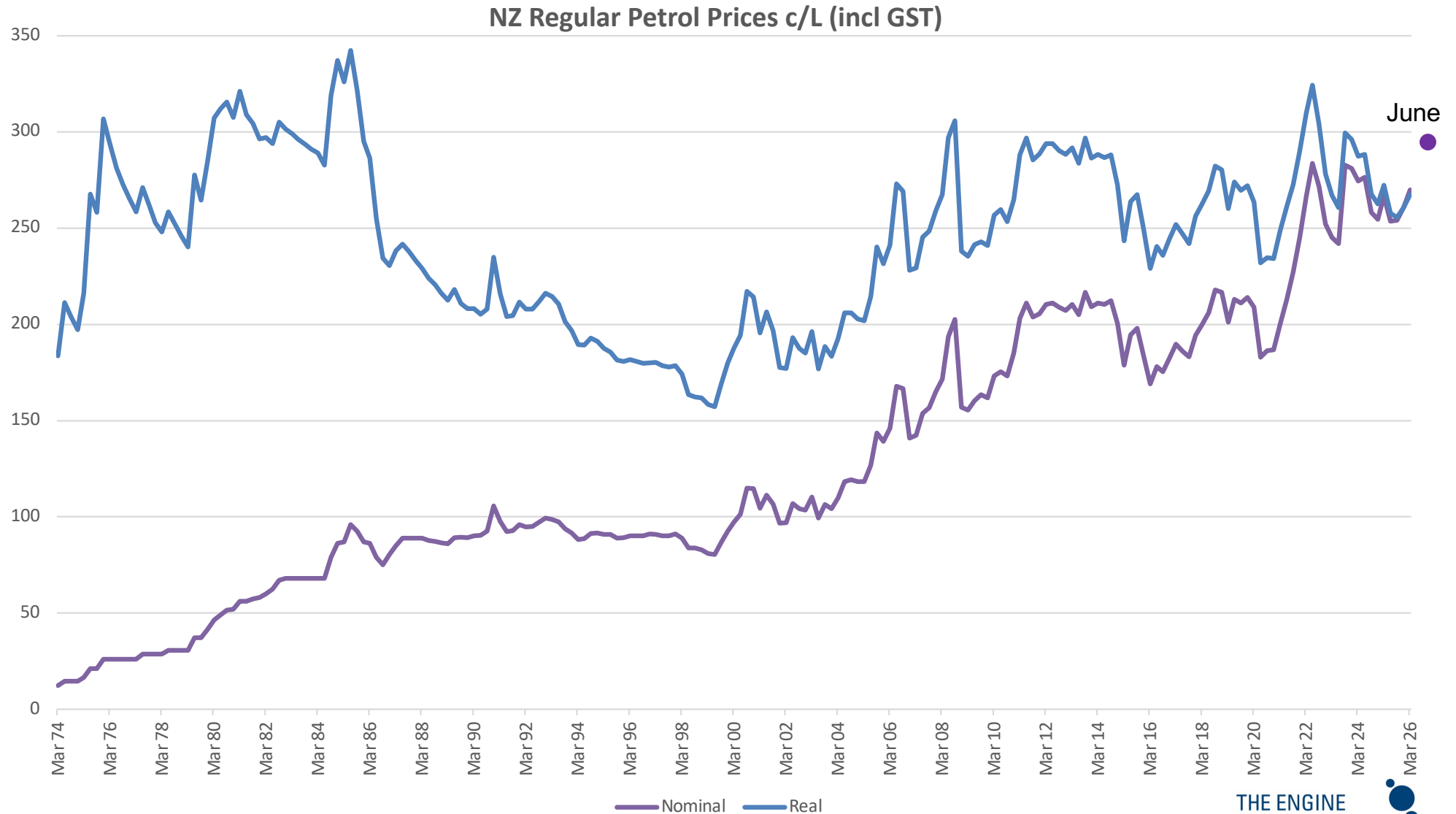
Line Chart

<https://www.mbie.govt.nz/assets/Data-Files/Energy/nz-energy-quarterly-and-energy-in-nz/energy-prices-2026-q1.xlsx>



Line Chart

<https://www.mbie.govt.nz/assets/Data-Files/Energy/nz-energy-quarterly-and-energy-in-nz/energy-prices-2026-q1.xlsx>



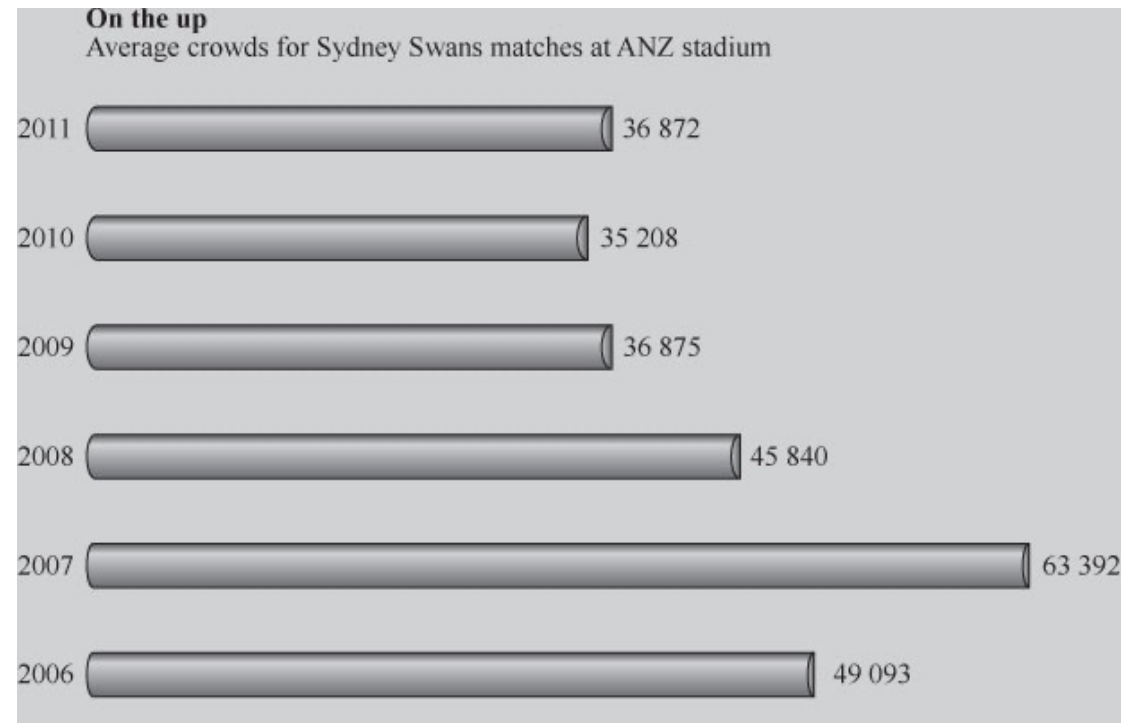
Bar Charts



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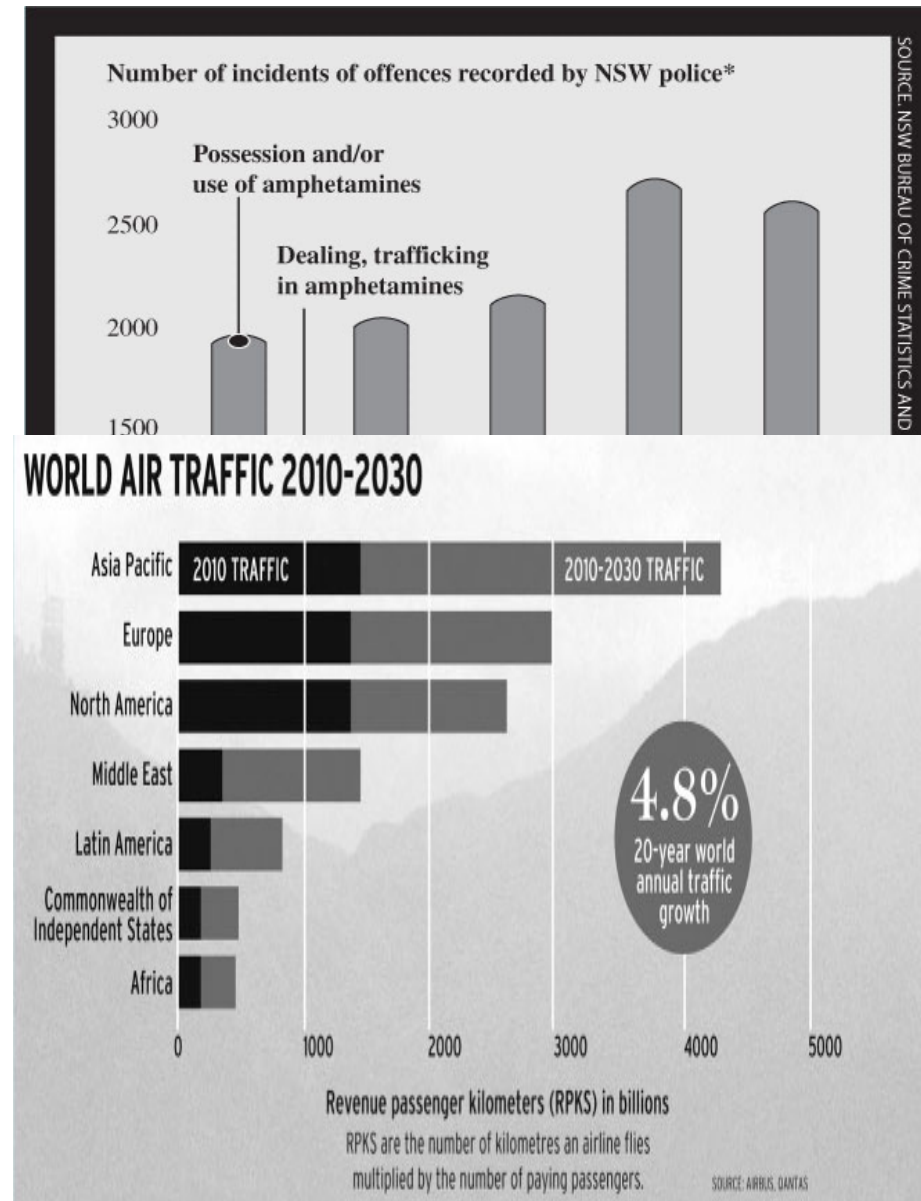
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- consists of a series of rectangular bars where the lengths of the bars represent the magnitudes of the respective quantities
- Guidelines:
 1. Make all bars the same *width* (*height if horizontal*)
 2. Use logical ordering if feasible or desirable
 3. Clearly label the axes
 4. Include key / legend to assist in the interpretation if necessary
 5. Include footnotes, tables and sources of data



Multiple Bar Charts

- *Multiple bar chart*: bars are displayed side by side, often in pairs or triples, in order to emphasise comparisons
 - Usually preferable to drawing two separate charts of the quantities
 - Usually no space between bars for data in same category
- *Compound bar chart*: bars are stacked on top of each other
 - often used to show growth



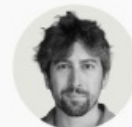
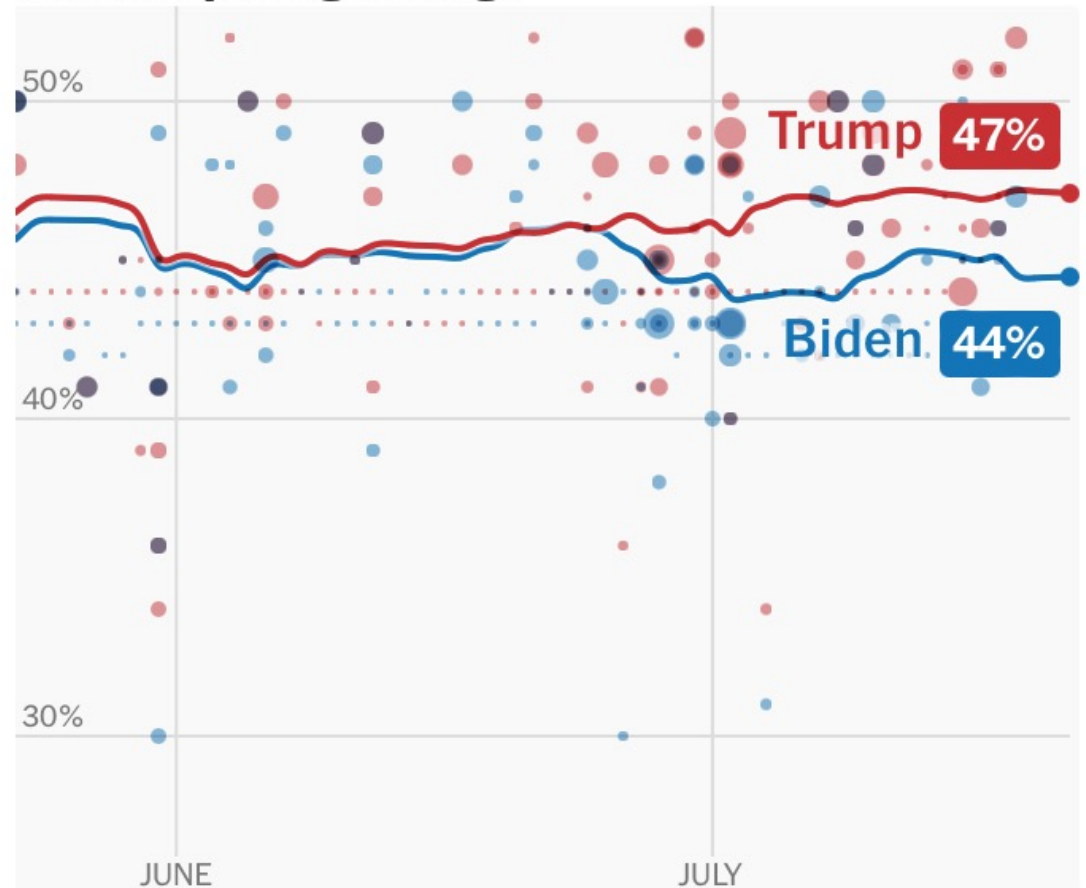
Compound Charts

Who's leading the polls? [🔗](#)

Biden vs. Trump

With Kennedy

National polling average

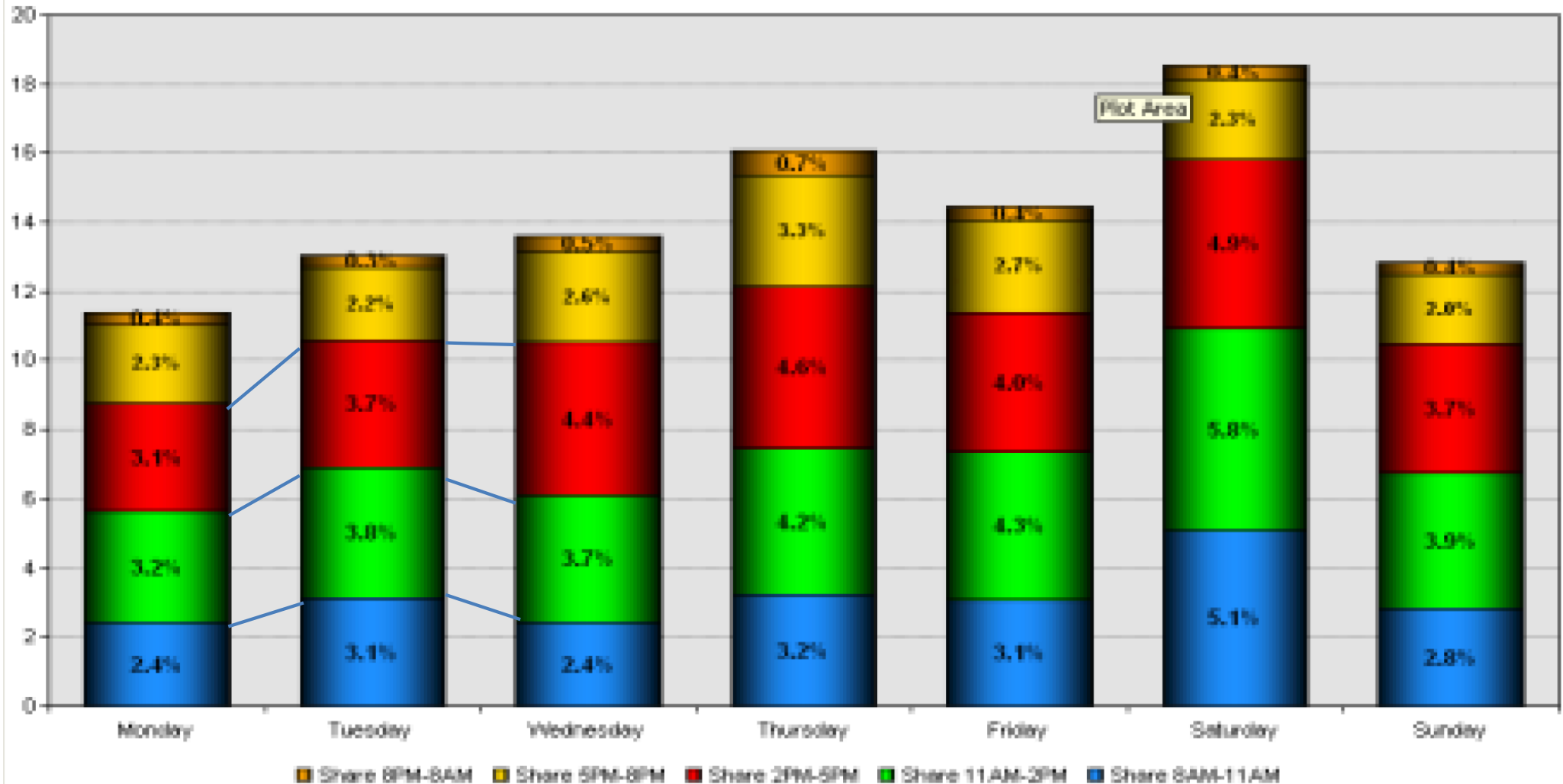


William P. Davis

Director, Election Data Analytics

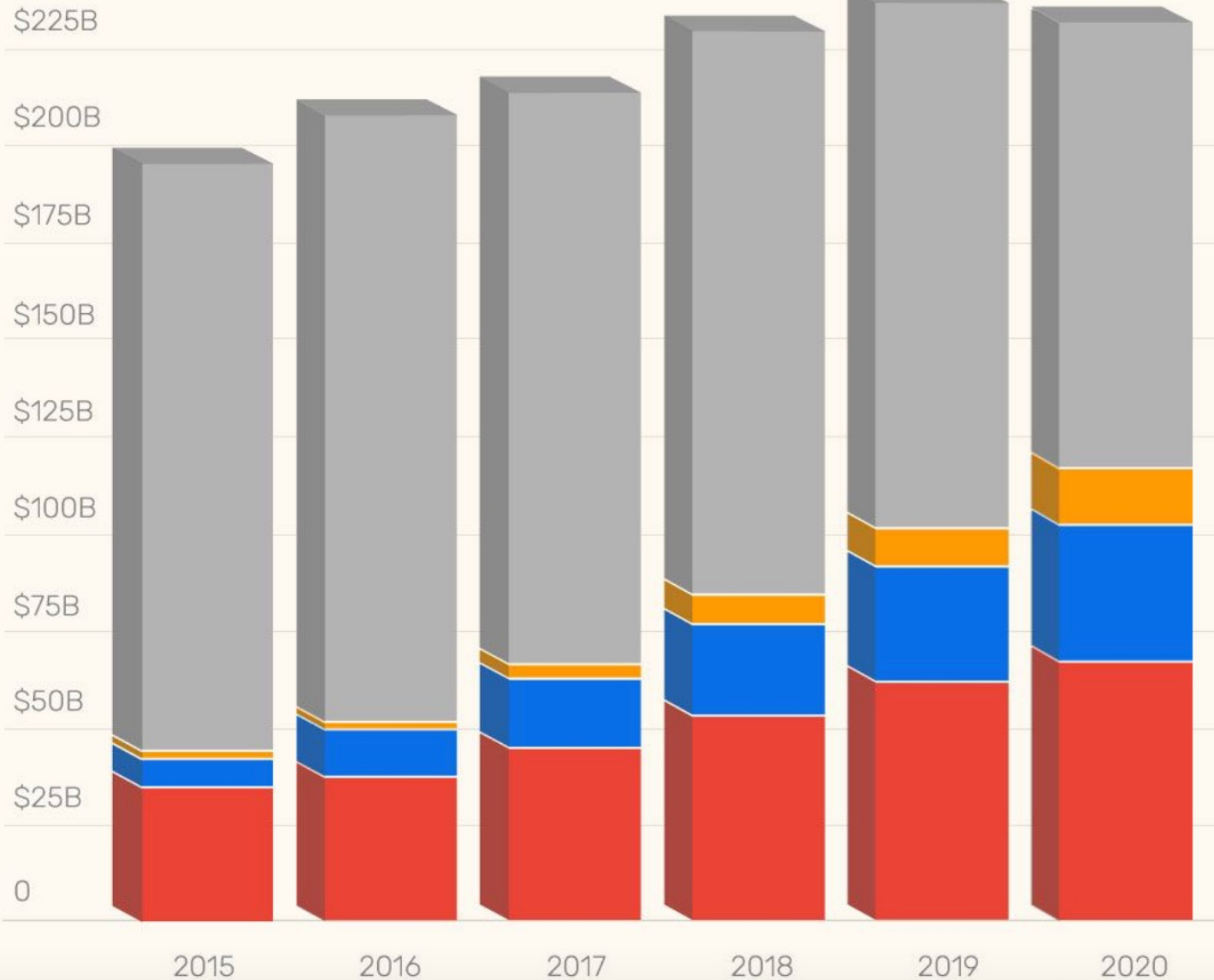
Bar Charts

When do households shop? Time of Day and Day of Week
TOTAL SUPERMARKETS - 13 WEEKS TO 17 JUNE 07

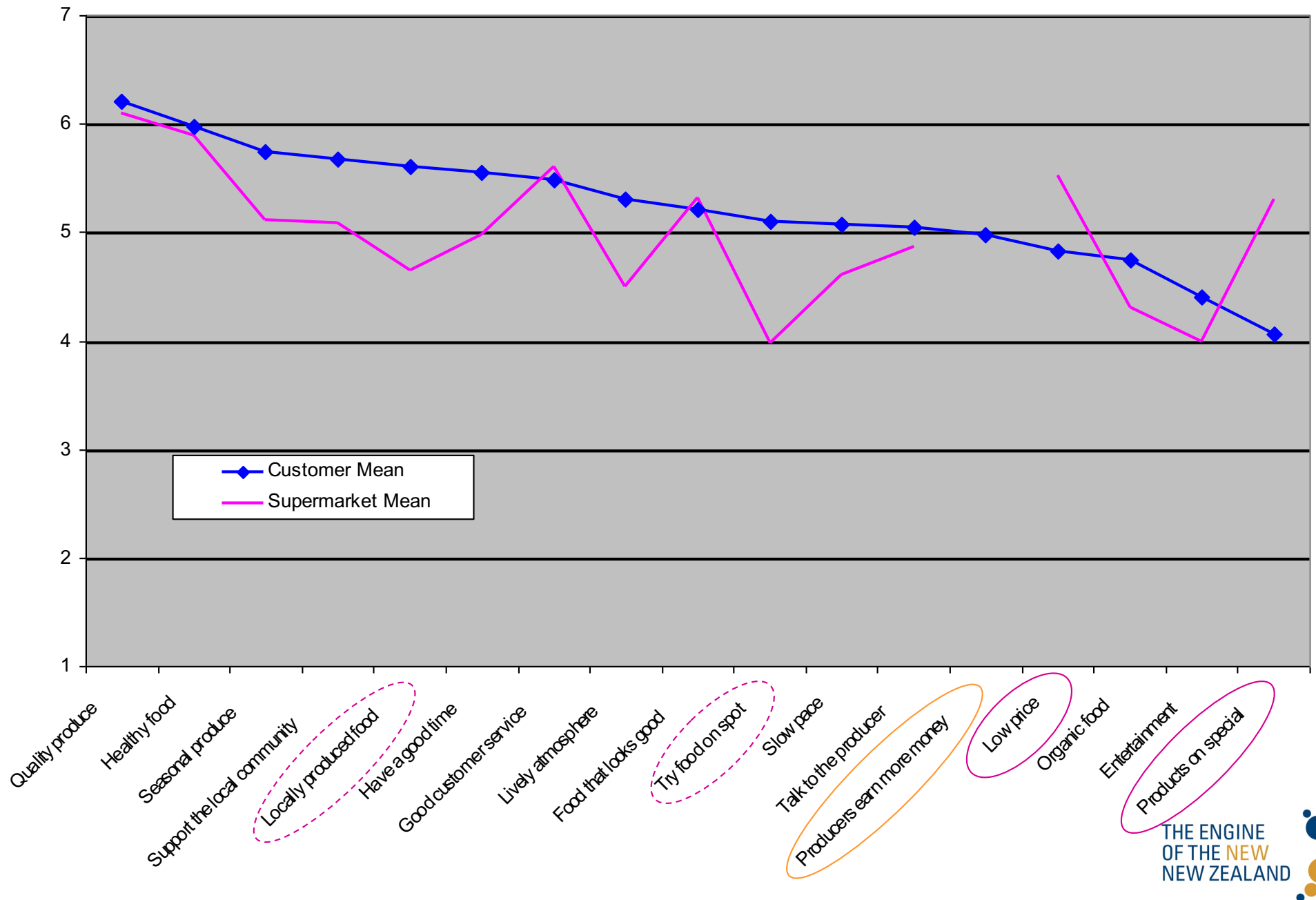


U.S. AD REVENUES

\$250 billion in ad revenue

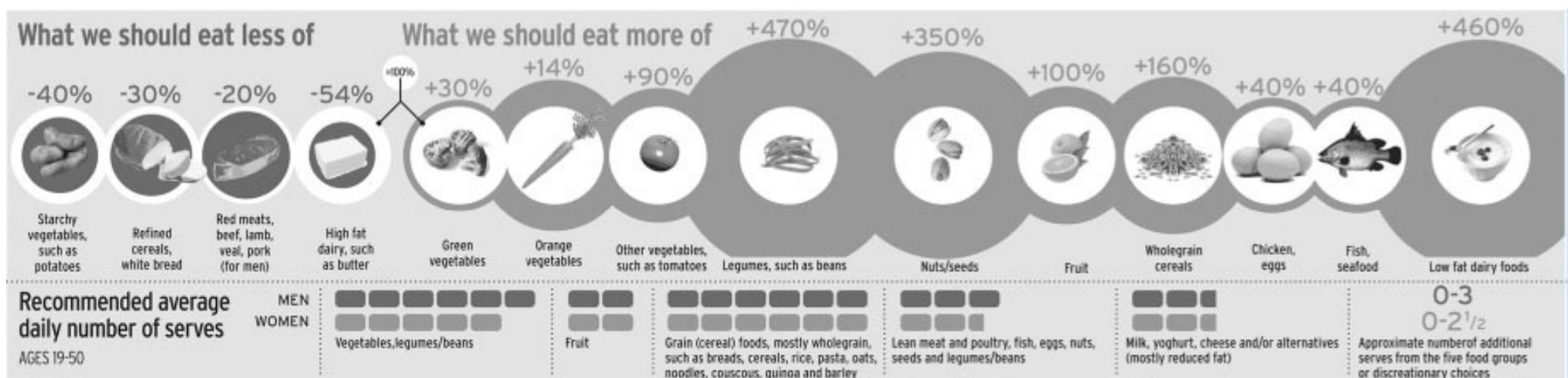
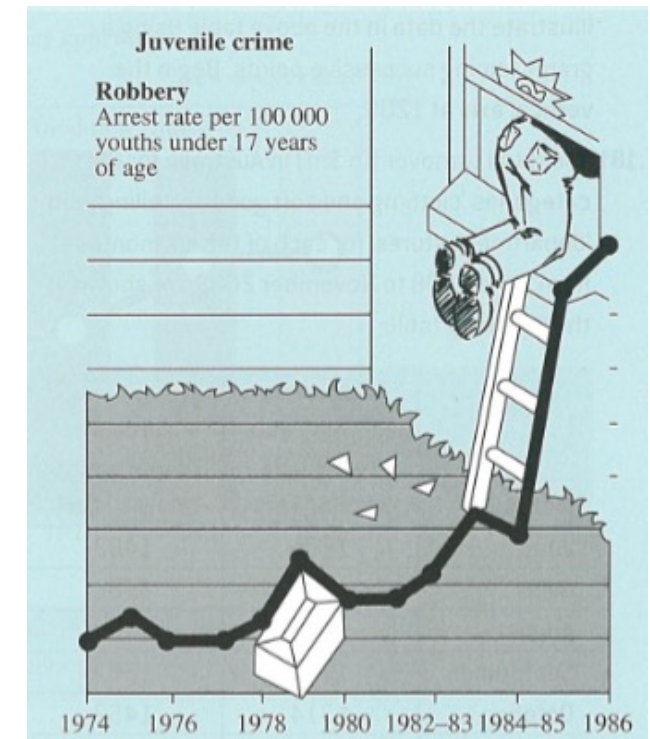


Farmers' Markets vs Supermarkets



Pictograms

- A graph in which data are displayed using pictures rather than traditional methods
- There are no real rules for drawing a pictogram – most often used in trade and public media



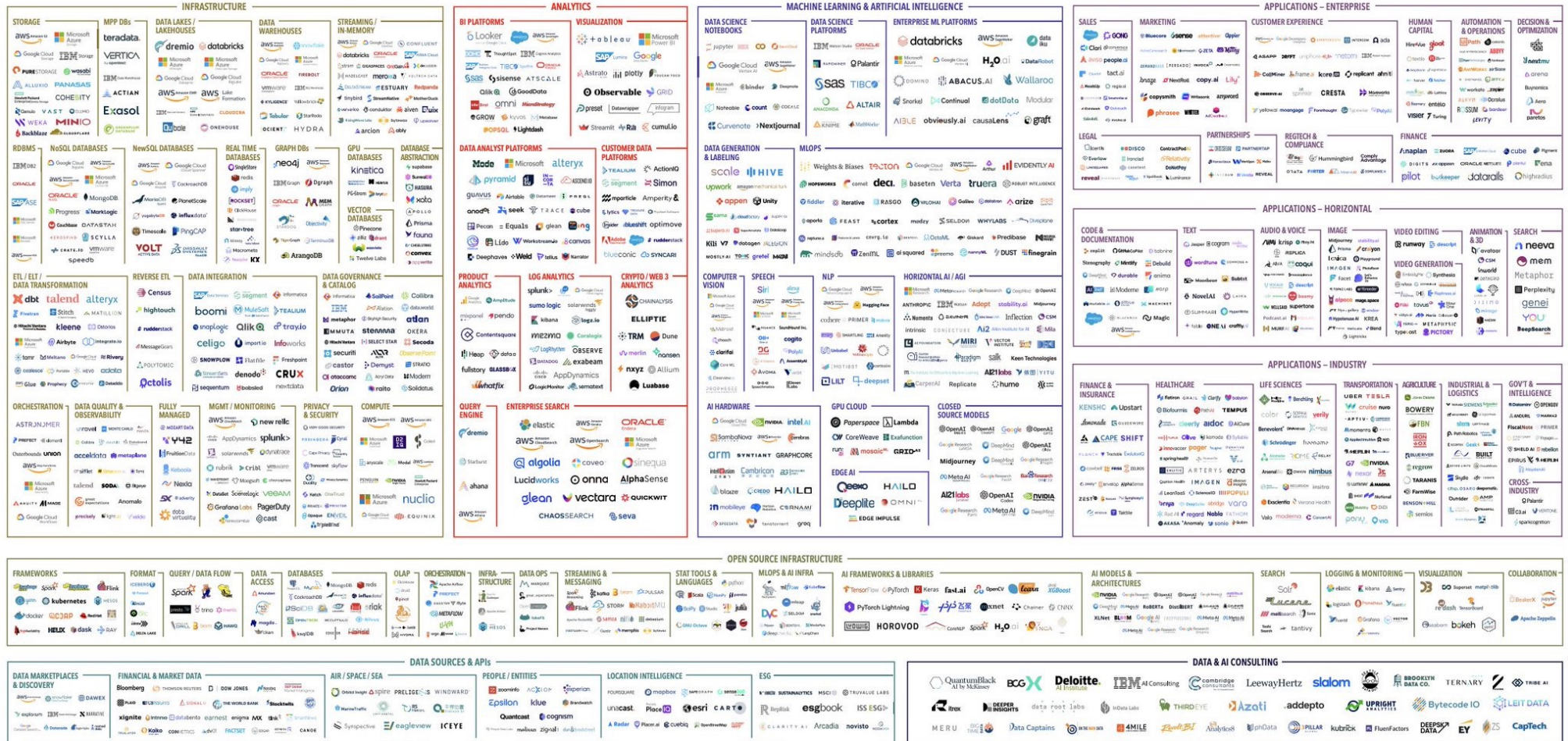
Pictograms



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THE 2023 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE



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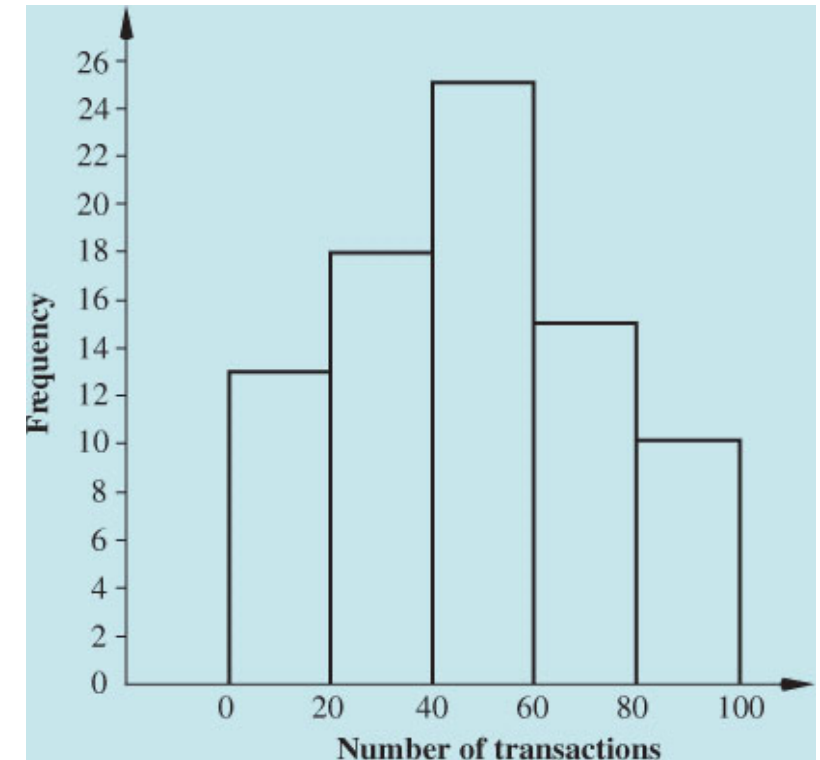


Frequency Distributions

- Gives a listing of different observations (in order of magnitude), each with the corresponding frequency alongside it
 - The data may be further reduced by grouping the observations into *class intervals* to form a *grouped frequency distribution*
 - A class interval is a range of values in which some of the observations may lie
 - A graph of this is called a *histogram*

Frequency Distribution

Class interval	Frequency
0 – under 20	13
20 – under 40	18
40 – under 60	25
60 – under 80	15
80 – under 100	9
	$\sum f = 80$



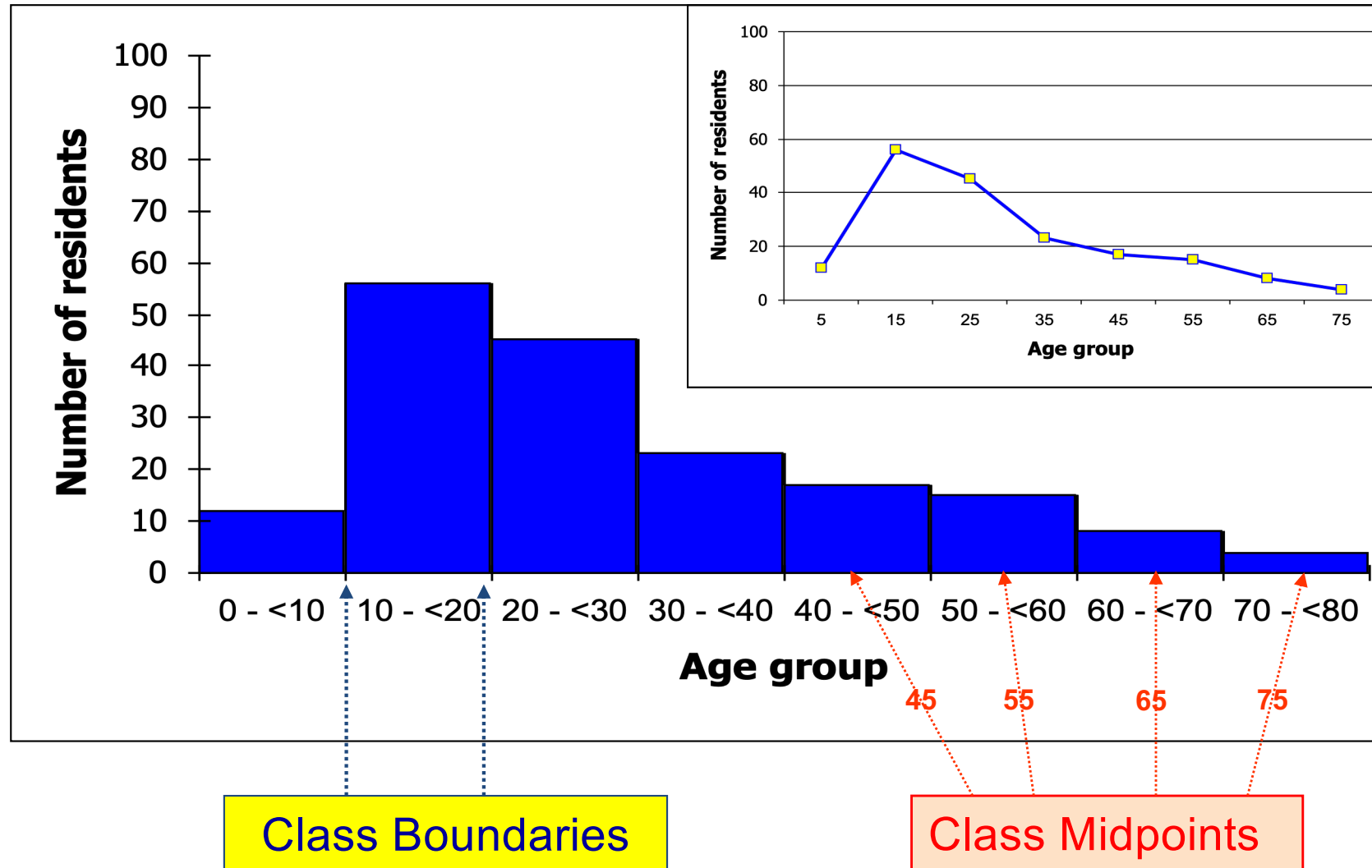
- A histogram is a simple (vertical) bar chart in which the frequency of observations within a class interval is represented by the corresponding bar

Histogram



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Meaningless Statements

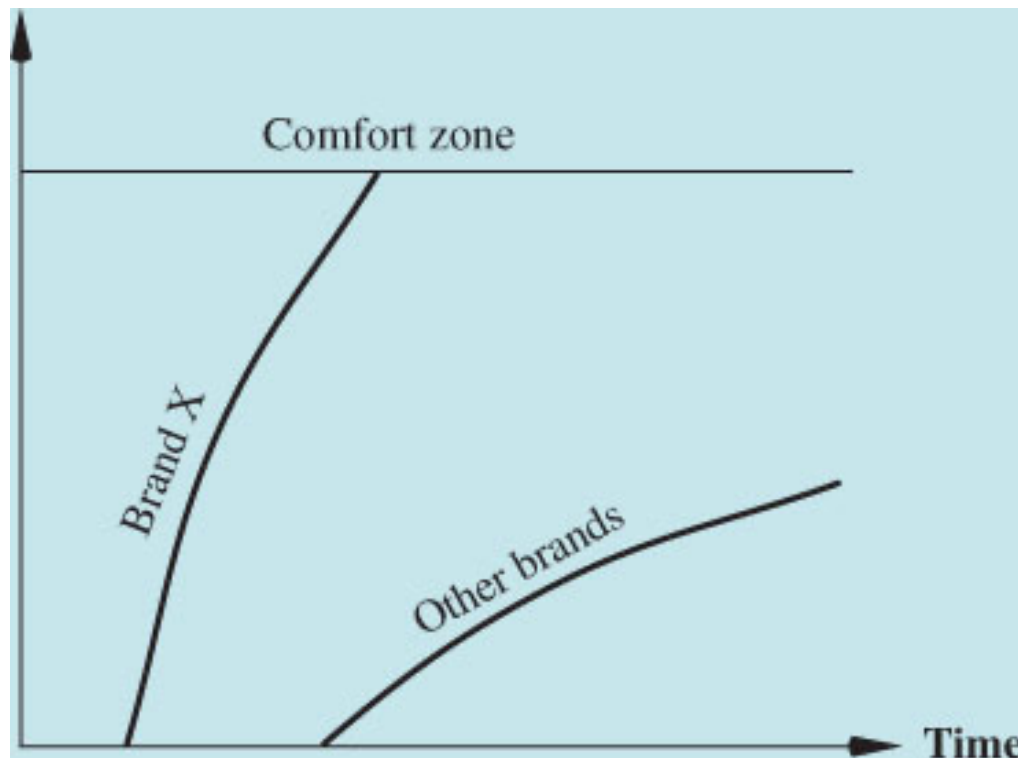
- There are many instances in practice where statistical information has been presented in such a manner as to purposefully (or otherwise) mislead the reader

E.g. 'Brand X washing detergent will wash your clothes 50% cleaner and brighter.'

Comment: What does '50%' mean? Cleaner and brighter than what? Nothing at all, another product or...? What is the basis for comparison?

Misleading graphs

- One of the most common ways to distort data is by means of a graph
 - graph has no scale, includes meaningless quantities and is quite useless in conveying *information*



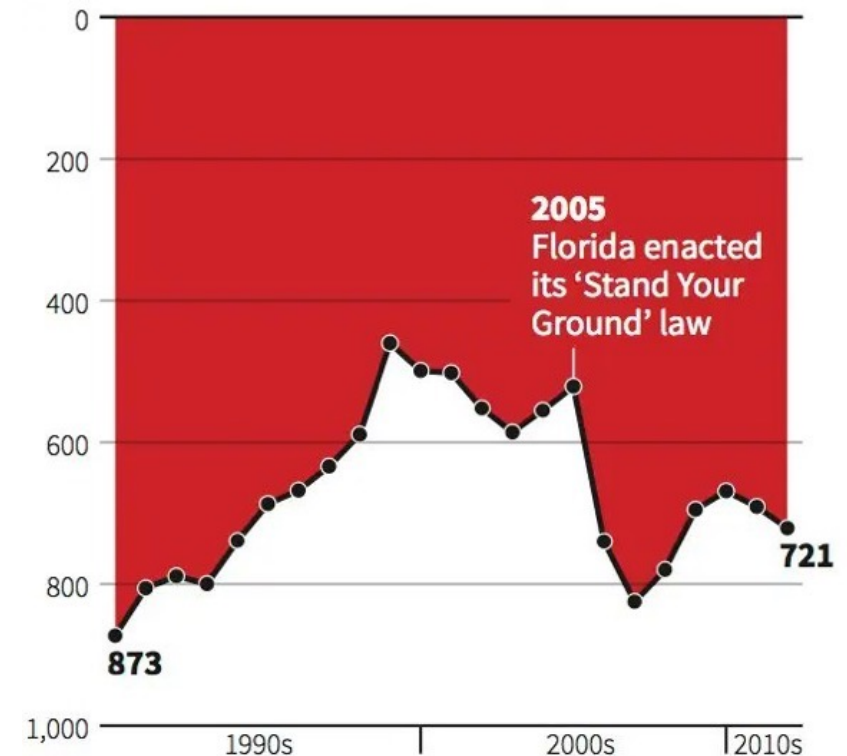
SAMPLING BIAS



sketchplanations

Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

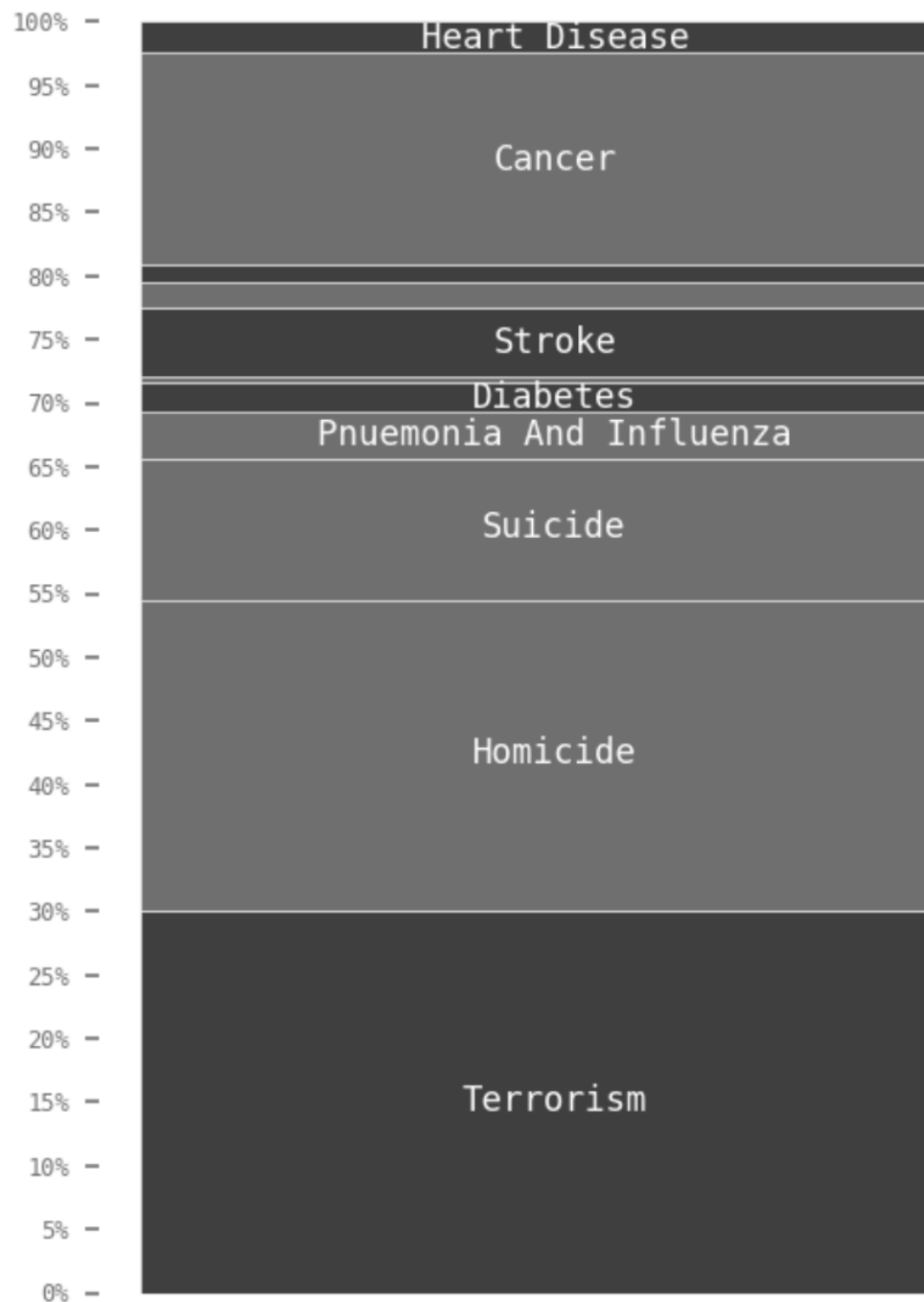
C. Chan 16/02/2014

REUTERS



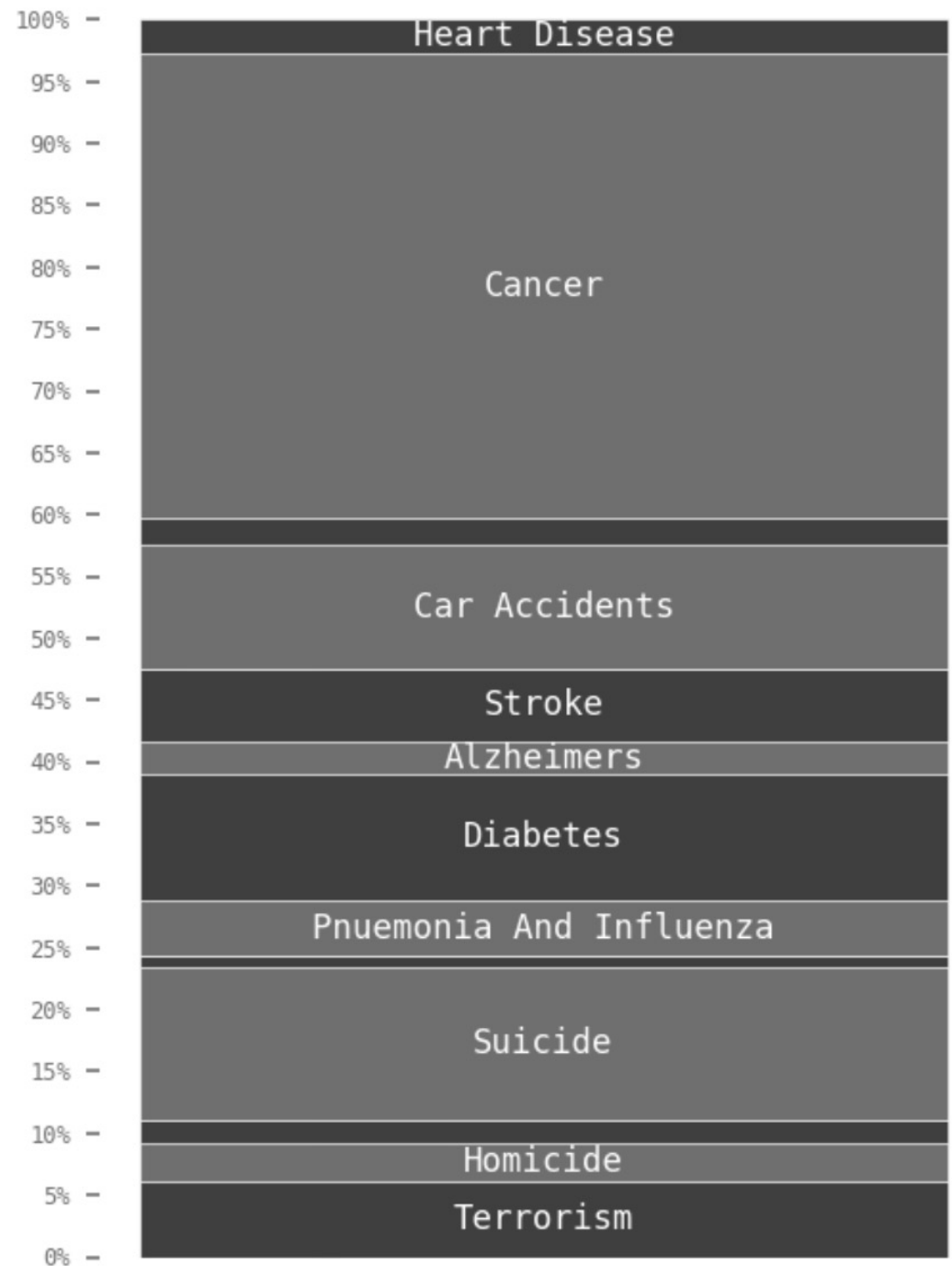
NYT & Guardian Headlines

"Which causes are in the media?"



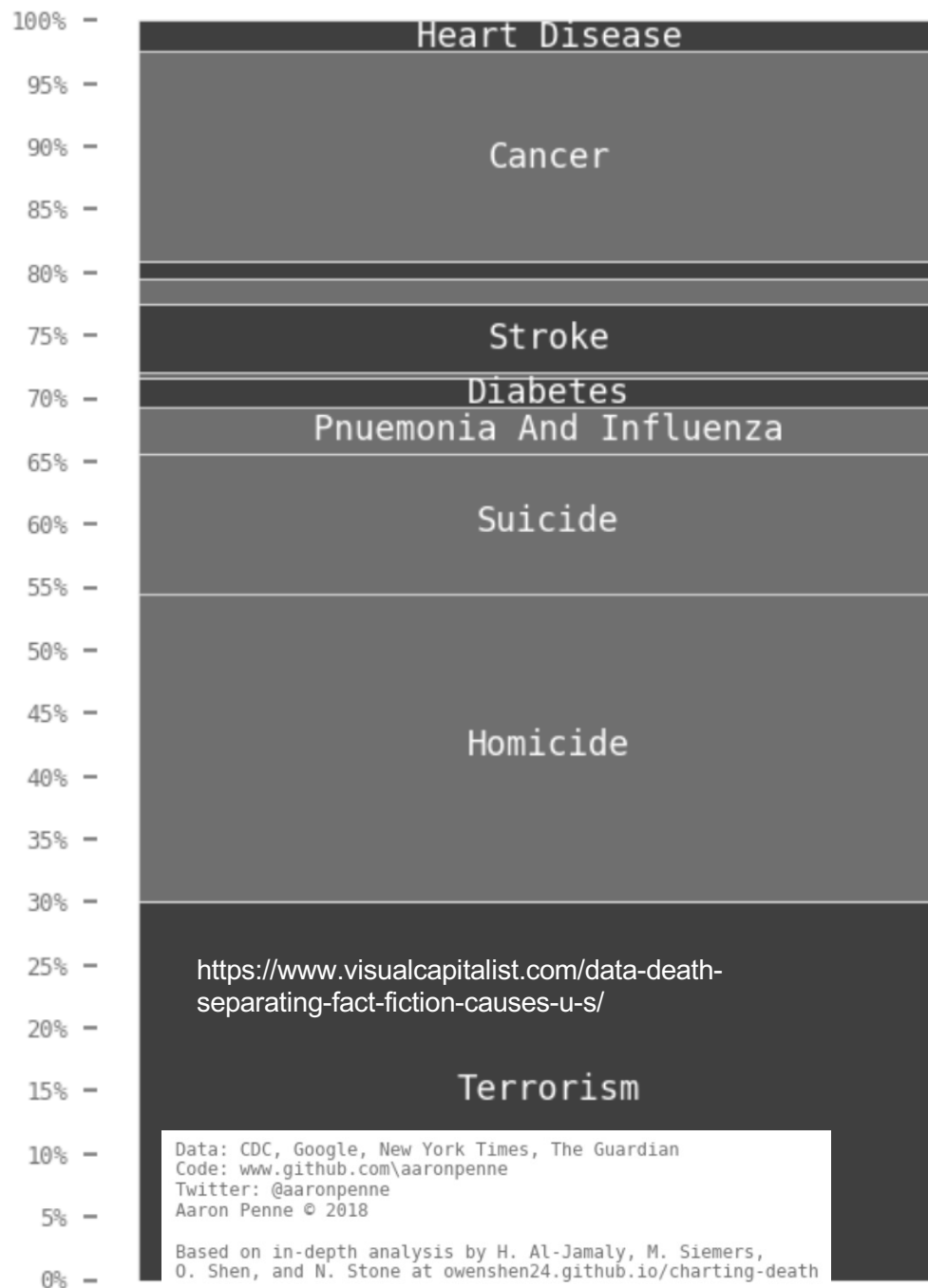
Google Search Trends

"Which causes do we worry about?"



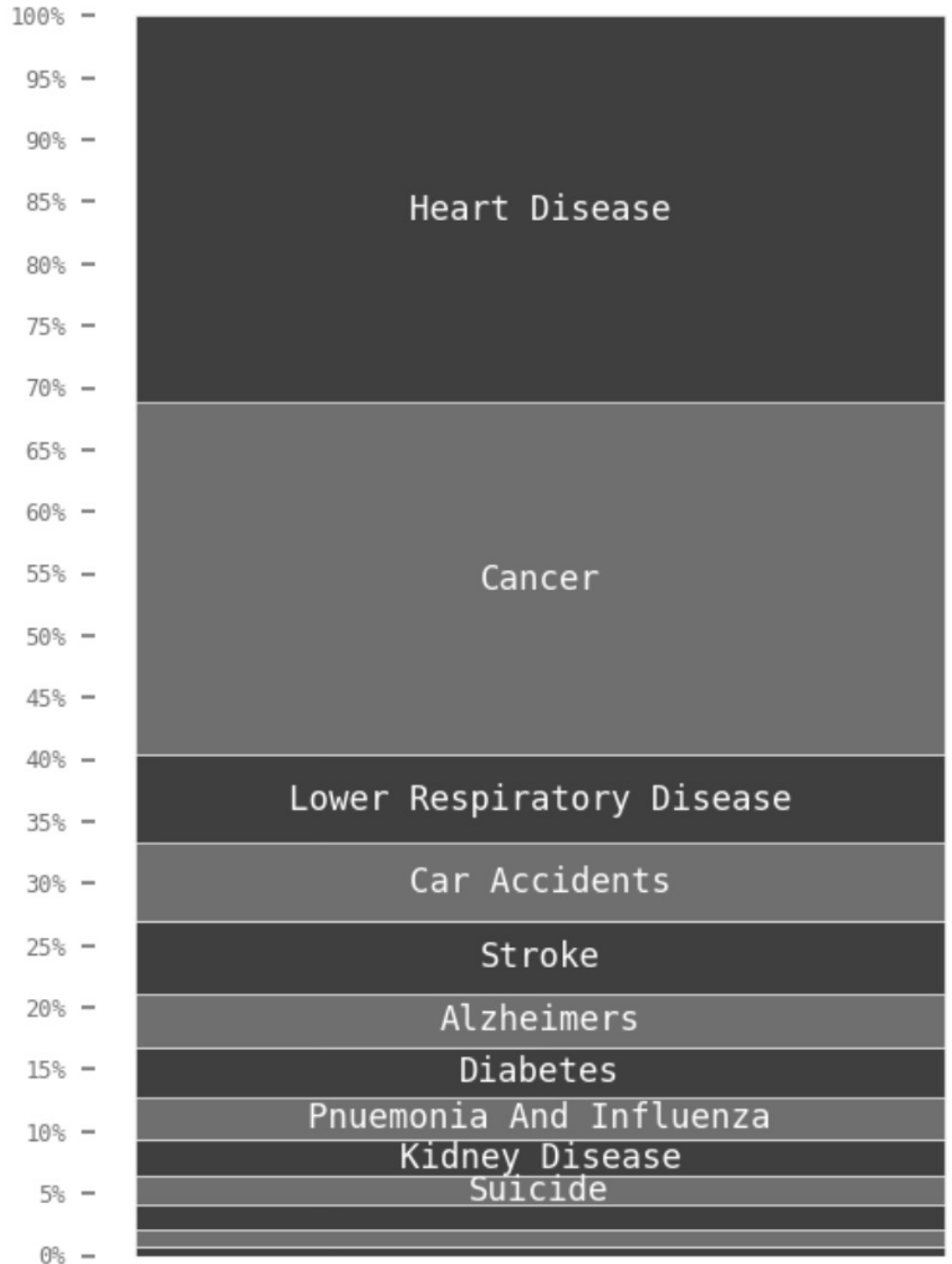
NYT & Guardian Headlines

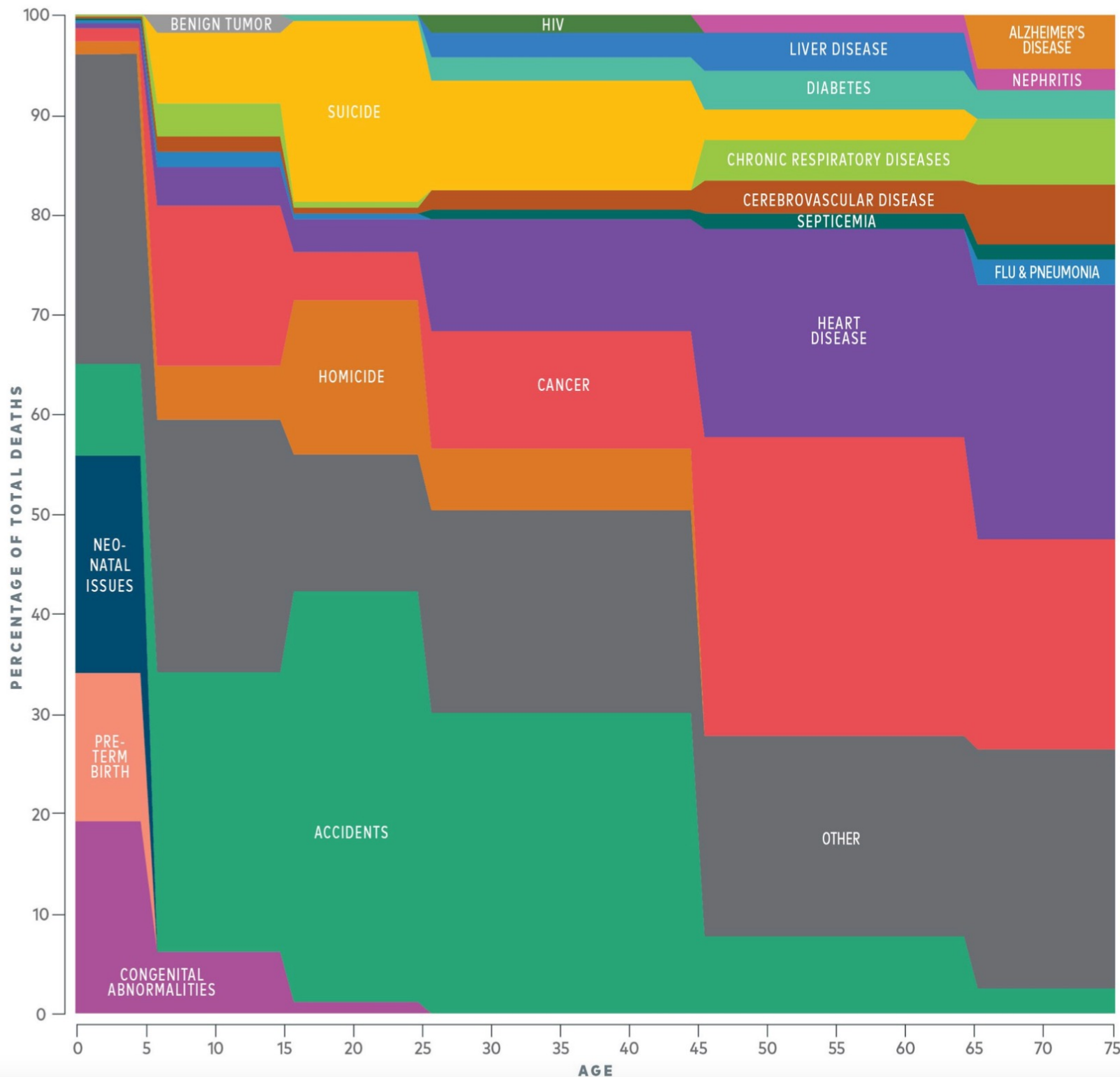
"Which causes are in the media?"



CDC Cause of Death in USA

"What actually causes death?"





US Causes of Death

<https://www.visualcapitalist.com/data-death-separating-fact-fiction-causes-u-s/>

Net Promoter Score

The Net Promoter Score (NPS) has attracted a lot of attention as the optimal measure of customer attitudinal loyalty. Based on a 'recommendation likelihood' rating, it's a relatively simple calculation containing three distinct steps:

1. Use an 11-point scale with a defined midpoint (5) is key. Based on a 0 to 10 scale where 0 means "not at all likely" and 10 means "extremely likely", how likely are you to recommend a [insert specific product/service] to a friend, colleague or family member desiring a similar [insert product/service]?
2. Next, collapse the recommend scale into three groups: 0-6 (labelled Detractors), 7-8 (Passives) and 9-10 (Promoters).
3. The third step is to subtract the percentage of the sample who are Detractors from the percentage who are Promoters.

Net Promoter Score

“Recommendation Likelihood” Net Promoter Scores

<i>Detractors</i>	0	25
<i>Passives</i>	60	10
<i>Promoters</i>	40	65
Total	100%	100%
NPS	40	40

How likely are you to recommend us to a friend or colleague?

0

1

2

3

4

5

6

7

8

9

10

Not likely at all

Extremely likely

Net Promoter Score

- Simple but large categories: those rating “0” (“not at all likely”) treated the same as those who give a 5 (neutral)
- A model of loyalty can best be built on multiple metrics opposed to a single measure
- Some questions are good measures of higher loyalty levels than others and should carry greater weight

Other measures of customer satisfaction:

- Overall repurchase likelihood
- Price sensitivity
- Willingness to consider competitive offerings
- Emotional attachment to the brand or service

<http://www.marketsource.com/blog/2015/06/16/customer-satisfaction-definition-still-using-net-promoter-score/>

Net Promoter Score

Net Promoter Score Dashboard

Customer Satisfaction Survey 08-2021

Replies

987



Detractors

73



Neutrals

45

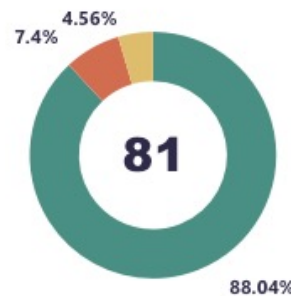


Promoters

869



Overall NPS



NPS Classification

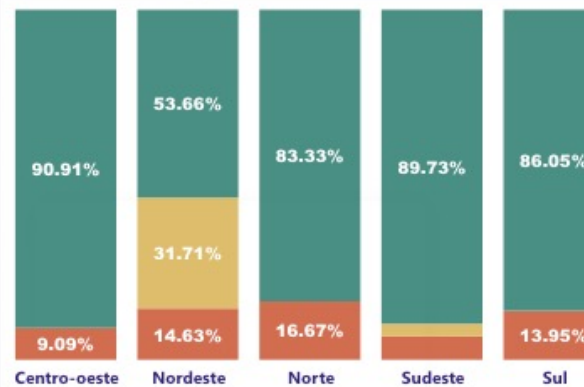
- Promoter
- Detractor
- Neutral

NPS Index NPS Classification

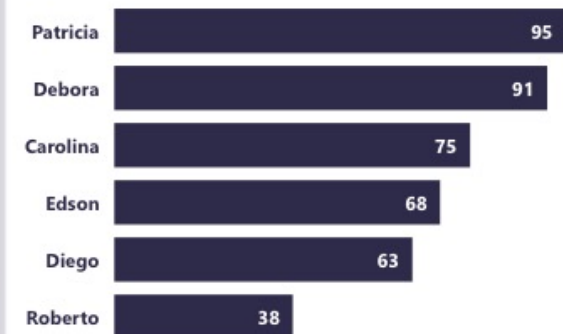
81 World Class



Satisfaction by Location



Average NPS by Managers



Filter Panel

Manager

Carolina	Edson
Debora	Patricia
Diego	Roberto

Region

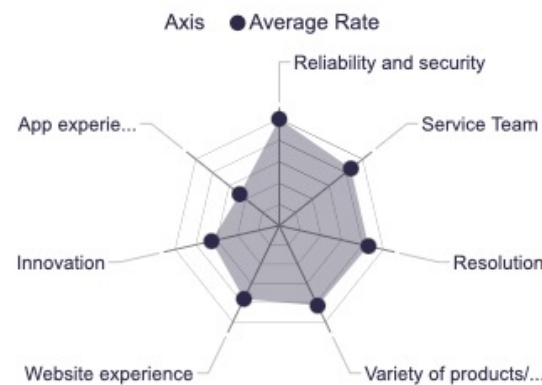
All

State

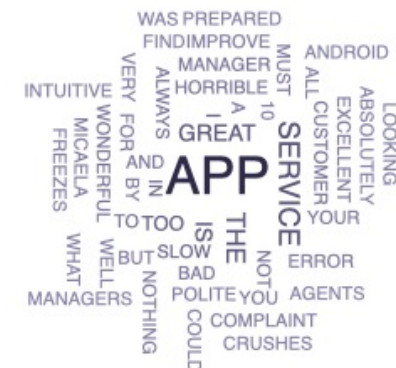
All

GET DASHBOARD

CSAT by Category



Customers Comments

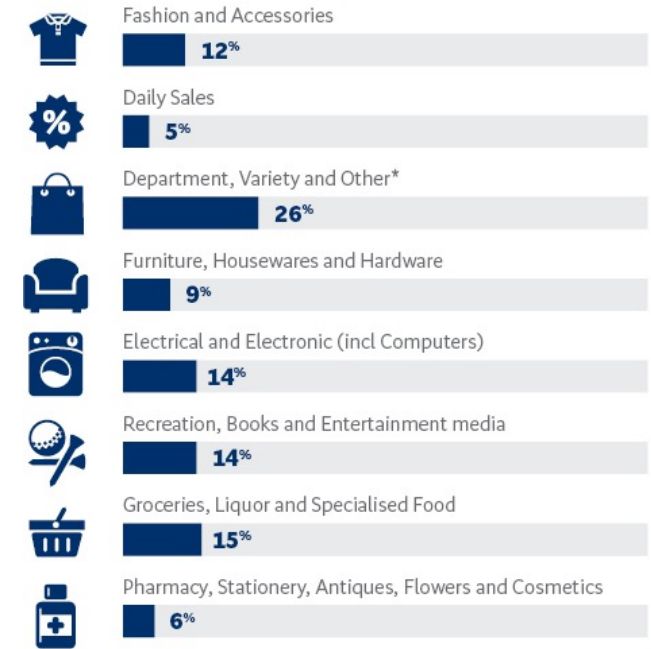


Dashboards/Scorecards

- Draw together a range of chart and table types to quickly 'take the pulse' of a unit or organisation
- Usual purpose is to compare performance to previous periods and/or against budget

Share of total online spend by sector

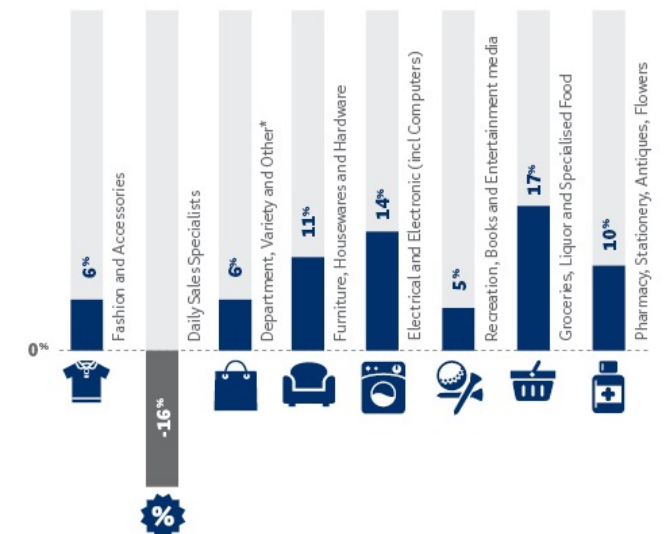
Year to January 2015 share of spend (%)



* "Other" includes unclassified stores and specialty stores that don't neatly fit into defined ANZSIC categories. Figures may not add to 100% due to rounding.
Trade Me spending (estimated "retail" component, as per the retail categories we include in the online index) is included with Department Store spending.

Annual growth in online retail sales

12 months to January 2015 (%)





FULL TIME EQUIVALENTS (FTE's)

	Current Month			Change		
	Permanent	Temps	Totals	Last Quarter	Last Year	% H/count
Stores	4,086.3	773.0	4,859.3	555.3	242.3	62%
Dist Centres	254.4	-	254.4	-4.5	-19.8	95%
SSO	559.8	24.2	584.0	-9.3	58.2	86%
Totals	4,900.5	797.2	5,697.7	541.5	280.8	65%

Movement from last year shows reinvestment in store labour. DC change reflects closure of Puhunui Road. Support Office reflects some of the Red/Blue transitions but move from last quarter is additional headcount

RETENTION

	Rolling 12 Months				Monthly	
	# leavers	Retention	Last Year	Change	This Year	Last Year
Stores	927	84.1%	84.7%	-0.6%	84	83
Dist Cent	13	95.1%		95.1%	0	
SSO	70	88.4%		88.4%	6	
Totals	1010	84.9%		84.9%	90	83

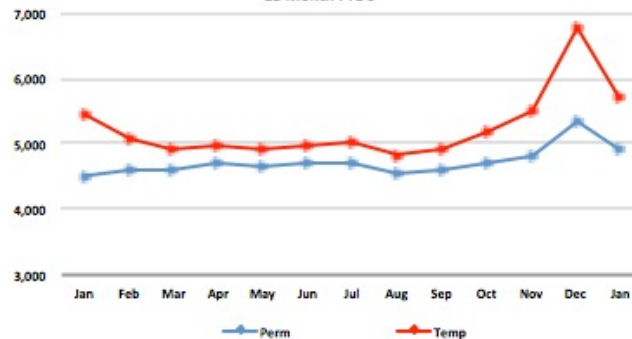
Note - retention rates for the as at January 2012 are not available. The company was going through a large period of change in Support Office and data could not be easily tracked as structures changed

HEALTH & SAFETY

	Lost Time Injuries			Total Injuries (Medical treated)		
	Number	Rate	v Last Year	Number	Rate	v Last Year
Stores	107	11.47	1.93	347	37.91	-1.33
Dist Cent	3	5.31	-0.03	10	17.69	1.66
SSO	0	0	-0.93	11	9.98	0.73
Totals	112	10.25	1.95	373	34.12	-0.25

The Lost Time Injury rate has begun to increase over the last 6 months after reaching a low in the middle of 2012. The Total Injury rate has remained constant across the company, although improvements are being seen in stores. January 2013 had a high incidents of claims compared to last year.

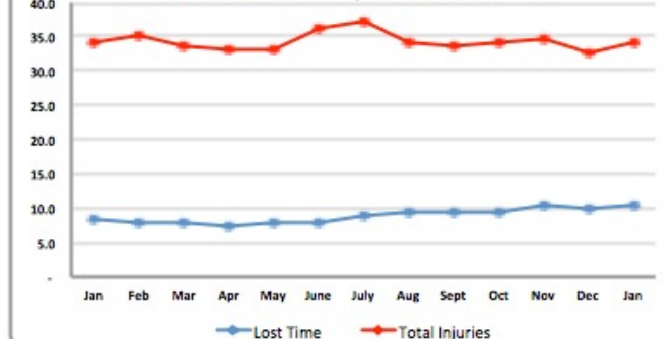
13 Month FTE's



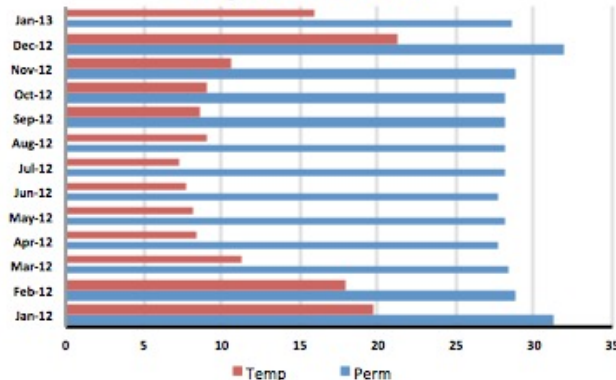
Rolling 13 Month Retention Rates - Stores



13 Month H&S Rates - per 1 million hours

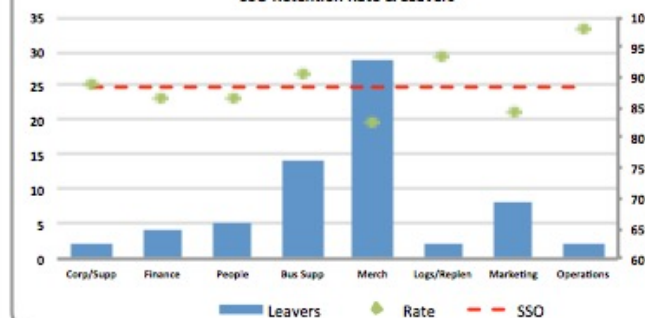


Average Hours Per Team Member



Above graph shows how the average hours per week increase during the year. The peaks of the red lines in Dec and Jan indicate that a large portion of additional store hours are allocated to temporary team members. Average week for a Perm TM increases from 28 to 31 hours per week on average during our peak periods

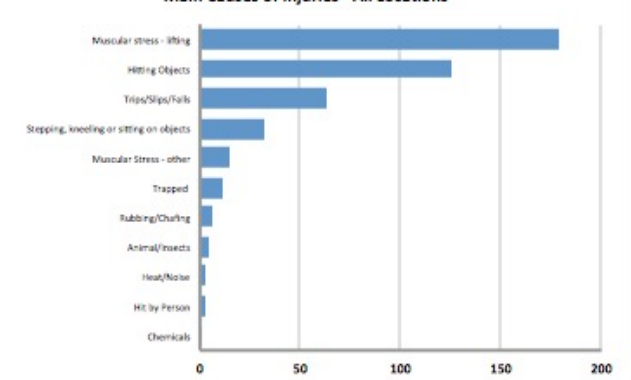
SSO Retention Rate & Leavers



	Leavers	Rate
Corp/Supp	2	88.9%
Finance	4	86.8%
People	5	86.6%
Bus Supp	14	90.7%
Merch	29	82.5%
Logs/Replen	2	93.7%
Marketing	8	84.1%
Operations	2	98.2%
Total SSO		88.4%

Overall retention rate for SSO is 88.4% with most functions about that level. Operations (incl. Store Dev) are currently tracking at 98% retention. Head 2 Toe has the lowest rates at around 66%

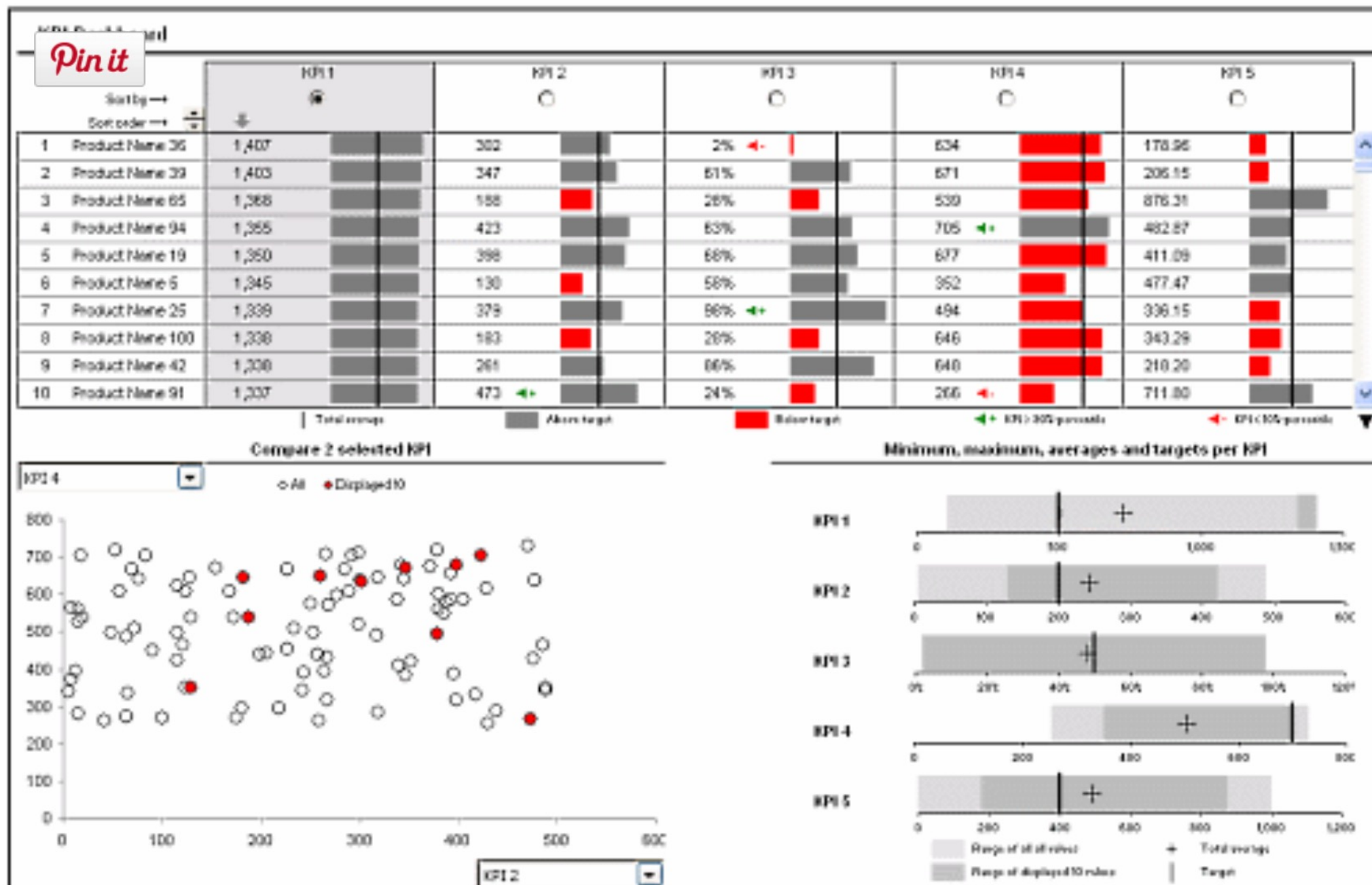
Main Causes of Injuries - All Locations



In Support Office the major source of injuries are Slips, Trips and Falls. Many are caused by wet floors. There has been a recent increase in muscular injuries from lifting. In stores our three main causes of injuries are Muscular Stress (manual handling issues), Hitting or being hit by Objects, and Slips, Trips and Falls. H&S prevention initiatives are based around these three causes.

Dashboards

	Year to Date								
	Actual	Budget		Var	% var	Last Year		Var	% var
Profit	Actual	Budget		Var	% var	Last Year		Var	% var
Sales	226.9	238.4	🔴	(11.4)	(6.1)%	221.5	🟢	5.4	3.2%
Gross Profit	119.5	124.4	🔴	(4.9)	(6.6)%	116.8	🟢	2.7	4.0%
%	39.3%	39.5%	🟡	(0.2)%		39.5%	🟢	0.3%	
CODB	109.2	113.3	🟢	4.1	6.4%	107.5	🟡	(1.8)	(3.1)%
%	33.5%	33.6%	🟢	0.1%		33.6%	🟢	0.0%	
EBIT	60.3	61.1	🔴	(0.8)	(7.5)%	59.3	🟢	0.9	9.8%
%	5.8%	5.9%	🟡	(0.1)%		5.9%	🟢	0.3%	
CAPEX spend	54.1	56.1	🟢	2.0	32.5%	55.2	🟢	1.1	20.7%
ROFE - 12 Months Rolling Average	27.6%	30.5%	🔴	(3.0)%	-0.09721	30.5%	🟢	0.2%	

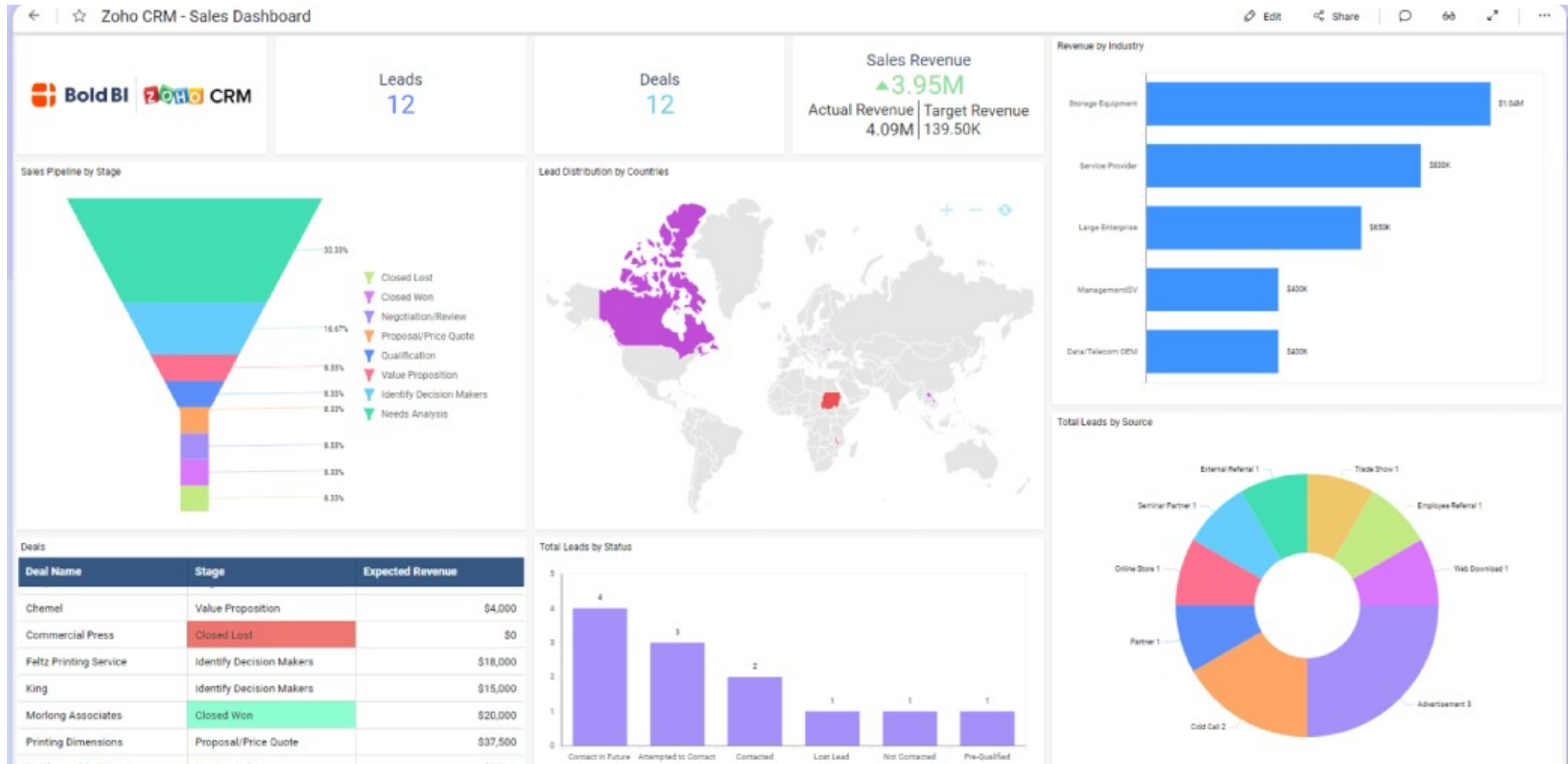


Dashboards: CRM



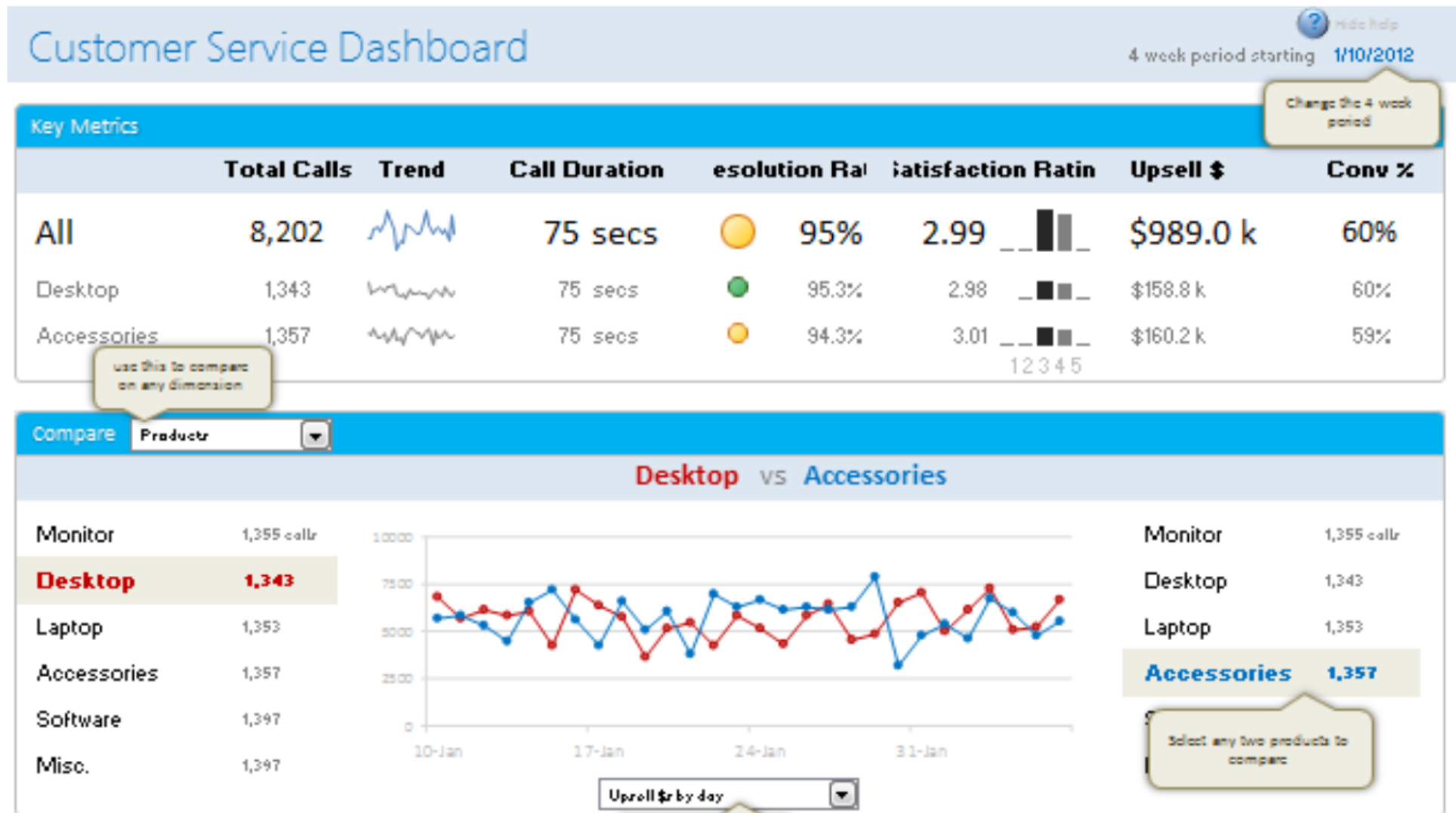
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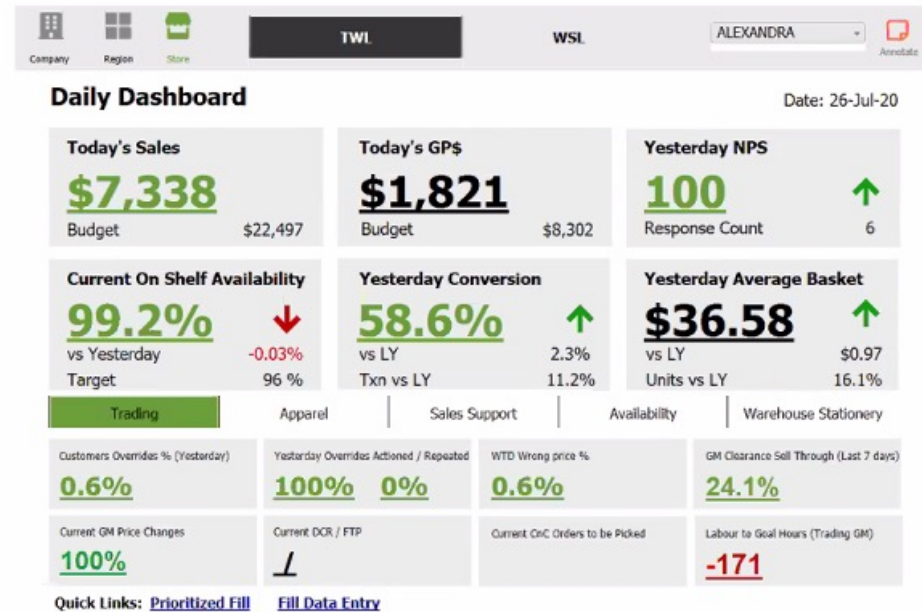
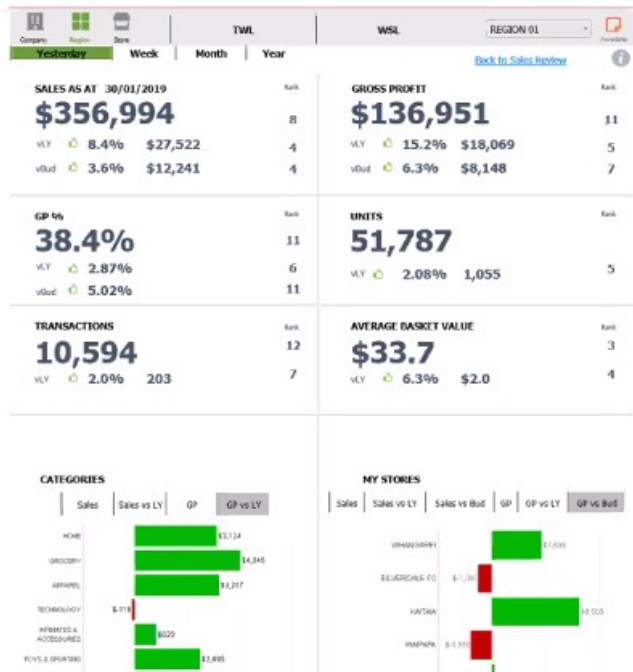


<https://www.softwaresuggest.com/blog/crm-dashboard/>

Dashboards



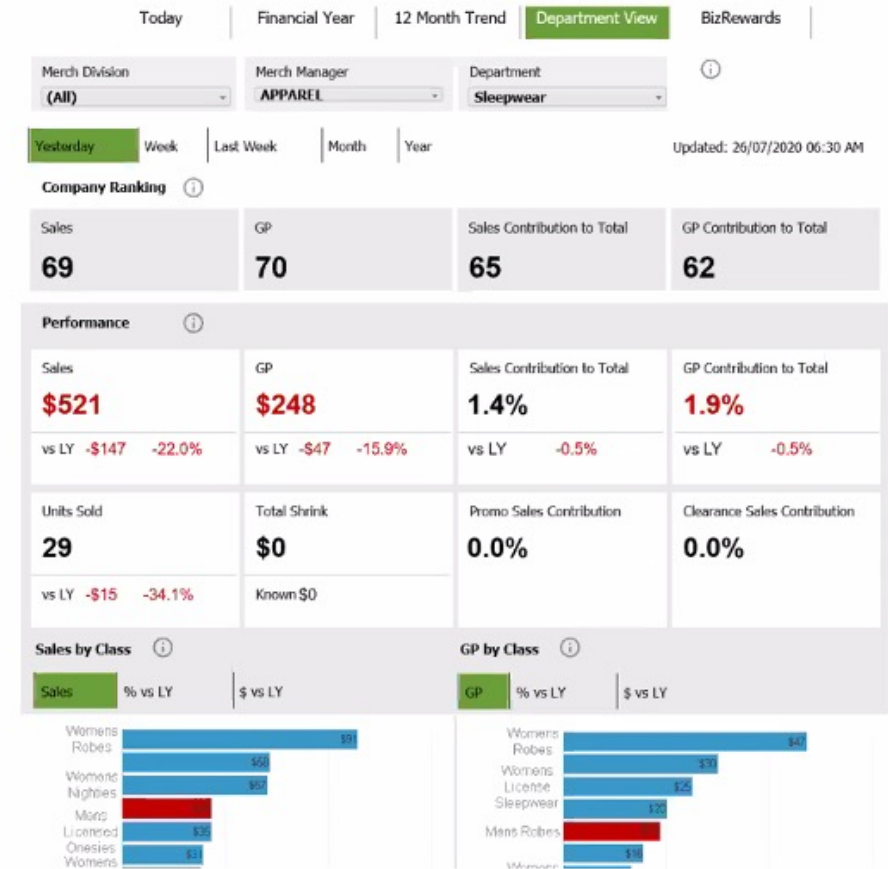
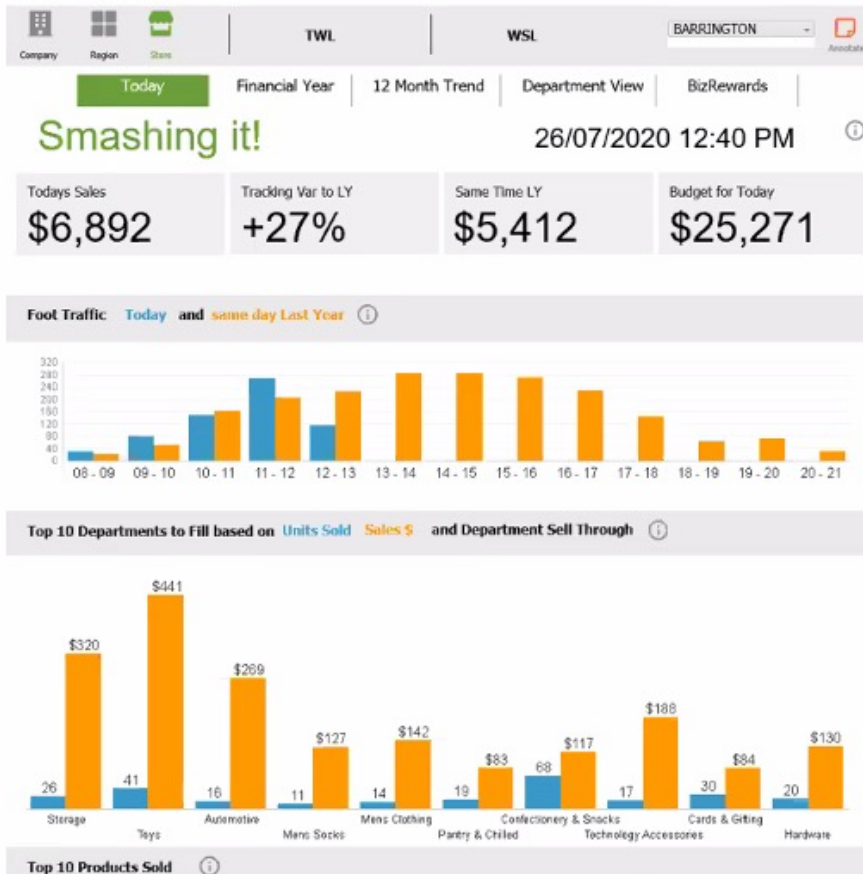
Dashboards: TWG



Please send us feedback



Dashboards: TWG



THE 

Dashboards: Lab 2

	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	115.109: Topic 2 Lab Worksheet				Your Name:							Rationale: First, we look at dashboards, using Income Statement for the financial 2023. This is a simple dashboard, but how we can use indicators to flag a budget or last year's performance). lights are configured, by clicking on and selecting 'Conditional Formatting More Rules.' Second, we analyse the knowledge and AK of the SustainabilityData-10 simpler way to do all this, which we PivotTables.' Third, we construct a column chart frequencies, and then a pie chart of from Lab 1. These are pretty simple graphing skills. It helps to label the chart titles, particularly for the columns of this in Lab 3.)		
2	Due date: August 3		Version: 1 (09.07)		Student ID:									
3														
4	Task 1:													
5	Locate the The Warehouse Group 2024 and 2025 Income Statements: extracts available in Week 1 lecture notes													
6	Links to financial statements:													
7	2025 Annual Report financial statements: https://www.thewarehousegroup.co.nz/download_file/force/83c5e7a5-091c-4 p. 36													
8	2024 Annual Report financial statements: https://www.thewarehousegroup.co.nz/download_file/force/5a624589-60db-4 p. 44 (includes 2023 info: both 52 week periods)													
9														
10	Complete the following table:													
11	Financial Year ending July	(Income statements reported in \$000s - paste into the formula bar not the actual cell, to retain format)										B14		
12		2025	2024		Var	% var	2023		Var	% var				
13		(restated in 2024: use the values from 2024 report)												
14	Sales (Total Revenue)	3,086,725		#DIV/0!	3,086,725	#DIV/0!		#DIV/0!	0	#DIV/0!				
15	COGS (Cost of Retail Goods Sold)		2,016,731		2,016,731	100.0%		#DIV/0!	(2,016,731)	#DIV/0!				
16	COGS % sales revenue	0.0%	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!				
17	Gross Profit \$	3,086,725	(2,016,731)		5,103,456	-253.1%	0	#DIV/0!	(2,016,731)	#DIV/0!				
18	GP % sales revenue	100.0%	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!				
19	CODB (Lease, Employee, Depreciation, Other expense)			#DIV/0!	0	#DIV/0!		#DIV/0!	0	#DIV/0!				
20	CODB % sales revenue	0.0%	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!				
21	EBIT (Earnings before Interest and	3,086,725	(2,016,731)		5,103,456	-253.1%	0	#DIV/0!	(2,016,731)	#DIV/0!				
22	EBIT % sales revenue	100.0%	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!				
23														
24	Interpret the table: what improved, and did not, between 2023-2024, and 2024-2025? {you can resize this box if you want}													
25	Answer:													

What do the dashboard colours mean?

Green: revenue is up, or costs are down, or a ratio is up/down/worse a lot

Orange: revenue is down a bit, or costs are up a bit

Red: up/down/worse a lot

To see how these are formatted: click on a colour ribbon click on 'Conditional Formatting' > Manage

What do the dashboard colours mean?
Green: revenue is up, or costs are down, or a ratio is down
Orange: revenue is down a bit, or costs are up a bit
Red: up/down/worse a lot
 To see how these are formatted: click on a colour ribbon click on 'Conditional Formatting' > Manage

Dashboards: Lab 2

	A	B	C	D	E	F	G	H	I
29	Task 2: Analyse sustainability knowledge (SustainabilityData-10%sample)								
30	1) The CountIf function used in Lab1 used a fixed criterion to determine what to count in a specified range of data (eg 2 or 3). The criterion can also refer to a cell, a								
31	Use the following table to determine the frequencies of each response level for the two 'Subjective Knowledge' questions (columns AI, AJ and AK in the Sustaina								
32	2) In cell C38 of this sheet, enter the formula: =COUNTIF('SustainabilityData-10%sample'!AI\$5:AI\$108,\$B38) [you can copy this, and paste into the formula bar fo								
33	(We use a \$ in front of the numbers so the specified range stays the same when you copy formula down the next six rows;								
34	AND in front of the letter B so that the criterion stays the same when you copy across to columns D and E)								
35	3) Now copy cell C38, and Paste Special > Formulas into C38:E42 (select the whole range when Pasting Special) [dragging will overwrite the green and red box form								
36									
37		Response level	Column AI: "I have a good knowledge about sustainability issues"	Column AJ: "I know a great deal about sustainability and sustainability matters"	Column AK: "I know a lot about sustainable products and brands sold in the marketplace"	% of respondents in column AI	% of respondents in column AJ	% of respondents in column AK	
38		1							Note Excel requires single quotes with : a different sheet. Ti current sheet. The desired cell (or rang
39		2							
40		3							
41		4							
42		5							
43		Total	0	0	0				
44									
45	Task 3: Contrast subjective knowledge of sustainability								
46	1) Calculate the % of total respondents in cells C38:C42 above. For cell F38, enter the formula =C38/C\$43*100								
47	2) What would happen if you copy that formula into cell E39 WITHOUT that dollar sign? (see hint in row 33 above)								
48	3) Do the equivalent calculations for G38:H42								
49	4) How knowledgeable do respondents claim to be?								
50	Answer:								
51									

Dashboards: Lab 2



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Task 4: Create a bar chart of the three % columns in Task 2 above

1) Select cells F37:H42 above, and Insert ribbon > insert Column > 2D Column.

2) Click on the first bar (mine is blue), and inspect the formula bar.

`=SERIES('Lab2'!$F$37,, 'Lab2'!$F$38:$F$42,1)`

This shows the information Excel uses to construct the chart

The first information (Lab2!\$F\$37) determines what goes into the legend.

The second bit is the X-axis info, and is currently missing (nothing shows between the commas)

The third bit is the data range; and the fourth is the series order

(change this number, and the series shift around in the legend, and may change colour)

3) In this case it doesn't matter that there is nothing in the X-axis range, as the default (1-5) corresponds to our data anyway (scale levels)

If you want to fix it anyway, click between the commas in the formula bar, and select B38:B42 above. The result should

Do this to any one series, and it fixes it for all three. Magic!

Please note: Excel is really picky about what information it needs: the apostrophes ('Lab2!'), exclamation mark and commas are vital!

4) Give the Chart a better Title, and add X-axis and Y-axis labels: Chart Design ribbon > Add Chart Element > Axis Titles > tu

Chart Design

Add Chart
Element

Axes

Axis Titles

Chart title

Data labels

Primary Horizontal

Primary Vertical

Place Chart here: Column ↓ (and scroll down for Task 5)

Task 5: Create a Pie Chart of the Gender split of the sample % Male vs Female vs Diverse

We created the table in Lab 1 (Task 5 Step 6), so generate a Pie Chart from that table, and cut and paste below (to shift it to this

Dashboards: Lab 2

117	Bonus Task 6: Create a Line Chart of the quarterly price of petrol from 1974 to 2026, in nominal and real prices										
118	The Ministry of Business, Innovation and Employment publishes quarterly fuel prices, in cents per Litre:										
119	https://www.mbie.govt.nz/building-and-energy/energy-and-natural-resources/energy-statistics-and-modelling/energy-statistics/energy-prices										
120	With such a long time series (something we will return to in Topic 11), there has been a lot of inflation along the way, so we can't just look at (nominal) prices at the time (eg 12.3 c/L in March										
121	The Reserve Bank publishes the Consumer Price Index going back to 1920, and we can 'inflate' historical prices to their current 'real' equivalent values.										
122	https://www.rbnz.govt.nz/statistics/series/economic-indicators/prices										
123	6.1: In the following data series, complete the CPI inflator series (E131:BB131), by continuing the pattern on from B131:D131. (How: select B131:D131, then drag the bottom-right corner drag handle across)										
124	6.2: Then complete the Real (inflated) petrol price (E129:BB129), and construct a Line chart of the real (inflated) and nominal petrol prices from 1974 to 2026										
125											
126	March Year:	Mar-74	Mar-75	Mar-76	Mar-77	Mar-78	Mar-79	Mar-80	Mar-81	Mar-82	Mar-
127	Petrol (c/l) regular (nominal)	12.30	16.40	26.10	26.10	28.70	30.70	46.50	56.00	60.00	68.0
128	Petrol (c/l) (real: MBIE)	183.66	216.30	293.80	258.49	247.97	240.28	307.42	321.26	297.22	299.0
129	Petrol (c/l) (real: inflated)	185.05	217.42	296.17							
130	CPI Index	89	101	118	134	154	170	201	231	268	30
131	CPI Inflator	15.045	13.257	11.347							
132											
133	6.3: Insert Line Chart here for bonus marks: include all three data series in rows 127 and 129										
134	{NB: your chart might be created over in columns AR onwards: cut it there, and paste it here, then resize										
135	NB: the lines should converge over time, and be the same at March 2026. I show a version of this chart in the Topic 2 lecture										
136											
137											
138											
139											
140											

Line Chart

<https://www.mbie.govt.nz/assets/Data-Files/Energy/nz-energy-quarterly-and-energy-in-nz/energy-prices-2026-q1.xlsx>

